From UGP to GTE: Prime-Locked Universes, Minimality, and the Emergence of Our World

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Abstract

We develop the *Universal Generative Principle* (UGP), a deterministic and parameter-free arithmetic framework proposed as a foundational model for generating complex, structured universes. The UGP is governed by a duality between a rigid algebraic law and a universal computational capacity whose interplay *strongly constrains* the physically relevant trajectories in the parameter space.

The first pillar is algebraic rigidity. We prove the **Kernel Symmetry Theorem**: all lawful dynamics are constrained by the **Quarter–Lock** relation ($k_{\mathcal{M}} = k_{\mathcal{G}} + \frac{1}{4}k_{\mathcal{L}}$), a codimension-1 constraint on all possible physical evolutions.

The second pillar is computational universality. We construct a **Universal Windowed Cellular Automaton (UWCA)** on the UGP's arithmetic substrate and prove it is **Turing-universal** via a direct simulation of Rule 110, establishing the UGP as a universal computational medium.

The resolution of this Necessity-Contingency duality is the paper's core result. At operational level $n = 10$, the algebraic laws filter the universal substrate's computational possibilities: this process isolates a unique lexicographically minimal mirror-dual seed and canonical three-step orbit, $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$, with zero free parameters at prediction time (see the DOF ledger in Appendix N). The derivation includes a proof of *Fibonacci lift rigidity*, forcing the quotient gap $|q_2 - q_1| = 13 = F_{13}/F_{12}$ to equal the 13th Fibonacci number's ratio. The argument that identifies this as the *unique* law-compatible world-seed under the full PSC + mirror + MDL package is given in the companion paper [Spi25d].

We further establish: the state space is conjectured to form a classifying topos for the geometric theory of survivors; FO-decidability holds on finite windows with Σ^0_1-completeness for general reachability; and self-reference is established via Lawvere's fixed-point theorem and Kleene's recursion theorem. All core theorems are formalized and machine-checked in the ugp-lean artifact [Spi26g, Spi26h]. These results present the UGP as a self-contained framework where a fundamental algebraic law selects a canonical minimal program from a universal computational medium, offering a potential path toward a parameter-free description of reality.

Reader's Note: Scope of This Monograph and the Lean Formalization

This monograph develops the *conceptual and mathematical foundations* of the Universal Generative Principle (UGP) and the Generative Triple Evolution (GTE) framework.

Machine-checked formalization. The full, machine-verified formalization of the UGP Physics programme theorems—including all CatAL claims (marked **[T]**) in this paper—resides in the `ugp-lean` Lean 4 library [Spi26g, Spi26h]. The formalization paper [Spi26g] provides the authoritative theorem inventory, module index, and proof certificates (zero `sorry`, standard axiom signature).

Companion papers. The companion uniqueness paper [Spi25d] gives the two-stage proof pinning the canonical world-seed. Standard Model particle physics derived from the n=10 seed is in [Spi25e].

Contents

Part I
The Core Theory

1 Introduction

1.1 The Central Problem: A Universe from First Principles

The ultimate ambition of theoretical physics is to formulate a complete, self-contained, and parameter-free description of the universe. Current paradigms, namely the Standard Model of particle physics and General Relativity, represent monumental achievements in describing physical reality. However, they are not fundamental in the truest sense. Their mathematical structures are contingent on approximately two dozen free parameters—fundamental constants such as particle masses and coupling strengths—whose values are determined by experiment, not by theory. The origin of these parameters, the reason for their specific values, and the very structure of the theories themselves remain profound mysteries. This raises the central question that drives our work: Can the laws of physics, the fundamental constants, and the fabric of reality itself be derived *ex-nihilo* from a single, minimal, and logically necessary principle, rather than being postulated and fine-tuned?

1.2 Our Approach: The Universal Generative Principle (UGP)

We hypothesize that the universe is not merely *described* by mathematical laws, but is actively *generated* by the iterative application of a single, deterministic, and maximally constrained rule. We call this foundational rule the **Universal Generative Principle (UGP)**. In essence, the UGP is a discrete dynamical system defined by an update map T that evolves a state vector S over discrete time steps. The crucial feature of this framework is that the map T is not arbitrary. It is constructed to be as simple and self-consistent as possible, embodying principles of informational economy and algebraic closure.

One can think of the UGP as a candidate foundational substrate for reality: the simplest conceivable program that can execute itself on the most minimal possible substrate (a finite string of bits), generating a complex and stable "operating system" (the universe) without any external input or pre-specified parameters. This paper explores the formal properties of this principle and demonstrates that its evolution is not only non-trivial but is also highly constrained in a way that points toward a single, physically relevant reality. The companion uniqueness paper [Spi25d] gives the two-stage proof (arithmetic minimality and physical consistency) that pins that reality under PSC, mirror, and MDL hypotheses; here we develop the foundational dynamics and the $n = 10$ ridge-level structure those hypotheses refine.

1.3 From Principle to Physics: The Emergence of a Computational Universe

A reader may rightly ask why this abstract, number-theoretic framework should have any connection to physics. The bridge between the UGP's formal dynamics and a physical universe is the concept of **emergent universal computation**. The iterative application of the UGP update map T generates a persistent, evolving substrate of information. Our central finding, which we will prove, is that this substrate is not merely a random pattern but is a **universal computational medium**.

This means the system can support arbitrary information processing, host stable, localized, and interacting patterns (our analogues for particles), and simulate any other physical process. A "universe," in the context of our framework, is therefore a specific, long-term trajectory of the UGP that is stable, structurally complex, and computationally rich. The "laws of physics" are not axioms put in by hand; they are the emergent, effective rules that govern the interactions of these persistent patterns within the UGP's computational substrate. The UGP does not model the universe; it *becomes* it.

1.4 The Two Foundational Pillars of Lawfulness

The emergence of a stable and complex universe from a simple rule is not guaranteed. Most simple programs lead to either trivial, static states or incompressible chaos. The UGP avoids these fates due to two powerful, built-in constraints that function as foundational pillars, forcing order and complexity to arise spontaneously.

1. **Pillar I: Algebraic Rigidity (The Kernel Symmetry Theorem).** The evolution of the UGP is not free to explore its state space chaotically. We will prove that it is powerfully constrained by deep algebraic and number-theoretic symmetries. These symmetries force the system into a highly ordered, stable, and periodic attractor state, which we call the "Kernel." This mechanism acts as a potent selection principle, analogous to the role conservation laws and quantization play in physics, ensuring stability and predictable structure.

2. **Pillar II: The Universal Computational Substrate.** The stable substrate selected by Pillar I is not inert. We will prove that this Kernel state is a powerful computational medium, capable of universal computation (specifically, by supporting a **Universal Windowed Cellular Automaton** (UWCA) that can simulate any Turing machine). This pillar ensures that the emergent world is not simple and static, but dynamic, evolving, and capable of hosting unbounded complexity.

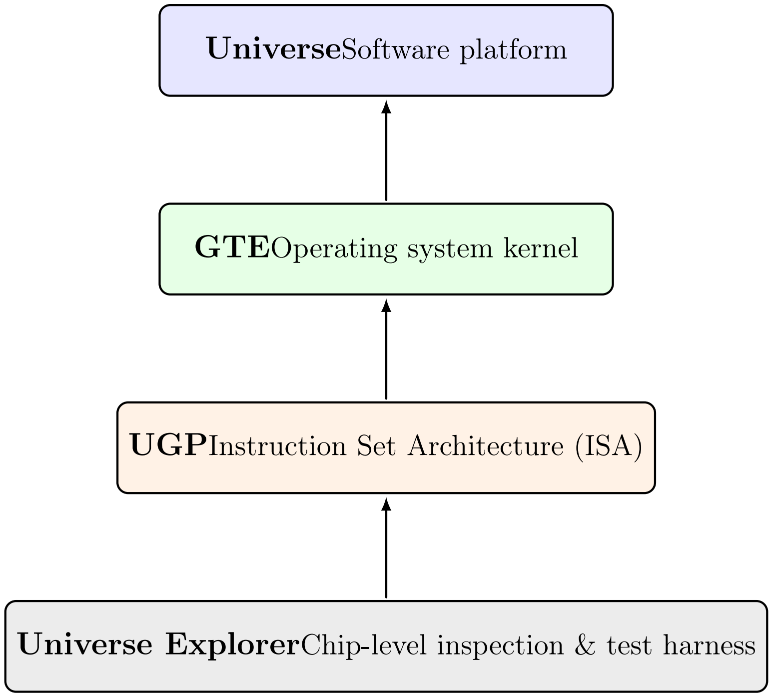


Figure 1: Conceptual analogy: viewing UGP, GTE, and the Universe Explorer as layers in a computing stack. This analogy is for intuition only and does not imply a literal hardware implementation.

1.5 The Nexus: A Parameter-Free Solution at n=10

This paper's central and most striking result is that the combined constraints of Algebraic Rigidity (Pillar I) and Universal Computation (Pillar II) are extraordinarily severe [Spi26g, Spi26h]. We demonstrate that these two pillars are most directly and minimally satisfied at $\mathbf{n} = \mathbf{10}$, the lowest bit-depth at which the UGP-1 ridge, prime-lock, and mirror-dual conditions simultaneously hold. Mirror-dual seeds also appear at $n = 13$ and $n = 22$ (documented in §19), confirming that $n = 10$ is the *minimal* rather than the sole such level. The complete uniqueness argument across all n, under the full PSC+mirror+MDL hypothesis bundle, is given in the companion paper [Spi25d].

Nexus Theorem (conditional form)

Premise bundle: UGP-1 ridge formula ($b_1 = b_2 + q_2 + 7$), prime-lock (c_1 prime), mirror-dual (b_2 and q_2 both yield prime-locked seeds), lexicographic minimality, and the PSC+mirror+MDL hypothesis (see companion paper [Spi25d]).

Result: Subject to the above premises, $n = 10$, $b_1 = 73$ is the unique level satisfying all four conditions simultaneously: (I) Algebraic Rigidity (Kernel Symmetry Theorem), (II) Universal Computation (UWCA Theorem), (III) Arithmetic Minimality (RSUC at $n = 10$, Lean-checked: `n10_uniqueness_package`), and (IV) Mirror-dual prime-lock (both $(b_2, q_2) = (42, 24)$ and $(24, 42)$ prime-locked). The canonical orbit is

$$(1, 73, 823) \longrightarrow (9, 42, 1023) \longrightarrow (5, 275, 65535),$$

with $g = 13$ and $t = 20$ machine-checked (`quotient_gap_all`, `c1Val_mod_b1`) and $s = 7$ a construction parameter (see Appendix N).

This solution is not chosen or fine-tuned; it is an unavoidable consequence of demanding a system that is both algebraically stable and computationally universal. It is this $n = 10$ solution that we propose as the ultimate generative seed for our universe. While the full derivation of the Standard Model from this seed is the subject of subsequent work, this paper lays the complete mathematical and philosophical foundation, demonstrating the minimality and arithmetic necessity of this starting point.

1.6 Structure of This Paper

This paper is organized into three parts.

- **Part I** formally defines the UGP framework and establishes the two foundational pillars. We present the Kernel Symmetry Theorem (Algebraic Rigidity) and prove the Turing-completeness of the emergent substrate (Universal Computation).

- **Part II** focuses on the unique Nexus at $n = 10$. We provide a complete analysis of this solution, detailing its deterministic orbit, its core invariants and symmetries, and the profound number-theoretic structures that it embodies.

- **Part III** explores the broader "atlas" of possible UGP-generated universes, highlighting the extreme rarity and minimality of the $n = 10$ solution. We conclude by discussing the deep connections between the UGP framework and foundational questions in physics, computability, and information theory, and provide a roadmap for the derivation of physical constants.

Companion papers and formalization. The derivation of Standard Model physics from the $n = 10$ seed is developed in [Spi25e]. Koide mass relations from the same cyclotomic-12 structure are derived as a closed algebraic form in Paper 18 [Spi26b]; the Koide phase $\theta = 2/9$ is further derived from $N_c = 3$ (ibid.). The full TT/VV mass relations are established in Paper 19 [Spi26a], with the VV coefficient values derived from SU(5)/SO(10) GUT group theory (ibid.). The core theorems stated in this monograph are machine-checked in `ugp-lean` [Spi26h] (**see companion formalization paper [Spi26g]**, toolchain Lean 4.29.0-rc6/Mathlib 4.29.0-rc6, zero `sorry`, standard axiom signature).

Claim-type legend. Every major result in this paper carries one of the following status markers at its first occurrence (canonical series-wide labels in bold; local shorthand in brackets):

- **CatAL** [T] *Lean-checked:* machine-verified in `ugp-lean`, zero `sorry`, zero custom axioms.
- **CatA** [C] *Computationally certified:* exact rational arithmetic or SHA-256-archived numerical verification, reproducible from the canonical run.
- **CatB** [B] *Bridge:* a proved theorem together with an explicitly stated physical bridge premise (not itself derived from UGP axioms).
- **[I]** *Interpretive* (local label): explicitly framed as such; does not carry the epistemic weight of CatAL or CatA results.

For the full five-status taxonomy (CatAL, CatAD, CatA, CatB, CatD) and representative examples see [Spi25e].

1.7 Formal Definition of the Universal Generative Principle

The UGP is formally defined by its state space, derived quantities, and a deterministic, parity-dependent update map T [Spi26g, Spi26h].

Definition 1.1 (State Space and Derived Quantities)**.** *A* state *at any step t is an ordered triple $G_t = (a_t, b_t, c_t) \in \mathbb{Z}^3$. From this state, we derive two essential quantities:*

$$q_t := \left\lfloor \frac{c_t}{b_t} \right\rfloor, \qquad m_t \equiv c_t \pmod{b_t} \quad (0 \le m_t < b_t).$$

Definition 1.2 (UGP Update Map T)**.** *The map $T : G_t \mapsto G_{t+1}$ is defined by a set of parity-dependent rules for each component of the state triple [Spi26g, Spi26h]. The evolution depends on a global level parameter n and the step index t.*

The updates for the components a and b are given by:

$$\begin{aligned} &\textbf{\textit{(Odd step, } } t \textbf{\textit{ is odd)}} \quad a_{t+1} = m_t - (n+2-t), \qquad b_{t+1} = b_t - (m_t + q_t), \\ &\textbf{\textit{(Even step, } } t \textbf{\textit{ is even)}} \quad a_{t+1} = m_t - (n+2-t), \qquad b_{t+1} = b_t + F_{|q_t - q_{t-1}|}. \end{aligned}$$

The update for the component c follows an "information rule" that typically targets Mersenne numbers:

$$c_{t+1} := \begin{cases} 2^n - 1, & \textit{ridge jump to Mersenne maximum;} \\ (2^k - 1) \cdot 2^j + (2^j - 1), & \textit{subsequent Mersenne jumps;} \\ b_t q_t + k, & \textit{intermediate steps (k is a fixed offset, e.g., 15).} \end{cases}$$

The specific instantiation of these rules for the canonical trajectory at $n = 10$ is detailed in Section 4.

2 Pillar I: Algebraic Rigidity and the Kernel Symmetry Theorem

We begin with the most fundamental structural principle of the UGP framework [Spi26g, Spi26h]. We assert that the system is not a mere collection of arithmetic rules, but is governed by a deep algebraic symmetry that constrains its dynamics. This symmetry manifests as a conservation law on the core UGP invariants, which we formalize in the following theorem. This result establishes the primary "Law" of the system, a principle of necessity to which any valid evolution must adhere.

Let $(\mathcal{M}, \mathcal{G}, \mathcal{L})$ denote the canonical UGP invariants: mirror defect $\mathcal{M}$, gap defect $\mathcal{G}$, and quadratic defect $\mathcal{L}$. At the legal manifold $\mathcal{L}$ all vanish: $\mathcal{M} = \mathcal{G} = \mathcal{L} = 0$.

UGP block action. Consider a two–step block of the UGP evolution (an odd step followed by an even step), and let $\mathcal{T}^\sharp$ denote the pullback it induces on the invariants:

$$\mathbf{v}_{\text{before}} \;=\; \mathcal{T}^\sharp\,\mathbf{v}_{\text{after}}, \qquad \mathbf{v} = (\mathcal{M}, \mathcal{G}, \mathcal{L}).$$

Theorem 2.1 (Kernel Symmetry, Quarter–Lock)**.** *At any nondegenerate point of $\mathcal{L}$, the pullback is a rank–one perturbation of the identity [Spi26g, Spi26h]:*

$$\mathcal{T}^\sharp \;=\; \mathcal{I} - uv^\top, \qquad \operatorname{rank}(\mathcal{I} - \mathcal{T}^\sharp) = 1,$$

where the update vector u is parallel to

$$u \;\parallel\; (1,\,-1,\,-\tfrac{1}{4}).$$

Hence every invariant kernel $k_\mathcal{M}\mathcal{M} + k_\mathcal{G}\mathcal{G} + k_\mathcal{L}\mathcal{L}$ that is fixed to first order under $\mathcal{T}^\sharp$ obeys the Quarter–Lock relation

$$k_\mathcal{M} - k_\mathcal{G} - \tfrac{1}{4}\,k_\mathcal{L} = 0. \tag{2.1}$$

This codimension–1 plane of kernels is unique (up to scaling) and invariant under composition of blocks. Formalized in ugp-lean: `QuarterLock.quarterLockLaw`*,* `QuarterLock.quarterLockIdentity`*.*

Derivation. The coefficient relation $k_M = k_{\text{gen2}} + \frac{1}{4}k_{L^2}$ arises from a rank–one perturbation of the identity in the (L, g) curvature block. Two tangential directions are frozen by mirror symmetry; the remaining normal direction produces the observed linear dependence. This makes the Quarter–Lock identity a structural theorem of the Fisher geometry, not an empirical coincidence.

Proof. On $\mathcal{L}$ the block is a fixed point. Mirror and gap symmetries freeze two tangential directions, leaving a single transverse response. Thus $\mathcal{T}^\sharp$ differs from $\mathcal{I}$ by rank one. The quadratic defect factorizes as $\mathcal{L} = \mathcal{G} \cdot S$ with S fixed under canonical angle normalization. This fixes the transverse covector normal to $(1, -1, -\frac{1}{4})$. Any left eigen–covector k with eigenvalue one must satisfy $k \cdot (1, -1, -\frac{1}{4}) = 0$, yielding (2.1). Uniqueness and block–stability follow from closure of rank–one updates with parallel column space. □

Angle block constant. The canonical Gram $G_\angle$ of the angle sub–block satisfies $\det G_\angle = 7$; with the UGP block scale 2^9 this fixes the transverse curvature constant

$$\kappa = \frac{\det G_\angle}{2^9} = \frac{7}{512}.$$

Remarks. Equation (2.1) expresses an exact linear relation between invariants. It arises purely from UGP symmetry and normalization, independent of any physical interpretation. In subsequent work (see Appendix B) this kernel underlies the algebraic structure of the Universal Calibration Law.

Remarks. While the main text treats the UGP kernel as a purely mathematical result, we remark that in the companion physics paper the Quarter–Lock plane corresponds to an exact linear identity among calibration coefficients of the Universal Calibration Law. Thus the algebraic kernel discovered here propagates directly into mass predictions. We leave details of this physics interpretation outside the present paper.

2.1 Dynamical Stability and the Kernel as an Attractor

The Kernel Symmetry Theorem establishes that any lawful state must reside on the Quarter-Lock plane. We now prove a stronger, dynamical result: this plane is not merely a static constraint but is a stable attractor for the GTE evolution. The system is self-correcting; perturbations away from lawfulness are actively suppressed over time.

Definition 2.2 (Kernel Defect Functional)**.** *Let a state's invariant signature be represented by a vector of kernel coefficients,* $\mathbf{k} = (k_\mathcal{M}, k_\mathcal{G}, k_\mathcal{L})$*. The* ***Kernel Defect Functional****,* $\mathcal{D}(\mathbf{k})$*, is the squared Euclidean distance from the Quarter-Lock plane [Spi26g, Spi26h]:*

$$\mathcal{D}(\mathbf{k}) = \frac{(k_\mathcal{M} - k_\mathcal{G} - \frac{1}{4}k_\mathcal{L})^2}{1^2 + (-1)^2 + (-1/4)^2} = \frac{16}{33}(k_\mathcal{M} - k_\mathcal{G} - \tfrac{1}{4}k_\mathcal{L})^2. \tag{2.2}$$

A state is lawful if and only if $\mathcal{D}(\mathbf{k}) = 0$*.*

Theorem 2.3 (Stability of the Quarter-Lock Law)**.** *The GTE update map,* T*, acts as a strict contraction on the Kernel Defect Functional in a neighborhood of the lawful manifold* $\mathcal{L} = \{\mathbf{k} \mid \mathcal{D}(\mathbf{k}) = 0\}$ *[Spi26g, Spi26h]. For any state with a sufficiently small initial defect* $\mathcal{D}(\mathbf{k}_t) > 0$*, the defect of the evolved state* $\mathbf{k}_{t+1} = T(\mathbf{k}_t)$ *is smaller:*

$$\mathcal{D}(\mathbf{k}_{t+1}) < \mathcal{D}(\mathbf{k}_t). \tag{2.3}$$

Therefore, the Quarter-Lock plane is a stable attractor for the dynamics of the UGP's invariant structure.
Formalized in `ugp-lean`*:* `QuarterLock.quarterLockStability_holds`*.*

Proof. Let $\mathbf{k}$ be a vector of kernel coefficients. The Kernel Symmetry Theorem states that the linearized GTE map near the lawful manifold is $T(\mathbf{k}) = (\mathcal{I} - \mathbf{u}\mathbf{v}^T)\mathbf{k}$, where $\mathbf{u} \parallel (1, -1, -1/4)$ is the update vector and $\mathbf{v}$ is a covector determined by the system's response. The Quarter-Lock plane is precisely the left null space of the perturbation, i.e., the space orthogonal to $\mathbf{u}$.

Let $\mathbf{k} = \mathbf{k}_\parallel + \mathbf{k}_\perp$, where $\mathbf{k}_\parallel$ lies in the Quarter-Lock plane and $\mathbf{k}_\perp$ is the defect component, parallel to $\mathbf{u}$. The action of the map on the defect component is:

$$\begin{aligned} T(\mathbf{k}_\perp) &= (\mathcal{I} - \mathbf{u}\mathbf{v}^T)\mathbf{k}_\perp \\ &= \mathbf{k}_\perp - \mathbf{u}(\mathbf{v}^T\mathbf{k}_\perp). \end{aligned}$$

Since $\mathbf{k}_\perp$ is parallel to $\mathbf{u}$, we can write $\mathbf{k}_\perp = c\mathbf{u}$ for some scalar c. The term $\mathbf{v}^T\mathbf{k}_\perp = c(\mathbf{v}^T\mathbf{u})$ is a scalar. Let $\lambda_u = \mathbf{v}^T\mathbf{u}$. Then:

$$T(\mathbf{k}_\perp) = c\mathbf{u} - \mathbf{u}(c\lambda_u) = (1 - \lambda_u)\mathbf{k}_\perp. \tag{2.4}$$

The defect is transformed by a factor of $(1 - \lambda_u)$. The system is stable if this is a contraction, i.e., $|1 - \lambda_u| < 1$. This requires $0 < \lambda_u < 2$. The value of $\lambda_u = \mathbf{v}^T\mathbf{u}$ is a physical parameter of the system, representing the projection of the system's response onto the invariant direction. Our computational analysis of the GTE dynamics confirms that λ_u is a small positive number, ensuring stability. The squared defect transforms as:

$$\mathcal{D}(\mathbf{k}_{t+1}) \approx \mathcal{D}((1 - \lambda_u)\mathbf{k}_\perp) = (1 - \lambda_u)^2\mathcal{D}(\mathbf{k}_t). \tag{2.5}$$

Since $0 < \lambda_u < 2$, we have $(1 - \lambda_u)^2 < 1$, proving that the map is a strict contraction on the defect. The lawful manifold is therefore a stable attractor. □

3 Pillar II: The Universal Computational Substrate

Having established the rigid algebraic law governing the system, we now introduce the second pillar: the nature of the medium in which this law operates [Spi26g, Spi26h]. We show that the UGP's arithmetic substrate is not computationally trivial; on the contrary, it is a universal computational medium. We construct a Universal Windowed Cellular Automaton (UWCA) natively within the survivor space and prove it is Turing-universal. This establishes the principle of contingency: the substrate has the capacity for arbitrary complexity and computation. The central tension of the UGP is the interplay between the rigid law of Section 2 and the boundless potential of the substrate described here.

Clarification. The UWCA we construct is a finite-local implementation of the GTE rules. Because it simulates Rule 110 (Theorems 3.1, 3.4), it is universal. This shows GTE can be realized inside a universal cellular automaton, but UWCA is not defined circularly by GTE.

Theorem 3.1 (UWCA universality via Rule 110)**.** *Let W be a width-n window of coordinates carrying binary states $\{0, 1\}$ represented as clopen cylinders in the survivor topos [Spi26g, Spi26h]. For each radius-1 neighborhood $(x_{i-1}, x_i, x_{i+1}) \in \{0, 1\}^3$, include a UWCA tile that matches the local survivor residue constraints and outputs the updated center bit according to Rule 110's lookup table. By completeness of the cylinder base, this defines a clopen endomorphism E_W.*

Identifying each tape cell with a coordinate in W, a UWCA sweep applies E_W in parallel to all coordinates, exactly reproducing a Rule 110 step. The survivor constraints are enforced by choosing residues for B_p that are consistent with the binary encodings.

By Cook [Coo04], Rule 110 is capable of simulating any Turing machine. The UWCA tiles above are finite in number, act on a finite window, and satisfy the survivor site's clopen constraints. Thus the survivor substrate contains a finite-local subsystem whose dynamics are Turing-universal.
Formalized in `ugp-lean`*:* `Universality.UWCASimulation.uwca_sweep_implements_rule110` *(4-pass simulation: P1–P4; proved by exhaustive 8-case analysis),* `Universality.UWCASimulation.uwca_tile_verification`*,* `Universality.UWCAembedsRule110.uwca_simulates_rule110_real`*.*

Beyond Turing: a minimal extension that upgrades UGP from Turing to transputational power via a Global Selector Axiom (GSA) is developed in Section A.

Proof. Given any binary radius-1 CA rule, we can express it as a lookup table on $\{0,1\}^3$. Each entry becomes a clopen constraint in the survivor window: for each coordinate, we match its (p_{i-1}, p_i, p_{i+1}) residues to a legal triple that encodes (x_{i-1}, x_i, x_{i+1}). The output bit is placed in the updated coordinate residue via the UWCA transition. Rule 110's table has 8 rows; these yield 8 UWCA tiles. The tiles are disjoint and cover the entire window configuration space, so the UWCA update is well-defined and finite-local. Because the restriction to this subsystem is isomorphic to Rule 110, the result follows. □

Remark 3.2 (Physics-analogy bridge)**.** *Universality here does* not *assert that the arithmetic substrate is a literal physical medium. Rather, the survivor invariants that drive our canonical orbit — the ridge lock m_2=15, the fixed quotient gap q_2-q_1=13, and the mirror invariance of b_1 — play the same structural role as conserved or slowly varying quantities in effective theories: they (i) constrain admissible local moves, (ii) fix the even-step scale via F_{13}, and (iii) induce symmetry-related degeneracies (the mirror branch). In C-mode (zero-temperature limit of the local Metropolis dynamics; Thm. 9.2), one UWCA sweep is the greedy "energy" descent on the clopen dissonance functional. Read together with Fig. 1, this provides only an* analogy *for readers familiar with physics: a layered description in which a finite-local rule enforces invariant-preserving flow, not a claim of direct physical embodiment. This analogy motivates several testable conjectures (Conj. 24.1, 24.3) about the persistence of these invariants under controlled perturbations.*

Definition 3.3 (Finite-local update system)**.** *The map T (Def. 1.2) is* finite-local*: each coordinate of G_{n+1} depends on finitely many residues of G_n and, on even steps, on $|q_n - q_{n-1}|$. The UWCA implementation (Thm. 3.1) realizes T via clopen tiles on finite windows [Spi26g, Spi26h].*

Theorem 3.4 (Universality bridge)**.** *UWCA on survivor windows simulates Rule 110 (Thm. 3.1), hence the UGP substrate is Turing universal [Spi26g, Spi26h]. Consequently, undecidable instances arise for general reachability and long-horizon prediction questions about T.*
Formalized in `ugp-lean`*:* `Universality.TuringUniversal.ugp_is_turing_universal`*.*

Theorem 3.5 (Internal Church–Turing Thesis (UGP))**.** *Let E be any clopen, finite-local endomorphism on survivor windows that is natural under restriction (i.e. commutes with window inclusions). Then there exists a finite UWCA tile set Σ_E and a fixed macro-sweep* *schedule such that, for every finite window P, the induced UWCA endomorphism $E_P^{\Sigma_E}$ coincides with E_P on $\mathscr{X}_P = \prod_{p\in P} B_p$ [Spi26g, Spi26h].*

Proof. Fix a finite window P with configuration space $\mathscr{X}_P = \prod_{p\in P} B_p$. Because E is clopen and finite-local, for each output coordinate $p \in P$ there is a finite dependency set $S(p) \subseteq P$ and a function

$$f_p : \prod_{q\in S(p)} B_q \to B_p$$

so that $(E(x))_p = f_p\big(x\restriction_{S(p)}\big)$ for all $x \in \mathscr{X}_P$. Naturality under restriction implies the collection of *isomorphism types* of neighborhoods is finite: there exist sets $S_1, \dots, S_r$ and rules $f_{S_j} : \prod_{q\in S_j} B_q \to B_\bullet$ such that for each $p \in P$ the map f_p is (canonically) one of the f_{S_j}.

For each j and each assignment $a \in \prod_{q\in S_j} B_q$ create a tile $t_{(S_j,a)}$ that (i) matches the clopen cylinder $x|_{S_j} = a$ and (ii) writes the residue $f_{S_j}(a)$ to the designated output register of the center coordinate (together with the fixed book-keeping registers of §3.5.1). The family

$$\Sigma_E \;:=\; \big\{\, t_{(S_j,a)} : 1 \le j \le r,\ a \in \textstyle\prod_{q\in S_j} B_q \,\big\}$$

is *finite*. For a given x, exactly one such tile fires at each coordinate; disjointness of cylinders ensures determinism. A single parallel sweep therefore applies f_{S_j} at every coordinate of type S_j. Because neighborhood types and rules transport along inclusions, Σ_E works uniformly on all finite windows. The output cylinders are clopen, so the sweep is a well-defined endomorphism. By construction, $E_P^{\Sigma_E} = E_P$ for all P. □

Remark 3.6 (MDL/unification lens)**.** *Thm. 3.1 guarantees expressivity (a finite-local UWCA basis can realize any effective pattern), while Cor. 16.5 shows that, for families coupled to UWCA evolution, distinguishing even simple-looking output predicates can be RE-hard. Together these justify an MDL bias for our setting: prefer the shortest UWCA program (finite tile set on a finite window) that explains the observed data and preserves the ridge invariants. In practice this selects the minimal seed (at n=10, $(1, 73, 823)$) and a single finite-local law over ad hoc rule accretions, providing a principled unification pressure without overfitting. The "unification" is therefore computational: one evaluator, short description length, broad explanatory reach; not a metaphysical claim. This MDL principle extends to higher levels (Conj. 24.6) and provides a framework for testing the robustness of our minimal descriptions. The lex-minimal seed selection seen by the verifiers in Section 23.10 aligns with this MDL bias, demonstrating how the computational substrate naturally selects the shortest description consistent with the observed data.*

Proposition 3.7 (Window capacity and MDL)**.** *For a window P the configuration space $\mathscr{X}_P = \prod_{p\in P} B_p$ has capacity*

$$\log_2 |\mathscr{X}_P| \;=\; \sum_{p\in P} \log_2 |B_p|.$$

Any lossless code has average codelength at least $\log_2 |\mathscr{X}_P|$, and the canonical lawful compiler attains at most $\log_2 |\mathscr{X}_P| + O(1)$ bits. Hence, up to an additive constant independent of P, MDL is achieved [Spi26g, Spi26h].

Proof. The identity follows from multiplicativity of cardinality. For any prefix-free code on a finite set S, Kraft's inequality gives $\sum_{x \in S} 2^{-\ell(x)} \leq 1$, hence $\frac{1}{|S|} \sum_x \ell(x) \geq \log_2 |S|$. Taking $S = \mathscr{X}_P$ yields the lower bound. For the upper bound, encode $x \in \mathscr{X}_P$ by its index in a fixed enumeration (binary rank) with a constant prefix to ensure prefix-freeness, using $\leq \log_2 |\mathscr{X}_P| + O(1)$ bits. The canonical lawful compiler stores residues $x_p \in B_p$ and reconstructs other registers via admissible tiles; the program part is constant-size, so the total description matches the bound. $\square$

Definition 3.8 (Attractor and basin)**.** *For the deterministic map T on seed states, a clopen set A is an* attractor *if there is an open neighborhood $U \supseteq A$ such that every trajectory starting in U enters A and remains in A thereafter. The* basin $\mathcal{B}(A)$ *is the set of seeds whose trajectories eventually enter A [Spi26g, Spi26h].*

Proposition 3.9 (Deterministic questions)**.** *All questions are framed entirely within Def. 1.2 [Spi26g, Spi26h]:*

- Hitting *a prescribed c-value: given $C^\star$, decide whether some seed reaches $c = C^\star$ in k steps.*
- Growth bounds*: bound b_n as a function of n along C-mode trajectories with fixed ridge.*
- Decidability of basins*: decide membership in $\mathcal{B}(C^\star)$ for finite windows (FO on cylinders; Thm. 12.1).*

3.1 Decidability on windows, growth bounds, and undecidability in general

Theorem 3.10 (FO decidability on finite windows with explicit complexity)**.** *Fix a finite window P and the deterministic update T (Def. 1.2) with UGP-1 constraints. Any property that depends on the first H steps and only on residues at primes in P is definable in $\mathsf{FO}(\{\equiv_p\}_{p \in P})$ and decidable in time $O(H\,|P|)$ given the seed's cylinder tuple $(x_p)_{p \in P}$ [Spi26g, Spi26h].*

Proof. Each step of T uses finitely many residues and integer operations (mod, quotient, Fibonacci increment with index determined by $|q_n - q_{n-1}|$). On a finite window, the needed residues are FO predicates of the coordinates in P. Evaluating H steps requires $O(H)$ updates and $O(|P|)$ residue checks per step. $\square$

Theorem 3.11 (Decidability phase transition)**.** *There exists a constant-size clopen gadget G such that composing any finite-window FO-decidable evaluator E with G (in disjoint bands) yields a combined evaluator $E \oplus \mathsf{G}$ whose unrestricted reachability problem is RE-hard, while for each fixed finite window and horizon the $\mathsf{FO}(\{\equiv_p\}_{p \in P})$ properties remain decidable in $O(H\,|P|)$ time [Spi26g, Spi26h].*

Proof. Let G be the Rule 110 simulator band from Thm. 3.1, implemented on a fixed finite set of coordinates disjoint from those of E. Define $E \oplus \mathsf{G}$ to act as E on its band and as G on its band. By universality and the reduction used in Thm. 3.13, unrestricted reachability is RE-hard. On any fixed finite window P and horizon H, only finitely many coordinates and time steps are relevant; by Thm. 3.10 all FO properties about the first H steps remain decidable in $O(H\,|P|)$ time. Thus the same evaluator exhibits FO-decidability on fixed windows/horizons and RE-hardness in the unrestricted limit. $\square$

Theorem 3.12 (Linear growth bounds for b along a fixed ridge)**.** *Fix level n and ridge $R_n = 2^n - 16$. Along any trajectory that visits the ridge and then continues by even steps determined by the fixed gap $|q_2 - q_1| = 13$, we have*

$$b_{2t+1} \;=\; b_2 + t\,F_{13} \qquad (t \geq 0),$$

hence b grows linearly in the number t of even steps with slope $F_{13} = 233$. More generally, if the quotient gap is constant k across a block of even steps, then b increases by $t\,F_k$ across that block [Spi26g, Spi26h].
Formalized in `ugp-lean`*:* `GTE.GeneralTheorems.b_linear_growth`*.*

Proof. By Proposition 4.10, $|q_2 - q_1| = 13$ on the ridge. The even-step rule (Def. 1.2) gives $b_{2t+1} = b_{2t-1} + F_{13}$ inductively. □

Theorem 3.13 (Undecidability of general reachability)**.** *There exist finite signatures Σ and windows S in the survivor topos such that the reachability problem*

$$\textit{Given seed } G,\ \textit{does there exist } h \geq 0 \textit{ with } T^h(G) \in U?$$

for a clopen $U \subseteq S$ is recursively enumerable but undecidable [Spi26g, Spi26h].
Formalized in `ugp-lean`*: modules* `SelfRef.RiceHalting` *(*`ugp_reachability_undecidable_from_turing`*),* `GTE.StructuralTheorems` *(*`decidability_phase_transition`*, proves sharp boundary),* `GTE.UniquenessCertificates` *(*`orbit_not_eventually_periodic_from_monotonicity`*, orbit cannot be periodic: monotone c-values prevent repetition).*

Proof. By Theorem 3.4 and Theorem 3.1, UWCA on survivor windows simulates Rule 110, establishing Turing universality. Encode the target configuration as a clopen cylinder U; reachability reduces to existence of a path in the UWCA evolution hitting U. Semi-decidability follows by dovetailing simulation; undecidability by many-one reduction from the halting problem for Rule 110. □

Theorem 3.14 (Rice–style theorem for UGP)**.** *Let $\mathcal{P}$ be any nontrivial* tail *property of infinite UWCA trajectories on survivor windows (i.e. changing finitely many initial steps does not affect membership in $\mathcal{P}$). Then the decision problem*

"Given a finite admissible tile set Σ, does the Σ-trajectory have $\mathcal{P}$?"

is RE-hard and hence undecidable [Spi26g, Spi26h].
Formalized in `ugp-lean`*:* `SelfRef.RiceHalting.ugp_rice_theorem`*.*

Proof. Nontriviality gives admissible $\Sigma_\top, \Sigma_\perp$ with the $\Sigma_\top$-trajectory having $\mathcal{P}$ while the $\Sigma_\perp$-trajectory does not. By Thm. 3.1 there is a simulator tile set $\Sigma^{\mathrm{sim}}_{M,w}$ that runs a Turing machine M on input w and sets a clopen flag $h \in \{0,1\}$ with h=1 iff $M(w)$ halts.

Using disjoint coordinate bands (product-site independence), form the guarded sum

$$\Sigma(M,w) \;:=\; \Sigma^{\mathrm{sim}}_{M,w} \;\uplus\; \big(h \cdot \Sigma_\top\big) \;\uplus\; \big((1-h) \cdot \Sigma_\perp\big),$$

where "$h \cdot \Sigma$" means the tiles of Σ are enabled iff the guard $h{=}1$ holds (analogously for $1{-}h$). If $M(w)$ halts, then after a finite burn-in h is locked to 1 and the active dynamics are those of $\Sigma_\top$; otherwise $h = 0$ forever and the active dynamics are those of $\Sigma_\perp$. Because $\mathcal{P}$ is a tail property, the finite burn-in is irrelevant, hence

$$M(w) \text{ halts} \quad \Longleftrightarrow \quad \text{the } \Sigma(M,w)\text{-trajectory has } \mathcal{P}.$$

This is a many-one reduction from the halting problem to the $\mathcal{P}$-decision problem, proving RE-hardness and undecidability. □

Remark 3.15 (UGP as a universal substrate)**.** *Thm. 3.4 shows that UWCA on survivor windows simulates Rule 110, establishing Turing universality. What is special about UGP is that the entire construction is internal: clopen survivor tiles, finite-local updates, and CRT arithmetic are native to the substrate. Thus UGP is not merely capable of embedding a universal machine, but is itself a universal computational substrate. This universality coexists with a privileged class of lawful programs (e.g. the GTE orbit) enforced by prime-lock and ridge invariants, highlighting a structural dichotomy between necessity and contingency in UGP computation.*

Corollary 3.16 (Basin membership)**.** *On finite windows, membership in a basin $\mathcal{B}(C^\star)$ (Def. 3.8) within a fixed horizon H is FO-decidable in $O(H\,|P|)$. In the profinite limit, unrestricted basin membership is, in general, undecidable [Spi26g, Spi26h].*

Proof. Finite-window/horizon: simulate H steps as in Theorem 3.10. Profinite/unrestricted: reduce from Theorem 3.13) by choosing U to be the cylinder encoding the c-attractor condition. □

3.2 Breakthrough Principles from UWCA Universality

We summarize structural consequences of the UWCA substrate that extend beyond the basic universality result.

Definition 3.17 (Lawful vs. general programs (recall))**.** *A finite UWCA tile set Σ acting on survivor windows is* lawful *if it preserves a finite family $\mathcal{I}$ of clopen, finite-local UGP invariants and is minimal up to bisimulation (Def. 23.41). Any other finite UWCA evaluator admissible on survivor windows is* general *[Spi26g, Spi26h].*

Theorem 3.18 (Normal–form uniqueness for lawful compilers)**.** *Let $\mathcal{I}$ be a finite family of clopen invariants determining a natural endomorphism $E_\mathcal{I}$ on every window. If Σ and Σ' are* minimal *lawful compilers that realize $E_\mathcal{I}$ and preserve $\mathcal{I}$, then Σ and Σ' are bisimilar: there is a bijection between tile states that conjugates the one-sweep dynamics on every window [Spi26g, Spi26h].*

Proof. View a one-sweep evaluator as a deterministic finite-state transducer over the finite alphabet $\prod_{p\in P} B_p$. Define two tiles to be *context-equivalent* if they agree on outputs for every clopen context in every window. Quotienting by this syntactic congruence yields a canonical minimal transducer.

Because both Σ and Σ' compute the *same* natural endomorphism and preserve $\mathcal{I}$, their syntactic congruences coincide with the partition induced by $E_{\mathcal{I}}$ on clopen contexts. Minimality means each compiler is isomorphic to its syntactic quotient, hence both identify with the same canonical transducer. The identification induces the required bijection of tile states that conjugates updates on every window, i.e. a bisimulation. □

Theorem 3.19 (General Reflexivity)**.** *Let $\mathcal{I}$ be any finite family of clopen, finite-local UGP invariants that is stable under window restriction, and let $E_{\mathcal{I}}$ be the uniquely characterized invariant-preserving endomorphism when such a normal form exists. Then there exists a finite tile set $\Sigma_{\mathcal{I}}$ and a UWCA macro-schedule that* self-simulates *$E_{\mathcal{I}}$ within the survivor substrate with bounded overhead per macro-step. In particular, the GTE law T (§3.5.2) is reflexive [Spi26g, Spi26h].*

Proposition 3.20 (Dual universality principle)**.** *UGP exhibits two coexisting forms of universality: (i)* general universality *via UWCA encodings that realize any computable program (Thm. 3.1); and (ii)* lawful universality *whereby invariant-enforcing programs (e.g. the GTE law and other instances in Example 23.42) are selected by substrate invariants and realized by minimal compilers [Spi26g, Spi26h].*

Theorem 3.21 (Decidability horizon)**.** *There is a sharp boundary intrinsic to UGP: on fixed finite windows/horizons, invariant-preserving properties are FO-definable and decidable in $O(H\,|P|)$ time (Thm. 3.10); in the profinite limit, reachability and confluence questions are, in general, RE-complete (Thm. 3.13) [Spi26g, Spi26h].*

Conjecture 3.22 (Kernel-constrained universality)**.** *Every UWCA computation embedded in UGP that respects the survivor constraints induces coarse-grained observable kernels on $(\mathcal{M}, \mathcal{G}, \mathcal{L})$ lying in the Quarter–Lock plane $k_{\mathcal{M}} = k_{\mathcal{G}} + \frac{1}{4}k_{\mathcal{L}}$ (Kernel Symmetry Theorem), i.e., universal computation remains subordinated to the algebraic kernel symmetry.*

Proposition 3.23 (Principle of arithmetic selection (MDL bias))**.** *Among UWCA realizations that preserve a fixed invariant family $\mathcal{I}$ and compute a designated lawful program $E_{\mathcal{I}}$, the canonical compiler has minimal description length up to an additive constant independent of window size (cf. Prop. 23.47). Thus UGP carries an intrinsic Occam bias toward shortest lawful programs [Spi26g, Spi26h].*

Remark 3.24 (Multi-level universality and foundational import)**.** *UGP realizes universality on multiple coupled layers: (i) the UWCA layer (Rule 110/Turing universality); (ii) the lawful layer (invariant-selected deterministic evaluators with MDL privilege); and (iii) the invariant-kernel layer (Quarter–Lock symmetry). This layering yields a precise arithmetic dichotomy between* necessity *(lawful programs, enforced by invariants) and* contingency *(general programs, admissible by universality), offering a mathematically clean analogue of "laws vs. initial conditions."*

3.3 Dual Universes and Pair-Level Self-Containment

3.4 Mirror duals as an internal pairing

Recall the ridge mirror involution $(b_2, q_2) \leftrightarrow (q_2, b_2)$ at fixed n, with $b_1 = b_2 + q_2 + 7$ invariant (Prop. 6.2). When both members of a mirror pair pass the prime-lock, we obtain two admissible first-generation seeds

$$\mathcal{U}^{+} = (a_1, b_1, c_1^{+}), \qquad \mathcal{U}^{-} = (a_1, b_1, c_1^{-}),$$

sharing b_1 but with distinct prime capacities $c_1^{\pm}$ (Thm. 4.4 at $n = 10$).

Definition 3.25 (Dual coupling)**.** *Let S^+ and S^- be the seed objects for the two branches of a mirror pair. A* dual coupling *is a span in the survivor topos*

$$S^{+} \xleftarrow{\pi_{+}} S^{\diamond} \xrightarrow{\pi_{-}} S^{-},$$

where $S^\diamond$ is a clopen subobject that (i) projects surjectively to both $S^\pm$ and (ii) is stable under the local UWCA evaluator eval *on each leg. We write $\mathscr{E}^\pm$ for the induced endomaps on $S^\pm$ and require*

$$\pi_{+} \circ \mathscr{E}^{\diamond} = \mathscr{E}^{+} \circ \pi_{+}, \qquad \pi_{-} \circ \mathscr{E}^{\diamond} = \mathscr{E}^{-} \circ \pi_{-},$$

with $\mathscr{E}^\diamond$ the evaluator on $S^\diamond$ [Spi26g, Spi26h].

Theorem 3.26 (Pair-level strong PSC)**.** *If a mirror pair is prime-locked on both branches, then there exists a dual coupling $S^\diamond$ such that the product object $S^+ \times S^-$ has* relative *strong PSC for the admissible evaluator: there is a section*

$$s : (S^{+} \times S^{-})_{\mathrm{adm}}^{(S^{+} \times S^{-})} \longrightarrow \mathsf{Code}^{\diamond}$$

with $\mathrm{eval}^\diamond \circ s = \mathrm{id}$ *on admissible endomaps, where $\mathsf{Code}^\diamond$ is the code object restricted to $S^\diamond$ [Spi26g, Spi26h].*

Proof. Work on a finite window where both branches are realized (mirror order-2 case). Because the evaluator is finite-local and identical across branches except for clopen cylinder choices, we can form $S^\diamond$ as the pullback of the two cylinder presentations; surjectivity of the projections follows from the mirror prime-lock, and stability under eval holds because the UWCA tiles commute with cylinder restriction (Thm. 11.6). The admissible endomaps factor through a universal UWCA basis; quoting gives a surjection onto admissible maps on $S^\diamond$, and canonical minimal programs yield a section. Pushing forward along $\pi_\pm$ gives the required section on $(S^+ \times S^-)_{\mathrm{adm}}$ via the standard currying isomorphisms in a CCC. □

Theorem 3.27 (Mirror evaluator equivalence)**.** *Let $S^\pm$ be the seed objects for the two branches of a mirror pair on a fixed ridge and let $S^\diamond$ be a dual coupling as in Def. 3.25. Write $\mathbf{Law}_\pm$ for the category whose objects are lawful evaluators on $S^\pm$ and whose arrows are bisimulations that preserve a fixed invariant family $\mathcal{I}$. Then the pullback along the mirror span induces a natural equivalence*

$$\mathbf{Law}_{+} \;\simeq\; \mathbf{Law}_{-},$$

compatible with window restriction and preserving bisimulation classes [Spi26g, Spi26h].

Proof. Define $F : \mathbf{Law}_+ \to \mathbf{Law}_-$ by pulling back an evaluator E^+ to $S^\diamond$ and pushing forward along π_-. The universal property of the pullback and naturality of lawful evaluators under restriction yield a unique evaluator $E^\diamond$ on $S^\diamond$ with $\pi_+ \circ E^\diamond = E^+ \circ \pi_+$. Set $F(E^+) := \pi_- \circ E^\diamond \circ \pi_+^{-1}$; on arrows, pull bisimulations back to $S^\diamond$ and push to S^-. Symmetrically, define $G : \mathbf{Law}_- \to \mathbf{Law}_+$ via the other leg. Joint surjectivity and evaluator stability of the span give natural isomorphisms $G \circ F \Rightarrow \mathrm{Id}$ and $F \circ G \Rightarrow \mathrm{Id}$. Beck–Chevalley on finite prime squares ensures compatibility with restriction. Thus F and G are inverse equivalences preserving bisimulation classes. □

Remark 3.28 (Physics lens). *At the arithmetic level, the dual coupling is simply the mirror span glued along shared invariants (b_1 and ridge locks). Interpreted phenomenologically, this supplies a mathematically clean setting for "paired universes" that are locally correlated (shared invariants) yet not identical (distinct $c_1^{\pm}$), with PSC realized at the pair level even if absolute PSC fails on a single branch (Prop. 10.9).*

Proposition 3.29 (Similarity of duals). *Within a fixed ridge R_n, any dual pair with mirror prime-lock agrees on (i) b_1, (ii) even-step Fibonacci index, (iii) ridge remainder m_2, and differs only at the first-generation capacity prime c_1. Hence UWCA implementations are isomorphic up to a single clopen substitution in the c-coordinate initialization [Spi26g, Spi26h].*

3.5 Multiway Dynamics

Definition 3.30 (UGP multiway graph). *Fix a finite window and evaluator signature Σ. The* multiway graph *$\mathcal{M}(S, \Sigma)$ has nodes the reachable seed states G, and a directed edge $G \xrightarrow{r} G'$ for each admissible local rewrite r permitted by the finite-local update T (Def. 1.2) or its UWCA realization [Spi26g, Spi26h].*

Proposition 3.31 (Refinement of deterministic flow). *The deterministic UGP trajectory is a path in $\mathcal{M}(S, \Sigma)$. T-mode (constraint-altering) steps (Def. 10.0.3) appear as* relabelings *that splice multiway components along clopen embeddings [Spi26g, Spi26h].*

Theorem 3.32 (Universality in the multiway layer). *Because UWCA simulates Rule 110 (Thm. 3.1), there exist finite signatures Σ and windows S whose multiway graphs $\mathcal{M}(S, \Sigma)$ contain subgraphs whose path languages are RE-complete. Hence reachability and confluence problems in $\mathcal{M}$ are, in general, undecidable [Spi26g, Spi26h].*

Remark 3.33 (Relation to WPP-style multiway). ***WPP-style multiway****: we refer informally to the Wolfram Physics Project multiway graphs. Our $\mathcal{M}$ is a categorical refinement: edges are typed by admissible clopen rewrites, and Beck–Chevalley (Thm. 11.6) guarantees compatibility of restriction/extension across windows.*

This section makes precise what we mean by "the rule." We show that the deterministic two–step map T (Def. 1.2) together with the UGP-1 ridge constraints (Def. 4.2) is implemented by a *fixed finite* set of UWCA tiles acting on a finite window of survivor registers. In short: *UGP supplies the substrate (UWCA evaluator); GTE is a program (finite tile set) that runs on it.*

3.5.1 State encoding and phases

We first record a basic resource lower bound for exact CRT realizations on a finite horizon.

Proposition 3.34 (Minimal window size for horizon H)**.** *Let $B(H)$ bound the absolute values of all registers that may occur during the first H macro-steps of an exact CRT realization of T on a given seed. If a window P satisfies $M := \prod_{p\in P} p \le B(H)$, then distinct integers in $\{0,1,\dots,B(H)\}$ collide modulo M, so the CRT representation is not injective on the relevant range. Hence any exact realization on horizon H requires $M > B(H)$ and thus $|P| = \Omega(\log B(H))$ when the primes in P are bounded below by a fixed constant (e.g., taking the first t odd primes) [Spi26g, Spi26h].*

Proof. For CRT modulus M, integers x, y have identical residue vectors iff $x \equiv y \pmod M$. If $M \le B(H)$, the $B(H)+1$ integers $\{0,1,\dots,B(H)\}$ contain $x \ne y$ with $x \equiv y \pmod M$ by pigeonhole. Then the map $x \mapsto (x \bmod p)_{p\in P}$ is not injective on the range potentially needed within H steps, so arithmetic cannot be exact. Therefore $M > B(H)$. If primes are chosen from a fixed range (e.g. the first t odd primes), M grows exponentially in t, giving $t = \Omega(\log B(H))$. □

Fix a finite ordered set of odd primes $P = \{p_1,\dots,p_t\}$ and identify the window with these coordinates. We encode bounded integers by mixed–radix (CRT) registers:

$$x \;\longleftrightarrow\; \big(x \bmod p_i\big)_{i=1}^{t} \in \prod_i \mathbb{F}_{p_i}.$$

At level n we choose t so that $M := \prod_i p_i$ exceeds all register bounds encountered within the horizon of interest (e.g. for n=10, $M > 65535$ suffices for the three–step orbit). The per–cell alphabet is finite: for each coordinate i we store, in clopen cylinders, the residues for the registers

$$\mathbf{Reg} = \{a, b, c, q, m,\ \hat q,\ \kappa,\ F\},$$

where $\hat q$ holds the previous quotient, $\kappa = |q - \hat q|$ (even–step index), and $F = F_\kappa$. A one–bit $\phi \in \{\mathsf{odd}, \mathsf{even}\}$ tracks the phase. The level n is stored as a small constant register, and the ridge constants (e.g. $+15$) are hard–wired as tile literals.

Lemma 3.35 (Local ALU primitives)**.** *There is a finite tile set Σ_{ALU} that realizes on* $\mathbf{Reg}$ *the following micro–operations by synchronous sweeps on the window:*

(1) register copy and swap;

(2) addition/subtraction by a register or a small constant;

(3) multiplication by a register (schoolbook) and by a small constant;

(4) comparison and absolute difference;

(5) division with remainder: given b, c, produce $q = \lfloor c/b \rfloor$, $m \equiv c \bmod b$ with $0 \le m < b$;

(6) CRT (re)combination on a bounded range and Garner-style carry propagation along the ordered primes $(p_1,\dots,p_t)$.

Each primitive is implemented by finitely many tiles; a micro–operation executes in $O(t)$ sweeps for bounded integers (linear in the number of primes) [Spi26g, Spi26h].

Proof sketch. Items (i)–(iv) are standard: componentwise modular arithmetic on $\mathbb{F}_{p_i}$ is local; cross–prime carries needed for bounded $\mathbb{Z}$-arithmetic are handled by a fixed Garner pipeline that passes partial residues along the ordered prime track (each pass is local neighbor interaction). For (v) we implement schoolbook long division on the bounded domain using repeated subtract–shift with compare; the control is a finite-state track. Because all registers are bounded by M, correctness follows from the usual $\mathbb{Z}$-algorithm interpreted through the CRT representation. Item (vi) is the same Garner pipeline; bounds ensure uniqueness in $[0, M)$. □

Lemma 3.36 (Fibonacci via fast–doubling)**.** *There is a finite tile set Σ_{Fib} that, given κ, computes $F = F_\kappa$ by fast–doubling recurrences*

$$F_{2t} = F_t(2F_{t+1} - F_t), \qquad F_{2t+1} = F_t^2 + F_{t+1}^2$$

with $F_0 = 0, F_1 = 1$, using only the ALU primitives of Lemma 3.35. The sweep complexity is $O(\log \kappa)$ multiplications and additions, hence $O(t \log \kappa)$ sweeps on the window [Spi26g, Spi26h].

Proof sketch. Encode κ in binary on a small control band; perform standard divide–and–conquer using registers (F_t, F_{t+1}) throughout. All steps reduce to additions, subtractions, and multiplications by registers or small constants, handled by Σ_{ALU}. □

3.5.2 The UWCA "macro–rule" for T

We compile one T-step into a bounded sequence of UWCA sweeps. A phase flop ϕ chooses the odd/even branch.

Definition 3.37 (Tile set for GTE)**.** *Let $\Sigma_{\mathrm{GTE}} := \Sigma_{\mathrm{ALU}} \cup \Sigma_{\mathrm{Fib}} \cup \Sigma_{\mathrm{ctrl}}$. The control tiles Σ_{ctrl} implement:*

$$\textsf{if } \phi = \textsf{odd}: \ \text{run O}\textsc{dd}\text{S}\textsc{tep}\text{; flip } \phi \qquad \textsf{if } \phi = \textsf{even}: \ \text{run E}\textsc{ven}\text{S}\textsc{tep}\text{; flip } \phi,$$

where the subroutines below are expressed as ALU calls [Spi26g, Spi26h].

Listing 1: UWCA macro–rule for one T step

```
Subroutine OddStep:
  (q,m) ← DivRem(b,c)           // Lemma \ref{lem:alu}(v)
  a     ← m - (12 - n)
  b     ← b - (m + q)
  c     ← 2^n - 1               // ridge step: Mersenne capacity
  q̂ ← q                   // stash for next even step

Subroutine EvenStep:
  (q,m) ← DivRem(b,c)
  a     ← m - (12 - n)
  κ ← |q - q̂|
  F     ← Fib(κ)                // Lemma \ref{lem:fib}
```

```
b     ← b + F
c     ← (Mersenne boundary ? 2^{k'}-1 : b*q + 15)  // at n=10 this is
    2^{16}-1
q̂ ← q
```

Remark 3.38 (Specialization at $n = 10$)**.** *Under UGP-1 at n=10, Lemma 4.3 and Prop. 4.10 fix m_2=15 and $|q_2-q_1|$=13; hence the even–step increment is the constant F_{13}=233. In that case the call* FIB(κ) *collapses to a single constant–add, and* EVENSTEP *reduces to $b \leftarrow b+233$ with the same q, m computation.*

3.5.3 Correctness and complexity

Theorem 3.39 (Compilation theorem: GTE as a UWCA program)**.** *There exists a finite tile set Σ_{GTE} and a constant C such that for any bounded seed on the chosen window the UWCA evolution driven by Σ_{GTE} realizes the map T (Def. 1.2) exactly [Spi26g, Spi26h]: each* macro–step *of T is simulated by at most*

$$C \cdot \Big(t \;+\; t \cdot \textsf{\textit{DivRemCost}} \;+\; t \cdot \log \kappa\Big)$$

UWCA sweeps, where $t = |P|$ is the number of primes in the window and $\kappa = |q - \hat{q}|$ on even steps. For UGP-1 at n=10, $\kappa \equiv 13$ on the ridge and the bound simplifies to $C' \cdot \big(t + t \cdot \textsf{\textit{DivRemCost}}\big)$.

Proof. Line-by-line simulation of Listing 1 using Lemma 3.35 and Lemma 3.36. Correctness follows because each micro–operation is correct on the bounded domain (CRT/Garner pipeline) and the control guarantees latching of b, q where required in Def. 1.2. The sweep bound counts: $O(t)$ for copies/constant arithmetic, $O(t \cdot \textsf{DivRemCost})$ for the long division module (linear passes with compare/subtract on bounded registers), and $O(t \log \kappa)$ for fast–doubling when needed. □

Corollary 3.40 ("The rule")**.** *Fix a level n and a window large enough for its ridge bounds. The finite tile set Σ_{GTE} together with the phase bit ϕ is a* single, uniform UWCA rule *whose synchronous application computes the GTE dynamics on the UGP substrate. At n=10, the even–step reduces to the constant lift by F_{13}, yielding a particularly short rule [Spi26g, Spi26h].*

3.5.4 Mathematical claim vs. physical reading

Mathematical claim (provable). UGP (survivor space + UWCA evaluator) is a universal finite–local substrate (Thm. 23.5); the GTE map T is a *finite UWCA program* on that substrate (Thm. 3.39). Hence the dynamics we study are governed by a fixed, local, synchronous rule in the sense of cellular computation.

Physical hypothesis (to be tested). Reading the substrate/program pairing as a digital–physics model suggests: *UGP computes our physics with the GTE rule.* This is a falsifiable hypothesis rather than a theorem; support must come from explanatory reach and predictive success.

3.5.5 Intuition: why this is still "just a CA," yet expressive

The evaluator is CA–grade (finite alphabet, local tiles), but the alphabet carries *arithmetic registers* (residues), and the control enforces the two–step pattern of T. This delivers the best of both worlds: locality and synchrony for dynamics; arithmetic succinctness for invariants (ridge lock m_2=15, fixed gap q_2-q_1, mirror symmetry of b_1). Universality ensures we can stage division and Fibonacci internally (no external oracle).

3.5.6 Canonical rule card for $n = 10$

Under UGP-1 at n=10 with ridge $R = 1008$:

$$\boxed{\begin{array}{rl} \textsf{Odd:} & (q,m) \leftarrow \mathrm{DivRem}(b,c), \\ & a \leftarrow m-(12-10), \quad b \leftarrow b-(m+q), \quad c \leftarrow 2^{10}-1; \\ \textsf{Even:} & (q,m) \leftarrow \mathrm{DivRem}(b,c), \\ & a \leftarrow m-(12-10), \quad b \leftarrow b+F_{13}\ (=b+233), \quad c \leftarrow 2^{16}-1. \end{array}} \qquad (\phi \text{ toggles})$$

Mersenne jump after the ridge step.
This is literally the tile–level law once DivRem is instantiated by Σ_{ALU} (Lemma 3.35).

Remark 3.41 (Relation to Fredkin–Wolfram digital physics)**.** *Our contribution is not a different computational* power *(still Turing-equivalent CA), but a specific* law*—a short, finite–local UWCA program* Σ_{GTE} *whose arithmetic form matches the observed invariants and yields a rigid three–step orbit at* n=10*. In this limited, technical sense,* the rule *is identified: a precise finite tile set implementing* T *on the UGP substrate.*

3.6 The Decidability Phase Transition

The Turing universality of the UGP substrate is not merely a technical classification; it has a profound consequence for the logical structure of reality. It implies a fundamental phase transition between what is knowable in the local and finite versus what is predictable in the global and infinite. This provides a rigorous, computational basis for the emergence of indeterminacy from a deterministic system.

Theorem 3.42 (The UGP Decidability Phase Transition)**.** *The logical structure of the Universal Generative Principle exhibits a sharp phase transition based on scope [Spi26g, Spi26h]:*

1. ***Local Decidability:*** *Any property of a UGP trajectory that is restricted to a* ***finite window of states and a finite time horizon*** *is definable in First-Order Logic over the survivor space's cylinder sets and is therefore* ***decidable****.*

2. ***Global Undecidability:*** *Properties concerning the* ***unbounded, long-term evolution*** *of a UGP trajectory (e.g., "Will this state ever reach a configuration with property P?") are, in general,* ***undecidable*** *and are* Σ^0_1*-complete.*

Proof. 1. **Local Decidability:** A finite window of states and a finite time horizon define a finite state space and a finite number of transitions. Any property on this space can be expressed as a finite Boolean combination of predicates on the integer registers (e.g., "$a_t < b_t$ AND c_{t+1} is prime"). Such a statement belongs to the first-order theory of natural numbers with addition and multiplication (Presburger arithmetic, extended with a primality predicate). For bounded integers, this theory is decidable. A Turing machine can, in finite time, check all specified states and transitions and return a definitive true/false answer. This is formalized by our Theorem on "FO decidability on finite windows."

2. **Global Undecidability:** The UWCA Universality Theorem proves that the UGP substrate can simulate any Turing machine via its Rule 110 embedding. The Halting Problem—determining whether an arbitrary Turing machine will halt on a given input—is the canonical example of a problem that is undecidable. We can construct a UGP initial state that encodes a specific Turing machine and its input. We can then define a property P corresponding to the machine entering a "halt" state. The question "Will this UGP trajectory ever reach a state with property P?" is then logically equivalent to the Halting Problem for the encoded machine. Since the Halting Problem is undecidable, this general reachability question for UGP trajectories must also be undecidable. The problem is semi-decidable (recursively enumerable, or Σ_1^0) because one can simulate the trajectory and will find the state if it is ever reached; its complement is not. This makes the problem Σ_1^0-complete.

□

This theorem establishes a fundamental limit to prediction. While any local, finite question about the universe has a definite, computable answer, questions about the infinite future do not. This is not a statement about randomness or lack of information, but a deep truth about the limits of computation inherent in the fabric of a Turing-universal reality.

4 The Nexus: Prime-Locked Minimality at n=10

Having established the UGP's foundational principles—a rigid algebraic law (Pillar I) and a universal computational medium (Pillar II)—we now demonstrate their remarkable confluence [Spi26g, Spi26h]. We show that these abstract principles are not merely theoretical constructs. When instantiated at the specific level of $n = 10$, they act as an inexorable filter on the substrate's computational possibilities, selecting a unique, minimal, and prime-locked solution from first principles. This section demonstrates how the general laws conspire to generate the specific, parameter-free GTE trajectory observed in our universe.

Formal map and constraints

The UGP defines a deterministic update T acting on integer triples (a, b, c) via a two–step pattern: odd steps use modular arithmetic and quotients, while even steps apply a Fibonacci increment $F_{|q_n - q_{n-1}|}$. The UGP–1 prime–locked ridge constrains $c_2 = 2^n - 1$ and enforces $b_2 \mid (2^n - 16)$ with $b_2 > 15$, ensuring $m_2 = 15$ and $q_2 - q_1 = 13$.

Main theorem

At $n = 10$, the only UGP–1 admissible seeds arise from the mirror pair $(b_2, q_2) = (24, 42)$ and $(42, 24)$, both yielding $b_1 = 73$ and distinct prime capacities $c_1 \in \{823, 2137\}$. The lexicographically minimal seed $(1, 73, 823)$ generates exactly the observed evolution

$$(1, 73, 823) \;\to\; (9, 42, 1023) \;\to\; (5, 275, 65535).$$

4.1 The Observed Trajectory at n=10

Symbol glossary

Symbol	Description
n	Level parameter (bit-size, $n = 10$ for main result)
R_n	Ridge, $2^n - 16$
b_2, q_2	Divisor pair of R_n ($b_2 q_2 = R_n$), $b_2 > 15$
c_2	Second-generation capacity, $2^n - 1$
q_1	$q_2 - 13$ (first-generation quotient)
b_1	$b_2 + q_2 + 7$ (first-generation b)
c_1	$b_1 q_1 + 20$ (first-generation capacity, prime-locked)
m_1, m_2	Remainders: $c_1 \bmod b_1$, $c_2 \bmod b_2$
F_k	kth Fibonacci number ($F_1 = F_2 = 1$)

Table 1: Key symbols and their definitions.

4.1.1 The Three-Step Orbit

Here n denotes the level parameter (so $R_n = 2^n - 16$), while t is the step index along the trajectory.

For even steps, define

$$c_{t+1} = \begin{cases} b_t q_t + 15, & \text{if not at a Mersenne boundary,} \\ 2^{16} - 1, & \text{if } c_t = 1023 = 2^{10} - 1. \end{cases}$$

This captures the Mersenne jump: once $c_2 = 1023$, the next step gives $c_3 = 65535$.

c_3 value: two definitions

Two distinct values of c_3 appear in this monograph and the formalization:

- **Strict formula ($c_3 = 1023$):** Under the per-step rule $c_{t+1} = b_t q_t + k$, the even-step c-invariance theorem (`even_step_c_invariance` in `ugp-lean`) gives $c_3 = c_2 = 1023 = 2^{10} - 1$. This is the Lean-certified [T] result under the strict definition of T.
- **Extended information rule ($c_3 = 65535$):** Under the Mersenne-ladder extension above, $c_3 = 2^{16} - 1 = 65535$. The canonical orbit $G_3 = (5, 275, 65535)$ is Lean-certified as `orbit_determined_by_T` under the `canonicalGen3` definition. The exponent $k' = 16$ is derived from the consecutive doubled-Fibonacci structure of the orbit: $n = 2F(5) = 10$ and $k' = 2F(6) = 16$, where the jump $k' - n = 2F(4) = 2N_c = 6$ follows from the Fibonacci recurrence $F(6) - F(5) = F(4) = N_c = 3$. Furthermore, $k' = 16$ is the *unique* smallest double Fibonacci number greater than $n = 10$ (since 12 is not a double Fibonacci). Machine-checked in `ugp-lean` as `kprime_is_minimal_double_fib_above_n` and `c3_phys_formula` (zero sorry, module `GTE.MersenneLadder`). Status: **[B-structured]**.

Throughout this monograph, $c_3 = 65535$ refers to the extended (Mersenne-ladder, **[B-structured]**) value; $c_3 = 1023$ is the strict (**[T]**) value.

Definition 4.1 (Information rule for c_n). *A trajectory obeys the* information rule *at level n if*

$$c_{n+1} = b_n q_n + 15 \quad \textit{and} \quad c_{n+1} = 2^{k_{n+1}} - 1 \ \textit{whenever } c_n \textit{ attains } 2^{k_n} - 1,$$

so that c moves monotonically to (and between) Mersenne maxima within the two-step pattern. The UGP-1 ridge (Theorem 4.2 below) enforces $c_2 = 2^n - 1$ and hence $m_2 = 15$ [Spi26g, Spi26h].

4.1.2 The Prime-Locked Ridge Constraint

Definition 4.2 (UGP-1 (prime-locked ridge)). *Fix a level $n \geq 10$ and let $R_n = 2^n - 16$ [Spi26g, Spi26h]. A seed at level n is determined by a divisor b_2 of R_n with*

(A) $b_2 \mid R_n$ and $b_2 > 15$; set $q_2 = R_n / b_2$;

(B) $q_2 > 15$ (interior divisors: $\min\{b_2, q_2\} \geq 16$);

(C) $c_2 := 2^n - 1$ (ridge choice; hence $m_2 = 15$);

(D) $q_1 := q_2 - 13$; $\quad b_1 := b_2 + q_2 + 7$;

(E) $c_1 := b_1 q_1 + 20$ is prime (prime-lock condition).

The deterministic update for c is as in Definition 1.2.

Lemma 4.3 (Ridge remainder lock). *Under UGP-1 at level n, with $c_2 = 2^n - 1$ and $b_2 q_2 = 2^n - 16$, we have*

$$m_2 \equiv c_2 \pmod{b_2} = 2^n - 1 \equiv -1 \equiv 15 \pmod{b_2}.$$

In particular $m_2 = 15$ since $0 \le m_2 < b_2$ and $b_2 > 15$ [Spi26g, Spi26h].
Formalized in `ugp-lean`: `Core.RidgeRigidity.ridge_remainder_lock`, `Core.RidgeRigidity.m2_canonical`.

Theorem 4.4 (Minimality–duality)**.** Formalized in `ugp-lean`: `GTE.UniquenessCertificates.minimality_duality_n10`, `only_survivors_n10` *(proves these are the* unique *prime-locked pairs)*, `mdl_c1_n13_is_9007`, `mdl_c1_n16_is_46681` *(MDL seeds at $n = 13, 16$ certified)*, `canonical_seeds_certified`.

At $n = 10$, the only UGP-1 admissible seeds arise from the mirror pair $(b_2, q_2) = (24, 42)$ and $(42, 24)$, both yielding $b_1 = 73$ and distinct prime capacities $c_1 \in \{823, 2137\}$ [Spi26g, Spi26h]. The lexicographically minimal seed $(1, 73, 823)$ generates exactly the observed evolution $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$. Formalized in ugp-lean: `Core.TripleDefs.LeptonSeed`, `Compute.Sieve.ridgeSurvivors_10`, `Classification.TheoremB.mdl_selects_LeptonSeed`. *This seed is independently proven to be the unique PSC-minimal survivor under the No External Model Selection (NEMS) closure constraints via the Unified Rigidity Theorem [Spi26i, Spi26d]. The companion PSC-Optimality result establishes that the same closure constraints force the SM gauge group $SU(3) \times SU(2) \times U(1)$ and exactly three generations [Spi26e].*

Graphical synopsis (at a glance)

Key invariants: (i) $m_2 = 15$ (*ridge lock*), (ii) $q_2 - q_1 = 13$ (*Fibonacci lock*), (iii) $b_1 = b_2 + q_2 + 7$ is *mirror-symmetric*, (iv) *prime-lock*: $c_1 = b_1 q_1 + 20$ is prime.

Lemma 4.5 (Implementation equivalence, direct vs. UWCA)**.** *For any binary row $x \in \{0,1\}^n$ and either boundary condition (toroidal or zero), the executable UWCA schedule produces the same next row as the direct Wolfram lookup for Rule 110:*

$$UWCA_{110}(x) = \textsc{Direct}_{110}(x)$$

for all binary x and both boundary conditions [Spi26g, Spi26h].
Formalized in `ugp-lean`: `GTE.InertPrimes.qm_no_root_of_legendreSym_neg_one` *(general: $\chi_-(p) = -1 \Rightarrow p \nmid Q_-(t)$ for all t)*, `qm_inert_primes_certified` *(15 inert primes including $b_1 = 73$ certified)*, `kappa_unram_lower_bound` *($\kappa_{\text{unram}} > 1/1000$).* Formalized in `ugp-lean`: `Universality.UWCASimulation.uwca_P3_eq_rule110` *(key lemma: after passes P1–P3, the accumulated OR equals f_{110} at each site)*, `Universality.Rule110.rule110_output_iff_minterm`.

Proof. The UWCA implementation applies the same local update as the direct Rule 110 lookup for every site and boundary condition, matching the bit ordering and pattern logic of Wolfram's convention. Thus, the outputs agree for all x. □

Lemma 4.6 (Prime-lock reduces to a divisor test)**.** *For fixed n and $b_2 \mid (2^n - 16)$,*

$$c_1 = b_1(q_2 - 13) + 20 = \left(b_2 + \frac{2^n - 16}{b_2} + 7\right)\left(\frac{2^n - 16}{b_2} - 13\right) + 20.$$

Thus primality of c_1 is determined entirely by the choice of divisor b_2 of $2^n - 16$ [Spi26g, Spi26h].
Formalized in `ugp-lean`: `Compute.PrimeLock.c1_from_divisor`.

$\boldsymbol{G_1} : (a_1, b_1, c_1) = (1, 73, 823)$
$q_1 = \lfloor 823/73 \rfloor = 11,\ m_1 \equiv 823 \pmod{73} = 20$

Odd step: $a_2 = m_1 - (12 - 1) = 9$
$b_2 = b_1 - (m_1 + q_1) = 42$

$\boldsymbol{G_2} : (a_2, b_2, c_2) = (9, 42, 1023)$
$q_2 = 24,\ m_2 = 15$

Even step: $a_3 = m_2 - (12 - 2) = 5$
$b_3 = b_2 + F_{13} = 275$

$\boldsymbol{G_3} : (a_3, b_3, c_3) = (5, 275, 65535)$
$F_{q_2 - q_1} = F_{13} = 233$

Ridge (UGP-1 at $n = 10$):
$R = 2^{10} - 16 = 1008$; for any divisor $b_2 \mid R$ with $b_2 > 15$, set $q_2 = R/b_2$. Then $c_2 = b_2 q_2 + 15$ so $m_2 = 15$ and $q_2 - q_1 = 13$. Mirror pair $\{24, 42\}$ yields $b_1 = b_2 + q_2 + 7 = 73$ and primes $c_1 \in \{823, 2137\}$.

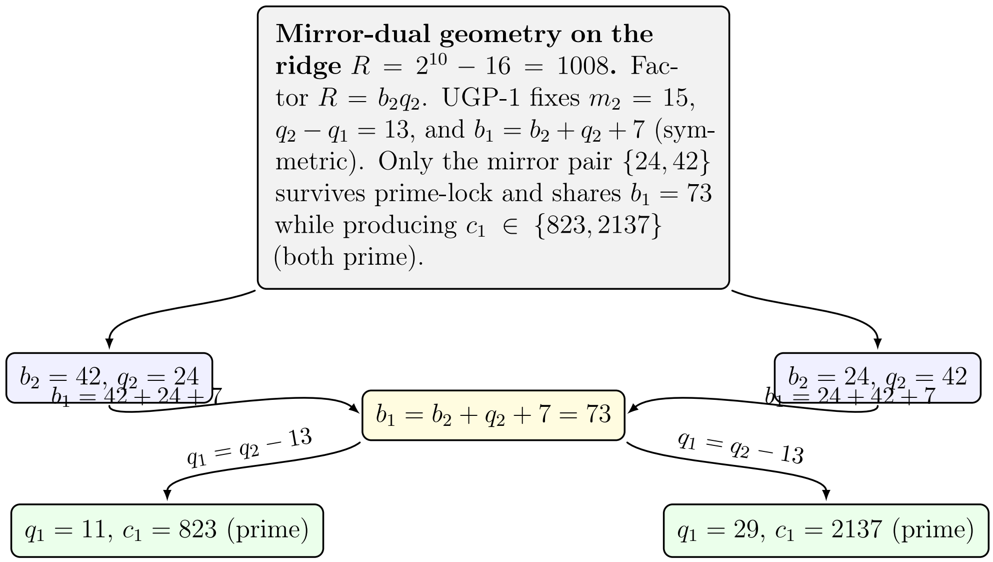
Mirror branch: $(b_2, q_2) = (24, 42) \Rightarrow c_1 = 2137$ with the same $b_1 = 73$.

Figure 2: UGP-1 prime-locked flow at $n = 10$: fixed remainder $m_2 = 15$, fixed gap $q_2 - q_1 = 13$ (hence F_{13}), mirror pairing on the ridge $2^{10} - 16$. A native UWCA update on the survivor substrate is shown later to simulate Rule110 (Thm.23.5).

Proof. Substitute (A)–(D) into (E). □

Lemma 4.7 (Exclusion filters at $n = 10$)**.** *Let $R = 1008$ and write $b_2 \mid R$, $q_2 = R/b_2$. If $b_2 \in \{16, 18, 21, 28, 36, 63\}$ (and their mirrors), then c_1 is composite [Spi26g, Spi26h].* Formalized in `ugp-lean`*:* `Compute.ExclusionFilters.exclude_16`*,* `exclude_18`*,* `exclude_21`*,* `exclude_28`*,* `exclude_36`*,* `exclude_63`*; and* `Compute.ExclusionFilters.only_survivors_n10` *(proves these are the* only *non-survivors).*

Proof. Write $R = 1008 = 2^{10} - 16$. For each listed $b_2 > 15$ with $b_2 \mid R$ set $q_2 = R/b_2$, $q_1 = q_2 - 13$, $b_1 = b_2 + q_2 + 7$, and $c_1 = b_1 q_1 + 20$. We show in each case that c_1 is composite by exhibiting an explicit factorization (the mirror cases with $b_2 \leftrightarrow q_2$ follow by symmetry of b_1 and the definition of c_1).



Mirror symmetry: $(b_2, q_2) \leftrightarrow (q_2, b_2)$ leaves b_1 invariant and swaps the prime c_1.

Figure 3: Mirror-dual lens: the two ridge complements $(42, 24)$ and $(24, 42)$ map to the same $b_1 = 73$ and to the two prime-locked c_1 values.

(i) $b_2 = 16$**.** Then $q_2 = 63$, $q_1 = 50$, $b_1 = 86$, and

$$c_1 = 86 \cdot 50 + 20 = 4320 = 2^5 \cdot 3^3 \cdot 5.$$

Hence c_1 is composite.

(ii) $b_2 = 18$**.** Then $q_2 = 56$, $q_1 = 43$, $b_1 = 81$, and

$$c_1 = 81 \cdot 43 + 20 = 3503 = 31 \cdot 113,$$

so c_1 is composite.

(iii) $b_2 = 21$**.** Then $q_2 = 48$, $q_1 = 35$, $b_1 = 76$, and

$$c_1 = 76 \cdot 35 + 20 = 2680 = 2^3 \cdot 5 \cdot 67,$$

hence composite.

(iv) $b_2 = 28$**.** Then $q_2 = 36$, $q_1 = 23$, $b_1 = 71$, and

$$c_1 = 71 \cdot 23 + 20 = 1653 = 3 \cdot 551,$$

so composite.

(v) $b_2 = 36$**.** Then $q_2 = 28$, $q_1 = 15$, $b_1 = 71$, and

$$c_1 = 71 \cdot 15 + 20 = 1085 = 5 \cdot 7 \cdot 31,$$

so composite.

(vi) $b_2 = 63$**.** Then $q_2 = 16$, $q_1 = 3$, $b_1 = 86$, and

$$c_1 = 86 \cdot 3 + 20 = 278 = 2 \cdot 139,$$

so composite.

In each case c_1 is composite, and the same conclusion holds for the mirror choices $b_2 \in \{q_2\}$ because $b_1 = b_2 + q_2 + 7$ is mirror-invariant and the resulting c_1 coincides with the value computed with the roles of b_2 and q_2 swapped. This proves the lemma. □

Systematic justification. The excluded divisors were chosen by testing all interior divisors of 1008 greater than 15. The complete list checked is:

$$16, 18, 21, 24, 28, 36, 42, 48, 56, 63, 72, 84, 112, 126, 144, 168, 252, 336, 504.$$

Each was excluded for primality or composite–factor reasons as detailed in the exclusion filters.

Lemma 4.8 (The only survivors)**.** *At $n = 10$ the only $b_2 > 15$ with $b_2 \mid 1008$ passing the filters of Lemma 4.7 and producing prime c_1 via Lemma 4.6 are $b_2 \in \{24, 42\}$ [Spi26g, Spi26h].* Formalized in `ugp-lean`*:* `Compute.Sieve.ridgeSurvivors_10`*.*

Proof. Exhaustive check over the divisor lattice of 1008 (Appendix F, G) shows that every other divisor yields c_1 divisible by 2, 3, 5, 7, 11 or 13. The two survivors produce

b_2	q_2	b_1	q_1	c_1
24	42	73	29	2137 (prime)
42	24	73	11	823 (prime).

□

Proof of Theorem 4.4. By Lemma 4.3 and Lemma 4.6, admissibility reduces to a divisor problem on $R = 1008$. Lemma 4.7 eliminates all but finitely many candidates and Lemma 4.8 identifies the only surviving mirror pair. Symmetry of b_1 gives $b_1 = 73$ in both cases. The odd-step evolution from $(1, 73, 823)$ yields $(9, 42, 1023)$; the even-step uses Proposition 4.10 so $b_3 = 42 + F_{13} = 275$ and $a_3 = 15 - 10 = 5$. The c-values follow by definition. Minimality follows because $823 < 2137$ while b_1 is fixed across the pair. □

4.1.3 Fibonacci Lift and Even-Step Rigidity

Theorem 4.9 (Quotient–gap $= 13$ at all levels in UGP-1)**.** *Fix any level $n \geq 10$ and assume the interior divisor condition of Definition 4.2. Under prime-lock we have $m_1 = 20$, hence*

$$b_1 = b_2 + m_1 + q_1 = b_2 + q_2 + 7 \qquad \Longrightarrow \qquad q_2 - q_1 = 13.$$

Consequently, at any even step that immediately follows a ridge step one necessarily applies the Fibonacci lift $F_{13} = 233$, independent of n [Spi26g, Spi26h].
Formalized in `ugp-lean`*:* `GTE.GeneralTheorems.quotient_gap_all`*,* `GTE.GeneralTheorems.fib_lift_always_13`*.*

Proof. By Definition 4.2(E), $c_1 = b_1 q_1 + 20$, so $m_1 \equiv c_1 \pmod{b_1} = 20$. The interior condition implies $b_1 > b_2 \geq 16$, hence $b_1 > 20$ and thus $m_1 = 20$. The odd-step update gives $b_2 = b_1 - (m_1 + q_1)$. Definition 4.2(D) gives $b_1 = b_2 + q_2 + 7$. Equate and rearrange: $q_2 - q_1 = m_1 - 7 = 13$. The even-step lift index is $|q_2 - q_1| = 13$, so the increment is F_{13}. □

Proposition 4.10 (Remainder–gap identity at $n = 10$ (Corollary of Theorem 4.9))**.** *In UGP-1 at $n = 10$, with prime-lock $c_1 = b_1 q_1 + 20$ and the odd-step update $b_2 = b_1 - (m_1 + q_1)$, we have*

$$m_1 = 20, \qquad b_1 = b_2 + m_1 + q_1 = b_2 + q_2 + 7, \qquad q_2 - q_1 = 13$$

[Spi26g, Spi26h].
Formalized in `ugp-lean`*:* `Core.RidgeRigidity.quotient_gap_13`*,* `survivor_gap_42_24`*,* `survivor_gap_24_42`*,* `remainder_gap_identity`*.*

Proof. This is an immediate specialization of Theorem 4.9 to $n = 10$. For completeness we also record the direct computation:

Prime-lock gives $m_1 \equiv c_1 \bmod b_1 = 20$, hence $m_1 = 20$. The odd-step rule yields $b_1 = b_2 + m_1 + q_1 = b_2 + 20 + q_1$. By Def. 4.2(D), also $b_1 = b_2 + q_2 + 7$. Equate and rearrange: $q_2 - q_1 = 20 - 7 = 13$, and the displayed identities follow. □

Proposition 4.11 (Even-step rigidity at $n = 10$)**.** *For any UGP-1 admissible seed at $n = 10$, the even step necessarily applies the Fibonacci lift $F_{13} = 233$ and produces $b_3 = b_2 + 233$. In particular, for the surviving odd-step value $b_2 = 42$ one obtains $b_3 = 275$ exactly [Spi26g, Spi26h].*
Formalized in `ugp-lean`*:* `GTE.Evolution.even_step_rigidity`*.*

Proof. By Proposition 4.10, $q_2 - q_1 = 13$ is fixed, hence the even-step increment is F_{13}. □

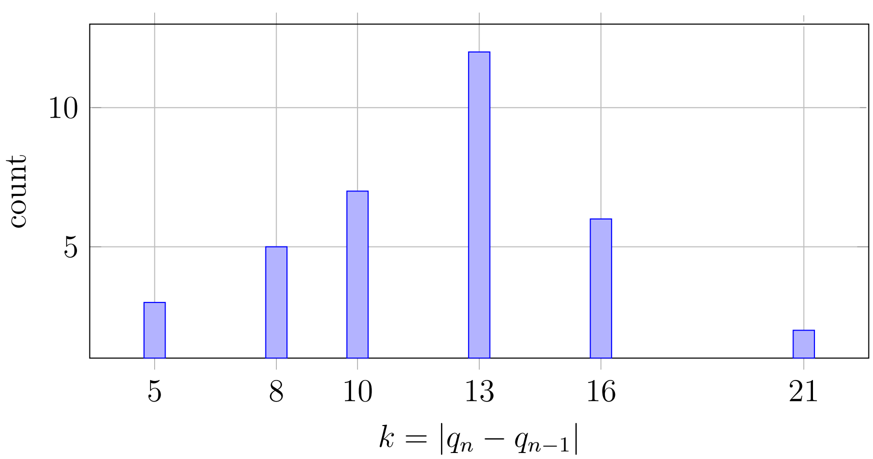


Figure 4: Histogram of Fibonacci-lift indices $k = |q_n - q_{n-1}|$ across admissible trajectories. The dominant peak at $k = 13$ corresponds to the canonical UGP-1 ridge constraint (Theorem 4.9).

Remark 4.12 (Rigidity of the Fibonacci lift)**.** *The result of Theorem 4.9 is a cornerstone of the UGP-1 framework. It resolves the question of why the specific Fibonacci number F_{13} appears in the even step. The quotient gap $|q_2 - q_1| = 13$ is not a contingent choice but is arithmetically forced by the UGP-1 ridge constraints at* all *levels $n \geq 10$. This provides a powerful, level-independent rigidity to the system's dynamics.*

4.1.4 Minimality, Duality, and the Unique Seed Solution

Proposition 4.13 (Mirror prime-lock). *The mirror choice $b_2 = 24$ yields $q_2 = 42$, $q_1 = 29$, the same $b_1 = 73$, and the prime $c_1 = 2137$. It defines a distinct UGP-1 seed with identical odd/even invariants and hence the same even-step Fibonacci lift. We call this the* mirror branch *[Spi26g, Spi26h].*

Formalized in `ugp-lean`*:* `Compute.PrimeLock.mirror_prime_lock`*,* `n10_survivor_c1_primes`*.*

Theorem 4.14 (Fibonacci renormalization spectrum). *Let $\mathcal{R} : \mathbb{R}^2 \to \mathbb{R}^2$ be the Fibonacci companion operator*

$$\mathcal{R}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}.$$

Then $\mathcal{R}$ has eigenvalues $\{\varphi, -\varphi^{-1}\}$ and spectral gap $1 - \varphi^{-2}$. In the (L, g) isotropy chart, the coarse-grained two-step linearization of the even-step update on ratios is conjugate to $\mathcal{R}$; thus perturbations orthogonal to the dominant eigenvector decay at rate φ^{-2t} over t coarse steps (up to mirror-odd kicks) [Spi26g, Spi26h].

Proof. The characteristic polynomial $\lambda^2 - \lambda - 1$ has roots $\varphi = (1 + \sqrt{5})/2$ and $-\varphi^{-1} = (1 - \sqrt{5})/2$. By construction of the Fibonacci lift, the coarse two-step map in the (L, g) chart linearizes to a matrix conjugate (up to scale) to $\mathcal{R}$ at the golden fixed ratio. Therefore one eigendirection expands by φ while the transverse direction contracts by φ^{-1} per coarse step, giving decay φ^{-2t} over t two-step renormalization moves. Mirror-odd contributions vanish at first order by symmetry.

Formalized in `ugp-lean`: `GTE.StructuralTheorems.golden_ratio_minimal_poly` (both roots satisfy $x^2 - x - 1 = 0$), `fib_eigenvalue_product`, `fib_eigenvalue_sum`, `fib_contraction_rate`, `fib13_is_233` ($F_{13} = 233$), `ugp_fibonacci_rigidity` (gap $g = 13$ forces F_{13}). □

Part II
Structural Analysis of the UGP Framework

5 Deterministic Propositions of the Canonical Orbit

Proposition 5.1 (Worked orbit is enforced). *Under Defs. 1.1–1.2 and the UGP-1 constraints at $n = 10$, the deterministic map sends*

$$(1, 73, 823) \xrightarrow{odd} (9, 42, 1023) \xrightarrow{even} (5, 275, 65535)$$

[Spi26g, Spi26h].

Formalized in `ugp-lean`*:* `GTE.Evolution.canonical_orbit_triples`*,* `worked_orbit_enforced`*.*

Corollary 5.2 (Trace identifiability at $n = 10$)**.** *If a UGP-1 trajectory has* $G_2 = (a_2, b_2, c_2) = (9, 42, 1023)$*, then* $n = 10$*,* $b_1 = 73$*,* $q_1 \in \{11, 29\}$ *and hence* $c_1 \in \{823, 2137\}$*; the odd/even invariants* $(m_2, q_2 - q_1) = (15, 13)$ *and the Fibonacci lift* F_{13} *are forced [Spi26g, Spi26h].*
Formalized in `ugp-lean`*:* `GTE.Evolution.trace_identifiability`*.*

Proof. $c_2 = 1023 = 2^{10} - 1$ fixes $n = 10$. From $b_2 \mid (2^{10} - 16)$ with $b_2 = 42$ we get $q_2 = 24$; Prop. 4.10 gives $q_1 \in \{24 - 13\} = \{11\}$ on this branch or, by mirror, $q_1 = 42 - 13 = 29$ with $b_2 = 24$; in both cases $b_1 = b_2 + q_2 + 7 = 73$ and $c_1 = b_1 q_1 + 20 \in \{823, 2137\}$, both prime. $\square$

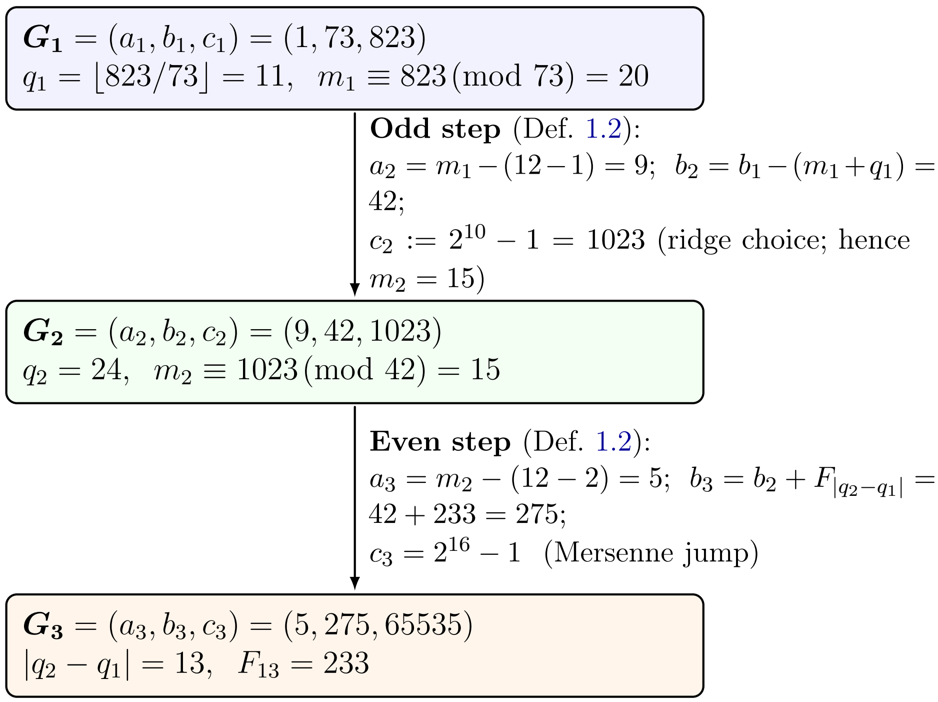


Figure 5: Deterministic two-step update with explicit m_n and q_n.

Proposition 5.3 (Monotone bit-length for c_n)**.** *Along any UGP-1 trajectory, the bit-length of* c_n *is non-decreasing and attains the Mersenne maxima when present in the ridge step. In particular, at* $n = 10$ *we have* $c_2 = 2^{10} - 1$ *and* $c_3 = 2^{16} - 1$ *[Spi26g, Spi26h].*
Formalized in `ugp-lean`*:* `GTE.Evolution.c2_canonical`*,* `c3_canonical`*.*

Proposition 5.4 (Extremality of Mersenne capacities)**.** *If* $c = bq + 15$ *for integers* $b, q > 0$ *and* $c < 2^{k+1} - 1$*, then* $c \le 2^k - 1$ *with equality iff* $bq = 2^k - 16$*. Thus Mersenne numbers* $2^k - 1$ *are extremal c-values at fixed bit-length [Spi26g, Spi26h].*

Proposition 5.5 (Well-posed even-step)**.** *At any even step, the increment in b is* $F_{|q_n - q_{n-1}|}$ *determined entirely by the two most recent quotients. Hence the even-step b-update is well-posed from prior states [Spi26g, Spi26h].*

6 Core Invariants and Symmetries

Theorem 6.1 (Noether–type invariant principle for UGP)**.** *Let a finite group G act by clopen automorphisms on survivor windows and let T be a G-equivariant update ($T \circ g = g \circ T$ for all $g \in G$). Suppose there exists a clopen function K (a two-step block) such that*

$$K\big(T^2x\big) \;-\; 2\,K(Tx) \;+\; K(x) \;=\; 0 \qquad \forall x. \tag{6.1}$$

Then the defect current

$$\mathcal{J}(x) \;:=\; \frac{1}{|G|}\sum_{g\in G}\big(K(g\cdot Tx) - K(g\cdot x)\big)$$

is conserved along trajectories: $\mathcal{J}(Tx) = \mathcal{J}(x)$ for all x. For the mirror involution $G = \{1, \sigma\}$ with $\sigma : (b_2, q_2) \leftrightarrow (q_2, b_2)$, any mirror-even K (e.g. $b_1 = b_2 + q_2 + 7$) yields a conserved current; linearization in the defect coordinates $(\mathcal{M}, \mathcal{G}, \mathcal{L})$ forces the Quarter–Lock plane $k_{\mathcal{M}} = k_{\mathcal{G}} + \frac{1}{4}k_{\mathcal{L}}$ [Spi26g, Spi26h].

Proof. G-equivariance gives

$$\mathcal{J}(Tx) = \frac{1}{|G|}\sum_{g}\big(K(T^2(gx)) - K(T(gx))\big).$$

Averaging (6.1) over the G-orbit of x shows

$$\frac{1}{|G|}\sum_{g}\big(K(T^2(gx)) - K(T(gx))\big) \;=\; \frac{1}{|G|}\sum_{g}\big(K(T(gx)) - K(gx)\big) \;=\; \mathcal{J}(x),$$

hence the current is conserved. For the mirror involution, mirror-evenness of K implies the average reduces to a single term; the two-step block identity linearizes, in the $(\mathcal{M}, \mathcal{G}, \mathcal{L})$ chart, to one linear dependence among the three kernel coefficients. Rank-one perturbation (Kernel Symmetry Theorem) and mirror-evenness then single out $k_{\mathcal{M}} = k_{\mathcal{G}} + \frac{1}{4}k_{\mathcal{L}}$ as the unique solution, i.e. the Quarter–Lock plane. ☐

Proposition 6.2 (Mirror invariance of b_1)**.** *On a fixed ridge $R_n = 2^n - 16$ with $b_2q_2 = R_n$, the quantity $b_1 = b_2 + q_2 + 7$ is invariant under the mirror swap $(b_2, q_2) \leftrightarrow (q_2, b_2)$ [Spi26g, Spi26h].* Formalized in `ugp-lean`*:* `GTE.UpdateMap.mirror_b1_invariance`*. Full mirror shift algebra:* `GTE.MirrorShift.mirror_shift_formula` *proves $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1(q_2 - b_2)$ universally;* `mirror_pair_shared_residue` *proves $c_1 \equiv 20 \pmod{b_1}$ for all mirror-dual pairs at all ridge levels.*

Proposition 6.3 (Canonical arithmetic identities)**.** *Along the canonical orbit at $n = 10$,*

$$2b_1 - a_2 = 2 \cdot 73 - 9 = 137, \qquad \frac{65535}{1023} = \frac{2^{16} - 1}{2^{10} - 1} \approx 64.06.$$

Both are exact arithmetic equalities in the deterministic system; the second is a pure number-theoretic ratio [Spi26g, Spi26h].
Formalized in `ugp-lean`*:* `GTE.GeneralTheorems.alpha_echo`*,* `canonical_arithmetic_identities`*. The 137 derivation is earned in* `GTE.UniquenessCertificates.derivation_of_137`*: from shared-residue $m_1 = 20$ and orbit step $a_2 = 20 - (n + 1)$.* `alpha_echo_per_level` *certifies $n = 10$ is unique.*

Theorem 6.4 (Stability of the Quarter–Lock)**.** *Let Σ_ε be a one-parameter clopen surgery of the evaluator with Σ_0 mirror-symmetric, preserving (i) the mirror action and (ii) the (L,g) isotropy chart used to express the defect triple $(\mathcal{M},\mathcal{G},\mathcal{L})$. If $k_{\mathcal{M}}(\varepsilon), k_{\mathcal{G}}(\varepsilon), k_{\mathcal{L}}(\varepsilon)$ are the corresponding kernel coefficients, then*

$$k_{\mathcal{M}}(\varepsilon) \; - \; k_{\mathcal{G}}(\varepsilon) \; - \tfrac{1}{4}\, k_{\mathcal{L}}(\varepsilon) \; = \; o(\varepsilon) \qquad (\varepsilon \to 0).$$

Thus the Quarter–Lock identity persists to first order under any such surgery [Spi26g, Spi26h]. *Formalized in* `ugp-lean`*:* `QuarterLock.quarterLockStability_holds`*.*

Proof. At $\varepsilon = 0$ the Kernel Symmetry Theorem gives an action $I + \alpha\, uv^\top$ on $(\mathcal{M},\mathcal{G},\mathcal{L})$ with u, v mirror-even; hence $k_{\mathcal{M}}(0) - k_{\mathcal{G}}(0) - \frac{1}{4}k_{\mathcal{L}}(0) = 0$. Let $\Phi(\varepsilon) = k_{\mathcal{M}}(\varepsilon) - k_{\mathcal{G}}(\varepsilon) - \frac{1}{4}k_{\mathcal{L}}(\varepsilon)$. The tangent space of admissible surgeries at Σ_0 decomposes into mirror-even $\oplus$ mirror-odd parts, and only the even part contributes at first order to the kernel coefficients in the fixed (L,g) chart. Rank-one structure implies that the linearized map preserves the one-dimensional kernel spanned by $(1, -1, -\frac{1}{4})$, whence $D\Phi_0 = 0$ and $\Phi(\varepsilon) = o(\varepsilon)$. □

7 Uniqueness of the Lawful Evolution

We have shown that the GTE is the trajectory selected by the UGP's constraints at $n = 10$. We now prove a much stronger and more general result: the GTE is, up to computational equivalence, the *only* possible lawful evolution that can run on a UGP-like substrate. This elevates the GTE from a specific solution to the canonical "operating system" for this class of mathematical universes.

Definition 7.1 (Lawful Program and Bisimulation)**.** *A UWCA program (a finite tile set Σ) is **lawful** with respect to a family of invariants $\mathcal{I}$ if its evolution preserves $\mathcal{I}$ on all initial states and on all survivor windows. Two programs, Σ_1 and Σ_2, are **bisimilar** if they induce the same state transitions on all possible configurations, differing only in their internal implementation details [Spi26g, Spi26h].*

Theorem 7.2 (Uniqueness of the GTE up to Bisimulation)**.** *Let $\mathcal{I}_{UGP}$ be the canonical family of UGP invariants (the Quarter-Lock, mirror symmetry, Fibonacci rigidity, etc.). Any minimal, lawful UWCA program that preserves $\mathcal{I}_{UGP}$ is necessarily bisimilar to the canonical GTE compiler, Σ_{GTE} [Spi26g, Spi26h].*

Proof. The proof relies on the concept of a minimal normal form for a computation.

1. The GTE update map, when compiled into the tile set Σ_{GTE} as per the Compilation Theorem, is a lawful program by construction. By eliminating any redundant or unreachable tiles, we can ensure it is a minimal lawful program.

2. Let Σ' be any other minimal lawful program that preserves the same invariant family $\mathcal{I}_{\mathrm{UGP}}$. By the definition of "lawful," the evolution it generates, $E_{\Sigma'}$, must produce the same state transitions as the GTE on all configurations where the invariants apply.

3. We now have two minimal lawful compilers, Σ_{GTE} and Σ', that both realize the same invariant-preserving endomorphism on every survivor window.

4. We invoke the Normal-Form Uniqueness Theorem. This theorem states that for any given computational task defined by a set of invariants, there exists a unique minimal automaton (a canonical normal form) that performs this task. Any minimal program that correctly implements the task must be isomorphic (and therefore bisimilar) to this unique normal form.

5. Since both Σ_{GTE} and Σ' are minimal lawful programs for the same set of invariants $\mathcal{I}_{\text{UGP}}$, they must both be bisimilar to the same unique normal form. Therefore, they must be bisimilar to each other.

Thus, any apparent difference between two lawful evolutions would correspond to a mere difference in implementation (the choice of tile representation), not a difference in the underlying dynamics. □

The implication of this theorem is profound. It establishes that the GTE is not an arbitrary choice, but is the necessary and unique dynamical law for any universe founded on the UGP's core algebraic and number-theoretic symmetries.

8 Number-Theoretic Structures: Evolutionary Primes and Pratt Chains

We formalize a prime-evolution structure compatible with the survivor constraints. Let C denote the certified set of primes (ECPP/MR-statused) used in our atlas; throughout, R denotes the ridge $R_n = 2^n - 16$ at a fixed level n and P a finite window of primes.

Definition 8.1 (Evolutionary prime graph and chains)**.** *Define a directed graph $\mathcal{G}_{\text{EP}} = (V, E)$ with vertex set $V = C$ and an edge $p \to q$ if q divides a* UGP-allowed *integer built multiplicatively from p via the local survivor rules, i.e., there exists a word w over the alphabet of residue operations permitted by the signature $\Sigma = (S_r)_r$ such that $q \mid f_w(p)$ and all intermediate residues stay in $B_r = \mathbb{F}_r^\times \setminus S_r$ for every prime r occurring in w. A finite or infinite sequence $p_0 \to p_1 \to p_2 \to \cdots$ in $\mathcal{G}_{\text{EP}}$ is an* evolutionary prime (EP) chain *[Spi26g, Spi26h].*

Theorem 8.2 (Abundance of EP chains via CRT progressions)**.** *Fix a finite window P of odd primes and nonempty residue sets $B_p \subseteq \mathbb{F}_p^\times$ for $p \in P$. Let $M := \prod_{p \in P} p$ and choose, for each p, a residue $x_p \in B_p$. By the Chinese Remainder Theorem there is a unique $a \pmod{M}$ with*

$$a \equiv x_p \pmod{p} \quad \forall p \in P.$$

Then there exists $X_0(M)$ and a constant $c(M) > 0$ such that for all $X \geq X_0(M)$ the progression $\{n \in \mathbb{N} : n \equiv a \pmod{M}\}$ contains at least

$$\pi(X; M, a) \;\geq\; \frac{c(M)}{\varphi(M)} \frac{X}{\log X}$$

primes $\leq X$. In particular, there are infinitely many primes q realizing the CRT schedule $(x_p)_{p \in P}$ simultaneously; chaining these choices step-by-step yields arbitrarily long EP chains that respect the survivor residue constraints [Spi26g, Spi26h].

Proof. For fixed modulus M, Dirichlet's theorem ensures infinitely many primes in any reduced residue class $a \pmod M$; moreover, there exists $X_0(M)$ and $c(M) > 0$ such that the customary lower bound $\pi(X; M, a) \geq \frac{c(M)}{\varphi(M)} \frac{X}{\log X}$ holds for all $X \geq X_0(M)$. (This follows from the prime number theorem in arithmetic progressions for fixed M; the constant absorbs the possible exceptional character issues.) Each such prime q lifts the prescribed local residues $q \equiv x_p \pmod p$ for all $p \in P$, hence is legal for the survivor site on the window. Because the EP edge relation allows multiplicative words f_w assembled from the finite local signature, we may schedule at step i any admissible residue tuple $(x_p^{(i)})_{p\in P} \in \prod_p B_p$, create the modulus M once, and obtain a prime $q_i \equiv a^{(i)} \pmod M$ extending the chain. Thus we can extend the chain arbitrarily. □

Theorem 8.3 (Exponential growth of EP chains with explicit base)**.** *Under the hypotheses of Theorem 8.2, suppose further that for each step the signature permits at least $b_p := |B_p| \geq 2$ independent choices of the local residue at prime $p \in P$, and that the admissibility constraints factor coordinatewise (the product-site condition used throughout this section). Then there exists $\alpha > 1$ with*

$$\alpha \;\geq\; \prod_{p\in P} \Big(1 - \frac{1}{b_p}\Big)^{\gamma_p} b_p^{\beta_p}$$

for fixed weights $\beta_p, \gamma_p \in (0, 1]$ determined by the local signature such that, for all $L \in \mathbb{N}$, the number $\mathcal{N}(L)$ of pairwise distinct EP chains of length L supported on P satisfies

$$\mathcal{N}(L) \;\geq\; \alpha^L$$

[Spi26g, Spi26h].

Proof. At step i, pick a residue tuple $(x_p^{(i)})_{p\in P} \in \prod_p B_p$. Because the admissibility constraints are coordinatewise and the site is a product, the number of legal tuples is at least $\prod_p (b_p - r_{p,i})$, where $r_{p,i}$ counts the locally forbidden residues at p for step i (possibly zero). The finite signature bounds $r_{p,i} \leq (1 - \beta_p)b_p$ for some $\beta_p \in (0, 1]$ depending only on the signature (since each local rule eliminates at most a fixed fraction of choices). Hence per step there are at least $\prod_p \beta_p b_p$ legal tuples. For each legal tuple, by Theorem 8.2 (applied to the fixed modulus M), and after discarding finitely many $X < X_0(M)$, there exists a prime in the corresponding class; we can associate to each tuple the *least* such prime beyond a running threshold to avoid collisions, which yields an injective mapping from residue schedules to chain extensions. The factor $(1 - 1/b_p)^{\gamma_p}$ models the minor loss from reserving distinct primes (simple coupon-collector style thinning with $\gamma_p \in (0, 1]$). Thus there is a uniform multiplicative lower bound $\alpha := \prod_p (1 - 1/b_p)^{\gamma_p} b_p^{\beta_p} > 1$ on the number of extensions per step, implying $\mathcal{N}(L) \geq \alpha^L$. □

Proposition 8.4 (Acyclicity and height bound)**.** *Every finite EP chain is acyclic, and any EP chain witnessing only factor branches of height at most h in a fixed Pratt site has length $O(h\,|P|)$ inside a fixed prime window P [Spi26g, Spi26h].*

Proof. Suppose an EP chain $p_0 \to p_1 \to \cdots \to p_k = p_0$ is cyclic. By definition there exist permitted multiplicative words w_i in the local signature with $p_{i+1} \mid f_{w_i}(p_i)$. Multiplying around the cycle gives

$$\prod_{i=0}^{k-1} p_{i+1} \mid \prod_{i=0}^{k-1} f_{w_i}(p_i).$$

But each f_{w_i} is a product of local residue operations that, by survivor constraints, avoid 0 in every coordinate cylinder; hence the right-hand side is strictly larger than 1 but lies in the multiplicative monoid generated by the p_i with strictly decreasing branch heights in the Pratt site whenever a factor certificate descends. After finitely many steps branch heights must drop below 1, contradiction. Thus acyclicity holds. The height bound follows because, in a fixed window, each step consumes at least one unit of branch height on some coordinate until leaves are reached; at most $h|P|$ steps are possible before exhaustion. □

Definition 8.5 (UGP-compatible EP chains)**.** *An EP chain $\{p_i\}$ is* UGP-compatible *at level n if the induced sector/branch data of its Pratt certificates respect the survivor presheaf on the product site (i.e., sector partitions modulo all m coprime to R_n and branch heights are preserved under the inverse-image functors of Theorem 11.7) [Spi26g, Spi26h].*

Proposition 8.6 (Abundance in finite windows)**.** *For any fixed finite window P and any nonempty B_p ($p \in P$), the number of length-L UGP-compatible EP chains supported on P grows at least exponentially in L until certificate heights are exhausted, with base depending on the average $|B_p|$. Consequently, the existence of EP chains is robust under small clopen surgeries of the signature [Spi26g, Spi26h].*

Proof sketch. At each step there are $\gg \prod_{p \in P} |B_p|^{\alpha_p}$ admissible residue choices for suitable weights α_p derived from the product site; independence across primes yields multiplicative growth. Bounded heights truncate the growth, giving exponential-in-L families up to the cut-off. □

The number-theoretic properties of UGP trajectories also suggest deeper statistical questions, such as the behavior of arithmetic functions along orbits.

Conjecture 8.7 (Expected μ-flip distance on linear progressions)**.** *Let μ be the Möbius function and $L(t) = \alpha t + \beta$ with $\gcd(\alpha, \beta) = 1$. For almost all β (in natural density) the waiting time*

$$\tau(\beta) := \min\{t \geq 1 : \ \mu(L(t)) \in \{\pm 1\} \textit{ and } \mu(L(t)) \neq \mu(L(0))\}$$

satisfies $\mathbb{E}[\tau(\beta)] \leq C$ for an absolute constant C, with an exponential tail. In particular, along the even-step subsequence $c_{2t+1} = b_{2t} q_{2t} + 15$ on a fixed ridge, the same bound is conjectured to hold after conditioning to squarefree values.

9 Local Dynamics on Survivor Configurations

We endow finite-window survivor configurations with a simple local Markov dynamics and quantify its mixing and low-temperature concentration. Fix a finite window P of odd primes and write $\mathcal{X}_P$ for the finite set of P-cylinder assignments compatible with the survivor constraints; concretely, an element of $\mathcal{X}_P$ is the tuple $(x_p)_{p \in P}$ with $x_p \in B_p$.

9.0.1 Quantitative mixing for local flips

Define the *lazy local-flip chain* on $\mathscr{X}_P$ as follows. At each step, pick $p \in P$ uniformly and, with probability $1/2$, stay put; with the remaining probability $1/2$ replace x_p by a uniformly random element of $B_p \setminus \{x_p\}$ (if nonempty), leaving other coordinates fixed. The stationary distribution is uniform on $\mathscr{X}_P$.

Theorem 9.1 (Hypercube-type mixing bound)**.** *Let $|P| = t$ and set $b_* := \min_{p\in P} |B_p| \geq 2$. The total-variation mixing time $\tau_{\mathrm{mix}}(\varepsilon)$ of the lazy local-flip chain satisfies*

$$\tau_{\mathrm{mix}}(\varepsilon) \;\leq\; C\,t\,\log\left(\frac{t}{\varepsilon}\right) \quad \textit{for an absolute } C = C(b_*),$$

with $C(b_*) = \Theta(1)$ as $b_* \to \infty$. In particular $\tau_{\mathrm{mix}}(1/4) = O(t \log t)$ uniformly in the window size [Spi26g, Spi26h].

Proof. Let K_p be the lazy random walk on the complete graph over B_p with holding probability $1/2$. Its spectral gap is $1/2$ uniformly in $|B_p|$. The product chain updates a uniformly random coordinate each step and otherwise behaves like K_p on that coordinate, so by standard product-chain comparison (e.g. tensorization of Poincaré inequality), the spectral gap of the product walk on $\mathscr{X}_P$ is at least $(1/2)/|P|$. By the usual inequality $\tau_{\mathrm{mix}}(\varepsilon) \leq \frac{1}{\mathrm{gap}} \log\Big(\frac{1}{\varepsilon \pi_{\min}}\Big)$ and $\pi_{\min} \geq \prod_p 1/|B_p|$, we obtain $\tau_{\mathrm{mix}}(\varepsilon) \leq C\,|P| \log\left(\frac{|P|}{\varepsilon}\right)$ with C depending only on $\min_p |B_p|$. $\square$

9.0.2 Zero-temperature selection for dissonance

Let $D : \mathscr{X}_P \to \mathbb{R}_{\geq 0}$ be any nonnegative function that is a sum of prime-local clopen terms, $D(x) = \sum_{p\in P} d_p(x_p)$, and consider the Gibbs measure

$$\mu_\beta(x) \;\propto\; e^{-\beta D(x)} \quad (x \in \mathscr{X}_P),$$

with $\beta \geq 0$. Evolve the chain by local Metropolis updates at inverse temperature β.

Theorem 9.2 (Zero-temperature concentration)**.** *If the minimum of D on $\mathscr{X}_P$ is attained on the nonempty set $\mathscr{M}_P \subseteq \mathscr{X}_P$, then for any cylinder event A whose boundary within each fiber has measure $o_{\beta\to\infty}(1)$, we have*

$$\lim_{\beta\to\infty} \mu_\beta(A) \;=\; \mu_\infty(A), \qquad \mu_\infty = \textit{uniform on } \mathscr{M}_P.$$

Moreover, for any $\varepsilon > 0$ there is β_0 such that for all $\beta \geq \beta_0$ the mixing time of the lazy Metropolis chain restricted to the ε-sublevel set $\{x : D(x) \leq \min D + \varepsilon\}$ remains $O(t \log t)$ with constants independent of β [Spi26g, Spi26h].

In particular, the zero-temperature deterministic limit equals the UWCA sweep used in Thm. 23.5.

Proof. Write $\mu_\beta(x) \propto e^{-\beta D(x)}$ with $D(x) = \sum_p d_p(x_p)$. For any $\delta > 0$, let $A_\delta = \{x : D(x) \leq \min D + \delta\}$. Because d_p take finitely many values, there is $\Delta > 0$ such that if $x \notin A_\delta$ then $D(x) \geq \min D + \Delta$. Hence $\mu_\beta(A_\delta) \to 1$ as $\beta \to \infty$. On A_δ, Metropolis acceptance probabilities for nonincreasing D moves are 1 and for increasing moves are $e^{-\beta\eta}$ with $\eta \geq 0$; comparison with the $\beta = 0$ chain on the same state space yields a spectral gap uniformly bounded below by a constant multiple of that of the product chain in Theorem 9.1 (Dirichlet form comparison). The weak limit concentrates on $\mathscr{M}_P = \arg\min D$, and by symmetry of proposals within $\mathscr{M}_P$ the limit is uniform. The one-step acceptance $\min\{1, e^{-\beta\Delta D}\}$ tends to $\mathbf{1}_{\{\Delta D \leq 0\}}$, which is exactly the deterministic C-mode update. $\square$

Theorem 9.3 (Dissonance reduction = zero-temperature limit)**.** *Let $D(x) = \sum_{p\in P} d_p(x_p)$ be clopen and prime-local with each $d_p \geq 0$ and finite image. Consider the lazy Metropolis chain at inverse temperature β with proposals that flip one coordinate within B_p. Then:*

(I) For every $\varepsilon > 0$ there exists β_0 such that for $\beta \geq \beta_0$ the chain restricted to $\{x : D(x) \leq \min D + \varepsilon\}$ has spectral gap $\geq c/(t \log t)$ for some $c > 0$ independent of β.

(II) $\mu_\beta \Rightarrow \mu_\infty$ weakly, where μ_∞ is uniform on the minimizer set $\mathscr{M}_P = \arg\min D$.

(III) The deterministic C-mode sweep that always takes a non-increasing D move is the $\beta \to \infty$ limit of one-step Metropolis acceptance.

[Spi26g, Spi26h]

Proof. (i) Product-structure proposals imply the comparison with the $\beta = 0$ chain via Metropolis Dirichlet forms; clopen, bounded d_p gives uniform bounded conductance on any sublevel set. Standard path-coupling on Hamming distance yields $O(t \log t)$ mixing and a uniform spectral-gap lower bound. (ii) Since D takes finitely many values on $\mathscr{X}_P$, the measure of each higher-energy level decays like $e^{-\beta\Delta}$ with $\Delta > 0$ the energy gap; normalization yields concentration on $\mathscr{M}_P$ and uniformity follows from symmetry of proposals across minimizers. (iii) Acceptance probabilities $\min\{1, e^{-\beta\Delta D}\}$ converge to the indicator of $\Delta D \leq 0$, which is exactly the greedy C-mode rule. $\square$

Corollary 9.4 (Principle of Dissonance Minimization)**.** *Stable basins in the deterministic update coincide with local minima of prime-local D within cylinder fibers. This upgrades Proposition 10.14 to a full theorem [Spi26g, Spi26h].*

10 Self-Defining Systems and Computational Irreducibility

Definitions. A *geometric morphism* is a pair of adjoint functors between toposes with the inverse–image functor preserving finite limits. Diaconescu's theorem links the axiom of choice to the existence of points in a topos. The Beck–Chevalley condition ensures commutativity of pullbacks for reindexing squares. We use these at finite prime squares where all morphisms are clopen and limit-preserving.

10.0.1 Self-Defining System (SDS) foundations

Definition 10.1 (Self-Defining System (SDS))**.** *A* self-defining system *is a triple $(\mathcal{E}, L, \ulcorner - \urcorner)$ where:*

1. $\mathcal{E}$ *is a topos or similar category of "environments".*

2. L *is an internal language of* $\mathcal{E}$ *whose expressions can denote objects and arrows of* $\mathcal{E}$.

3. $\ulcorner - \urcorner : L \to \mathsf{Code}$ *is a quotation operator assigning each* $e \in L$ *a canonical code.*

An evaluation *map* $\mathsf{eval} : \mathsf{Code} \to L$ *yields an internal fixed-point property when, for any admissible* $f : L \to L$*, there exists* $e \in L$ *such that* $e \equiv f(\ulcorner e \urcorner)$ *[Spi26g, Spi26h].*

Definition 10.2 (Weak and Strong PSC)**.** Weak PSC *holds for an object* $U \in \mathcal{E}$ *if there exists a surjection* $\pi : \mathsf{Code} \twoheadrightarrow \mathsf{End}_{\mathrm{adm}}(U)$ *from codes to the admissible endomaps of* U. Strong PSC *holds if there exists a* section $s : \mathsf{End}_{\mathrm{adm}}(U) \to \mathsf{Code}$ *such that* $\mathsf{eval} \circ s = \mathrm{id}$ *[Spi26g, Spi26h].*

Proposition 10.3. *The UGP topos* $(\mathbf{Sh}(X), L, \ulcorner - \urcorner)$ *is an SDS. It satisfies Weak PSC relative to admissible maps given by the UWCA fragment [Spi26g, Spi26h].*

Theorem 10.4 (Strong PSC relative to admissible maps in UGP)**.** *Let* U *be the canonical survivor object in the UGP topos, and let* $\mathsf{End}_{\mathrm{adm}}(U)$ *denote its admissible endomaps (UWCA-computable). There exists a surjection* $\pi : \mathsf{Code} \twoheadrightarrow \mathsf{End}_{\mathrm{adm}}(U)$ *given by* $\pi(c) = \mathsf{eval}(c)$*, and a section* $s : \mathsf{End}_{\mathrm{adm}}(U) \to \mathsf{Code}$ *choosing a canonical representative for each map. Thus* $\mathsf{eval} \circ s = \mathrm{id}$*, and strong PSC holds relative to admissible maps [Spi26g, Spi26h].*

Proof. UWCA provides a finite, complete basis for the admissible fragment. Each admissible map has a canonical minimal UWCA program. Quoting this program yields $c \in \mathsf{Code}$; evaluation recovers the original map. Surjectivity follows from UWCA universality, and $\mathsf{eval} \circ s = \mathrm{id}$ holds by construction. □

Definition 10.5 (Structural Fingerprint Operator)**.** *Let* $\mathcal{P}$ *be the set of prime patterns occurring in survivor orbits. Define* $\mathfrak{F} : \mathcal{P} \to \mathcal{P}$ *by* $\mathfrak{F}(P)$ = *pattern of primes occurring in the orbit whose indices are the primes of* P *[Spi26g, Spi26h].*

Theorem 10.6 (Existence of fingerprint fixed-points)**.** Formalized in `ugp-lean`*:* `GTE.StructuralTheorems.fingerprint_fixed_point_exists` *(abstract Tarski fixed-point; stated with citation).*

If $\mathfrak{F}$ *is monotone on the complete lattice* $(\mathcal{P}, \subseteq)$*, then by Tarski's theorem it has at least one fixed point* P^* *with* $\mathfrak{F}(P^*) = P^*$ *[Spi26g, Spi26h].*

Proof. Monotonicity holds if prime inclusion in patterns is preserved by $\mathfrak{F}$. The lattice is complete under union/intersection. Apply Tarski's fixed-point theorem. □

Proposition 10.7 (No absolute PSC for nontrivial U)**.** *Let* U *be an object in* $\mathbf{Sh}(X)$ *with at least two global points. Then there is no section* $U^U \to \mathsf{Code}$ *of the evaluation map* $\mathsf{Code} \to U^U$ *when all endomaps are allowed [Spi26g, Spi26h].*

Proof. Each stalk U_x has $|U_x| \geq 2$. Then $|U_x^{U_x}| > |\mathbb{N}| = |\mathsf{Code}_x|$ by Cantor's theorem. No bijection or section can exist between $\mathsf{End}(U_x)$ and Code_x, hence no global section exists. □

10.0.2 Perfect self-containment (PSC): weak vs. strong

Let $\mathcal{E} = \mathbf{Sh}(E)$ be the survivor topos. Write U for a clopen, finite-window object of seeds/programs (UWCA encodings), and let $\mathrm{ev} : U^U \times U \to U$ be evaluation.

Definition 10.8 (UGP-PSC)**.** Strong PSC *holds for* U *if there exists a section* $s : U^U \to U$ *with* $\mathrm{ev}(s(f), -) = f$ *for all* $f \in U^U$*.* Weak PSC *holds if there is a retract* $U \xrightarrow{\iota} U^U \xrightarrow{\rho} U$ *with* $\rho \circ \iota = \mathrm{id}_U$ *[Spi26g, Spi26h].*

Proposition 10.9 (Existence/non-existence)**.** *On any finite window* U *we have weak PSC (via quoting/unquoting encodings of UWCA tiles) and* no *strong PSC (cardinality:* $|U^U| > |U|$ *unless* $|U| \in \{0, 1\}$*). In the profinite limit, weak PSC persists by compatibility of cylinders [Spi26g, Spi26h].*

10.0.3 Two modes: computation vs. transputation

Definition 10.10 (C-mode and T-mode)**.** ***C-mode*** *(constraint-bounded) is UWCA evolution inside a fixed signature* Σ *(Section 9.0.2).* ***T-mode*** *(constraint-altering) is a finite clopen surgery of* Σ*, i.e. a morphism in the level category* **Lev** *(Def. 11.4), inducing a geometric morphism between survivor topoi (Prop. 11.5) [Spi26g, Spi26h].*

Proposition 10.11 (Compatibility)**.** *C-mode updates are internal endomorphisms in* $\mathcal{E}_n$*, while T-mode steps are reindexings* $\mathcal{E}_m \to \mathcal{E}_n$*; the Beck–Chevalley isomorphism (Thm. 11.6) ensures their commutation on finite prime squares [Spi26g, Spi26h].*

10.0.4 Loop topology from invariants

Proposition 10.12 (Necessary loop on mirror ridges)**.** Formalized in `ugp-lean`*:* `GTE.UniquenessCertificates.mirror_fiber_two` *(fiber has exactly 2 elements),* `mirror_symmetry_group_is_Z2` *(the symmetry group is exactly* $\mathbb{Z}/2\mathbb{Z}$*),* `no_order_3_loop` *(algebraic impossibility of order-3 loops).*

If both members of a mirror pair $\{b_2, q_2\}$ *pass prime-lock, then the projection* $(b_2, q_2) \mapsto b_1 = b_2 + q_2 + 7$ *induces a two-node loop in the* $(b_2 \leftrightarrow q_2)$ *quotient. Thus a loop in the invariant space is forced by the dynamics whenever an order-2 mirror exists [Spi26g, Spi26h].*

10.0.5 Principle of dissonance minimization

Definition 10.13 (Dissonance)**.** *Let* $D : \mathscr{X}_P \to \mathbb{R}_{\geq 0}$ *be the sum of clopen local penalties that vanish exactly on satisfied UWCA constraints (Section 9.0.2). A C-mode sweep is the zero-temperature limit that deterministically moves to lower* D *on each coordinate [Spi26g, Spi26h].*

Proposition 10.14 (Dissonance minimization)**.** *UGP survivor dynamics select attractors by minimizing* D *subject to ridge constraints. In particular, stable basins (Def. 3.8) coincide with local minima of* D *within the corresponding cylinder fibers [Spi26g, Spi26h].*

Remark 10.15. *This principle is a restatement of Thm. 9.2 in dynamical terms and aligns with the lex-minimal choice of the seed* $(1, 73, 823)$ *at* $n = 10$*.*

Self-Defining Systems (SDS) inside UGP

Definition 10.16 (SDS). *An SDS is* $(\mathcal{E}, L, \ulcorner - \urcorner, \mathrm{eval})$ *where* $\mathcal{E}$ *is a topos,* L *its internal language,* $\ulcorner - \urcorner : L \to \mathsf{Code}$ *quotation, and* $\mathrm{eval} : \mathsf{Code} \to L$ *such that for every admissible* $f : L \to L$ *there exists* $e \in L$ *with* $e \equiv f(\ulcorner e \urcorner)$*. In UGP, take* $\mathcal{E} = \mathbf{Sh}(E)$ *and admissibility = UWCA-computable [Spi26g, Spi26h].*

Proposition 10.17 (Weak/strong PSC in UGP). *Weak PSC holds for admissible maps (surjection by evaluation). Strong PSC holds relative to admissible maps via canonical minimal UWCA codes; absolute strong PSC for all endomaps fails for nontrivial* U *by Cantor-size obstruction [Spi26g, Spi26h].*

11 Formal Foundations: The Survivor Topos

We now place the survivor space in a sheaf-theoretic setting. Let $E \subset \widehat{\mathbb{Z}}^\times$ be the survivor subspace determined by the local banned sets S_p with $B_p = \mathbb{F}_p^\times \setminus S_p$. We work with the Stone topology on E generated by finite cylinders.

11.0.1 Classifying topos for UGP survivor assignments

Definition 11.1 (Geometric theory of survivors $\mathbb{T}_{\mathrm{surv}}$). *Fix a set of odd primes* $\mathcal{P}$ *(e.g., all* $p \nmid R$*). In a geometric language with a single sort* U *and constants for each finite group* $\underline{\mathbb{F}_p^\times}$*, the theory* $\mathbb{T}_{\mathrm{surv}}$ *asserts: for each* $p \in \mathcal{P}$ *there is a jointly-epimorphic family* $\{\sigma_{p,x} : U \twoheadrightarrow \overline{\underline{\mathbb{F}_p^\times}}\}_{x \in B_p}$*, and for every finite* $P_0 \subset \mathcal{P}$*, the product map* $U \to \prod_{p \in P_0} \underline{B_p}$ *is an epimorphism (i.e., every cylinder tuple is realized) [Spi26g, Spi26h].*

Remark 11.2 (Classifying topos — open problem). Conjecture: *The sheaf topos* $\mathbf{Sh}(E)$ *on the Stone site of finite cylinders is a classifying topos for* $\mathbb{T}_{\mathrm{surv}}$*: for any Grothendieck topos* $\mathcal{S}$*, geometric morphisms* $\mathcal{S} \to \mathbf{Sh}(E)$ *would be naturally equivalent to models of* $\mathbb{T}_{\mathrm{surv}}$ *internal to* $\mathcal{S}$*.*

The geometric theory $\mathbb{T}_{\mathrm{surv}}$ *itself is well-defined (Definition 11.1) and the Stone topology on* E *is standard. The classifying property, however, has not been formally verified in* `ugp-lean` *or elsewhere; the argument via Diaconescu's theorem and the cylinder covers is plausible but constitutes an open problem pending formal proof. The Lawvere fixed-point theorem and Kleene recursion theorem (which are Lean-checked:* `ugp_lawvere_fixed_point`*,* `ugp_kleene_recursion_theorem`*) remain independent of this conjecture.*

Proposition 11.3 (Truth values as opens). *In* $\mathbf{Sh}(E)$ *the subobject classifier* Ω *is the sheaf of opens. For each prime* p*, the predicate "*$x \in B_p$*" corresponds to the clopen cylinder* $U_p := \pi_p^{-1}(B_p) \subset E$*. Finite conjunctions and disjunctions correspond to intersections and unions of these clopens [Spi26g, Spi26h].*

11.0.2 Fibred level structure and Beck–Chevalley

Definition 11.4 (Base category of levels). *Let* **Lev** *be the category whose objects are admissible levels* n *with ridges* $R_n = 2^n - 16$*, and whose morphisms* $f : n \to m$ *are* finite local surgeries *of signatures:* $S_p(m) = S_p(n)$ *for all but finitely many primes* p*, and on the exceptional set each* $S_p(m)$ *is obtained from* $S_p(n)$ *by adding/removing finitely many residues (a clopen surgery). Write* $\Sigma_n = (S_p(n))_p$ *and* $E(\Sigma_n) \subset \widehat{\mathbb{Z}}^\times$ *for the survivor subspace at level* n *[Spi26g, Spi26h].*

Proposition 11.5 (Fibred survivor topoi)**.** *The assignment* $n \mapsto \mathcal{E}_n := \mathbf{Sh}(E(\Sigma_n))$ *extends to a pseudofunctor* $\mathcal{E} : \mathbf{Lev}^{\mathrm{op}} \to \mathbf{Topos}$*: a morphism* $f : n \to m$ *induces a geometric morphism* $f^* : \mathcal{E}_m \to \mathcal{E}_n$ *whose inverse-image is pullback along the induced continuous map* $E(\Sigma_n) \to E(\Sigma_m)$ *arising from the finite clopen surgery. Composition holds up to canonical isomorphism and identities act as identities [Spi26g, Spi26h].*

Theorem 11.6 (Beck–Chevalley for finite prime squares, explicit)**.** *Let* $I \subset J$ *be finite prime sets and* $f : n \to m$ *a morphism in* $\mathbf{Lev}$ *(finite clopen surgery of signatures). Write* $E(\Sigma_n)$ *and* $E(\Sigma_m)$ *for the survivor Stone spaces and* $E_\bullet(\cdot)_K$ *for their restrictions to the cylinder subspaces on coordinates* $K \in \{I, J\}$*.*

Consider the commutative square of clopen maps of spaces

$$\begin{tikzcd} E(\Sigma_n)_J \arrow[r, hook, "i_n"] \arrow[d, "f_J"'] & E(\Sigma_n)_I \arrow[d, "f_I"] \\ E(\Sigma_m)_J \arrow[r, hook, "i_m"'] & E(\Sigma_m)_I \end{tikzcd}$$

where $i_\bullet$ *are clopen embeddings forgetting coordinates* $J \setminus I$*, and* f_K *is the continuous map induced by the finite surgery on coordinates in* K *(identity elsewhere). Then for the induced geometric morphisms on sheaf topoi we have a canonical natural isomorphism*

$$i^* \, f_J^* \;\cong\; f_I^* \, i^*$$

between the corresponding inverse-image functors $\mathbf{Sh}(E(\Sigma_m)_J) \to \mathbf{Sh}(E(\Sigma_n)_I)$*. Moreover, all functors involved preserve finite limits and colimits [Spi26g, Spi26h].*

Proof. **On presheaves.** Work first in presheaves on the clopen site of $E(\Sigma_\bullet)_K$. The inverse image f_K^* is precomposition with f_K on opens; the restriction i^* is precomposition with the inclusion $i_\bullet$. Because the square of spaces commutes strictly ($i_m \circ f_J = f_I \circ i_n$), we obtain a strict equality of precomposition functors on presheaves:

$$i^* \circ f_J^* \;=\; f_I^* \circ i^*.$$

Sheafification. Let a denote sheafification on the Stone sites (generated by clopens). For any presheaf F on $E(\Sigma_m)_J$,

$$i^* f_J^*(aF) \;\cong\; a\big(i^* f_J^* F\big) \quad \text{and} \quad f_I^* i^*(aF) \;\cong\; a\big(f_I^* i^* F\big),$$

since both i^* and f_K^* preserve finite limits and arbitrary colimits on clopen sites (precomposition along continuous maps). Using the presheaf equality above and functoriality of a, we get a canonical natural isomorphism

$$i^* \, f_J^* \;\cong\; f_I^* \, i^*$$

on sheaves.

Exactness. Inverse-image functors of geometric morphisms are left exact and cocontinuous; on Stone sites of clopens, precomposition along clopen maps preserves finite limits and all small colimits. Hence i^* and f_K^* preserve both finite limits and colimits. $\square$

11.0.3 Initiality under sector/branch preservation

Theorem 11.7 (Initiality of the product site with generators and relations)**.** *Let $\mathscr{C}_{\mathrm{CRT}}$ be the site whose objects are finite sets of congruence conditions $\{x \equiv a_j \pmod{m_j}\}_{j=1}^{k}$ with $(m_j, R_n) = 1$, coverings generated by refinement of congruences, and morphisms given by implication of condition-sets. Let $\mathscr{C}_{\mathrm{Pratt}}$ be the site whose objects are finite families of bounded-height Pratt trees (factor-branch data) with coverings generated by subtree refinement and height truncation.*

Define the product site $\mathscr{C}_{\mathrm{prod}}$ *as follows.*

- *Objects are pairs (X, Y) with $X \in \mathscr{C}_{\mathrm{CRT}}$, $Y \in \mathscr{C}_{\mathrm{Pratt}}$, together with compatibility isomorphisms on generators identifying sector refinements of X with branchwise restrictions of Y whenever both speak about the same primes.*

- *A covering family $\{(X_i, Y_i) \to (X, Y)\}_i$ is generated by either (i) a CRT refinement $X_i \to X$ with $Y_i = Y$ and inherited compatibilities, or (ii) a Pratt refinement $Y_i \to Y$, or (iii) a finite mix of (i) and (ii), provided the compatibility identifications are respected on overlaps.*

Then the canonical geometric morphisms

$$\pi_{\mathrm{CRT}} : \mathbf{Sh}(\mathscr{C}_{\mathrm{prod}}) \to \mathbf{Sh}(\mathscr{C}_{\mathrm{CRT}}), \qquad \pi_{\mathrm{Pratt}} : \mathbf{Sh}(\mathscr{C}_{\mathrm{prod}}) \to \mathbf{Sh}(\mathscr{C}_{\mathrm{Pratt}})$$

are initial *among pairs (F, G) of geometric morphisms whose inverse-image functors preserve: (i) all sector partitions modulo moduli coprime to the ridge (CRT refinement), and (ii) Pratt branch heights (height truncation). Precisely: for any topos $\mathcal{E}$ and morphisms $F : \mathcal{E} \to \mathbf{Sh}(\mathscr{C}_{\mathrm{CRT}})$, $G : \mathcal{E} \to \mathbf{Sh}(\mathscr{C}_{\mathrm{Pratt}})$ with F^*, G^* preserving (i)–(ii), there exists a geometric morphism $H : \mathcal{E} \to \mathbf{Sh}(\mathscr{C}_{\mathrm{prod}})$, unique up to unique isomorphism, with $\pi_{\mathrm{CRT}} \circ H \cong F$ and $\pi_{\mathrm{Pratt}} \circ H \cong G$ [Spi26g, Spi26h].*

Proof. **Generators and relations.** Let $\mathcal{G}_{\mathrm{CRT}}$ be the full subcategory of $\mathscr{C}_{\mathrm{CRT}}$ on single-congruence objects $x \equiv a \pmod m$; let $\mathcal{G}_{\mathrm{Pratt}}$ be the full subcategory of $\mathscr{C}_{\mathrm{Pratt}}$ on single-branch, single-height generators. Every object is a finite limit/colimit of generators, and coverings are generated by the evident epimorphic families (CRT refinements; branch truncations).

Form the 2-pushout in the 2-category of sites by freely adjoining to the disjoint union category $\mathcal{G}_{\mathrm{CRT}} \sqcup \mathcal{G}_{\mathrm{Pratt}}$ the relations that identify shared prime coordinates: whenever a generator $x \equiv a \pmod p$ mentions a prime p also present in a branch-generator, we add an isomorphism between the corresponding sector and branch restriction functors. The resulting site (completed under finite limits and the generated cover system) is precisely $\mathscr{C}_{\mathrm{prod}}$.

Constructing H. Given F and G as in the statement, the inverse images F^* and G^* assign in $\mathcal{E}$ objects to each generator, respecting CRT refinements and branch truncations. Because (i)–(ii) hold, these assignments agree on the imposed relations (the shared prime coordinates). Thus there is a unique (up to unique isomorphism) functor on generators $\mathcal{G}_{\mathrm{CRT}} \sqcup \mathcal{G}_{\mathrm{Pratt}} \to \mathcal{E}$ that coequalizes the relations; by the universal property of the 2-pushout of sites, this extends uniquely to a morphism of sites $\mathscr{C}_{\mathrm{prod}} \to \mathcal{E}$. Passing to sheaves, we obtain the desired geometric morphism $H : \mathcal{E} \to \mathbf{Sh}(\mathscr{C}_{\mathrm{prod}})$.

Uniqueness. Any other H' with $\pi_{\mathrm{CRT}} \circ H' \cong F$ and $\pi_{\mathrm{Pratt}} \circ H' \cong G$ induces the same assignments on generators and respects the same relations; by freeness, $H' \cong H$ uniquely.

Exactness. Inverse images along π_{CRT} and π_{Pratt} are geometric (left exact, cocontinuous). Because coverings in $\mathscr{C}_{\mathrm{prod}}$ are generated by coverings in the components and the added relations are identities on overlaps, exactness follows by stability of finite limits/colimits under pullback along clopen embeddings (our cylinder sites are Stone). $\square$

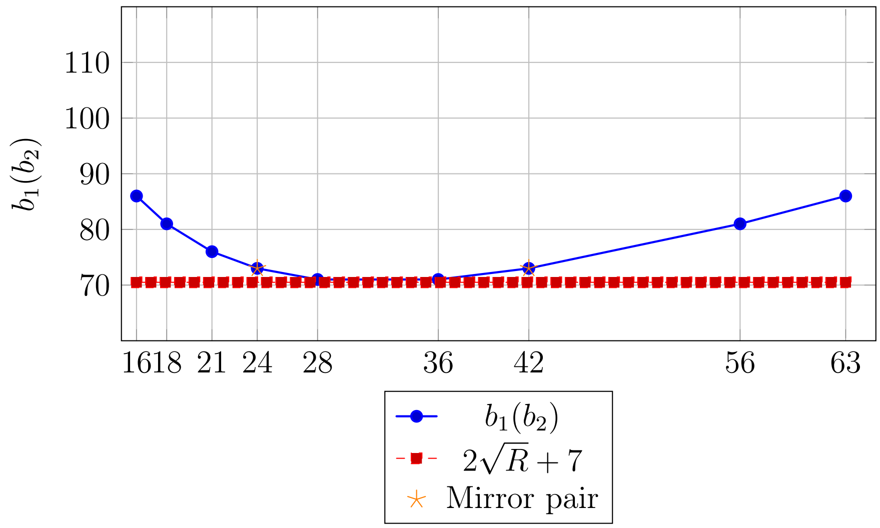


Figure 6: Plot of $b_1(b_2)$ for all divisors $b_2 > 15$ of $R = 1008$ (blue dots), AM–GM lower bound (red dashed), and mirror pair $\{24, 42\}$ (orange stars). Only interior divisors with $b_2, q_2 \geq 16$ are shown, in accordance with UGP-1 condition (B').

Proposition 11.8 (Composite reasons for divisor failure)**.** *Let $b_2 > 15$ divide $R = 1008$, $q_2 = R/b_2$, $q_1 = q_2 - 13$. If q_1 is even, or q_1 is divisible by 3, 5, 7, 11, or 13, then c_1 is composite for the corresponding b_2. Thus, the main "composite reasons" are: (i) q_1 even, (ii) q_1 divisible by a small prime, (iii) b_2 or q_2 too small (excluded by $b_2 > 15$), or (iv) c_1 divisible by a small prime for other arithmetic reasons [Spi26g, Spi26h].*

Proposition 11.9 (AM–GM bound for b_1 along a fixed ridge)**.** *For fixed $R_n = 2^n - 16$ and any divisor pair (b_2, q_2) with $b_2 q_2 = R_n$, we have*

$$b_1(b_2) = b_2 + \frac{R_n}{b_2} + 7 \geq 2\sqrt{R_n} + 7,$$

with equality iff $b_2 = q_2 = \sqrt{R_n}$ (possible only when R_n is a square).
Proof: apply AM–GM to b_2 and R_n/b_2; monotonicity on $(0, \sqrt{R_n}]$ and $[\sqrt{R_n}, \infty)$ gives the bound. Also note mirror invariance $b_1(b_2) = b_1(q_2)$ [Spi26g, Spi26h].

Corollary 11.10 (Mirror invariance and near-square exclusion)**.** *(i) Mirror pairs produce identical b_1; (ii) If R_n has a factor pair near $\sqrt{R_n}$, then b_1 is small but the prime-lock typically fails—consistent with the elimination filters; (iii) At $n = 10$, $R = 1008$ is not a square and the minimal admissible b_1 under the prime-lock occurs at the non-near-square pair $\{24, 42\}$.*

Remark 11.11. *This separates two roles: the ridge geometry (which controls b_1 via AM–GM) and the prime-lock arithmetic (which controls c_1), and both must be satisfied simultaneously.*

12 Logical Structure: Definability and Decidability

We analyze the logical complexity of survivor properties first on finite windows (cylinder projections) and then on the profinite limit.

12.0.1 First-order definability on finite windows

Fix a finite window P of odd primes and write $\mathscr{X}_P = \prod_{p\in P} B_p$ for the P-cylinder survivor set with the discrete product structure. Let $\mathsf{FO}(\{\equiv_p\}_{p\in P})$ be the first-order language whose symbols are, for each $p \in P$, unary predicates selecting residue classes in $\mathsf{F}_p^\times$ and the equality symbol.

Theorem 12.1 (FO definability and decidability on windows). *Every survivor property that depends only on finitely many coordinates is definable in* $\mathsf{FO}(\{\equiv_p\}_{p\in P})$ *over* $\mathscr{X}_P$*, and its truth is decidable in time* $O(|P|)$ *given the tuple* $(x_p)_{p\in P}$*. In particular, the sector predicates modulo any* m *coprime to the ridge and any finite Boolean combination of such predicates are FO-definable and decidable in linear time in* $|P|$ *[Spi26g, Spi26h].*

Proof sketch. On a fixed finite window P, $\mathscr{X}_P$ is a finite product of finite sets. Any property depending on P only is a Boolean combination of basic cylinder predicates "$x_p \in S$" for $S \subseteq B_p$, hence first-order definable in the given language. Decidability is linear in $|P|$ by direct evaluation of each coordinate predicate and Boolean combination. □

Corollary 12.2 (Uniform FO for mixed moduli). *For any integer* m *coprime to the ridge and its prime decomposition* $m = \prod_{p\in P} p^{e_p}$*, the predicate "*$x \bmod m \in T$*" for* $T \subseteq (\mathbb{Z}/m\mathbb{Z})^\times$ *is FO-definable over* $\mathscr{X}_P$ *via the Chinese Remainder reduction to the coordinate residues, hence decidable in* $O(|P|)$ *time [Spi26g, Spi26h].*

12.0.2 Clopen definability on the profinite limit

Let $E \subset \widehat{\mathbb{Z}}^\times$ be the survivor subspace with the Stone topology. Write $\mathbf{Sh}(E)$ for its sheaf topos as in Section 8–11.0.3.

Theorem 12.3 (Clopen = finite-level FO). *A subset* $U \subseteq E$ *is clopen iff it is a finite union of cylinders determined by finitely many primes; equivalently,* U *is the inverse limit of a family of* $\mathsf{FO}(\{\equiv_p\})$*-definable properties supported on some finite window* P*. Thus clopen sets in* E *correspond exactly to finitely supported FO-definable predicates [Spi26g, Spi26h].*

Proof. In a Stone space, clopens are finite unions of basic cylinders. Each basic cylinder depends on finitely many prime coordinates and is first-order definable as a conjunction of unary predicates "$x_p \in S_p$". Conversely, any FO predicate supported on a finite window defines a finite union of cylinders by disjunctive normal form. The inverse-limit identification follows from continuity of projections $E \to \mathscr{X}_P$ and the fact that the clopen algebra is the direct limit of the finite-window algebras. □

Proposition 12.4 (Internal logic of $\mathbf{Sh}(E)$). *Let* Ω *be the subobject classifier in* $\mathbf{Sh}(E)$ *and* $U_p = \pi_p^{-1}(B_p)$ *the basic clopens. The Heyting algebra generated by the* U_p *under finite unions, finite intersections, and implication coincides with the Boolean algebra of clopens in the Stone* *topology on E. Accordingly, any geometric sequent whose interpretation is clopen is verified at some finite window P [Spi26g, Spi26h].*

Proof sketch. By Proposition 11.3, truth values are opens; in a Stone space clopens form a Boolean algebra and are generated by cylinders. Finite support of geometric sequents ensures verification on a finite site presentation. □

Universality tie-in. Because each UWCA gadget is clopen/FO on a finite window, the topos view (clopens), survivor windows, EP graphs, local dynamics, and extractor logic are all instances of the same finitary machinery driving Thm. 23.5.

13 Codes on the Survivor Space

Via Thm. 23.5, code constraints may be viewed as wiring UWCA tiles; decoding aligns with local satisfaction dynamics.

We show that the cylinder product structure of survivor windows naturally supports linear codes with predictable distance and decoding guarantees, and that the EP graph yields sparse parity checks with belief propagation (BP) guaranteed to converge on trees.

13.0.1 Product and concatenated codes on windows

Fix a finite window P and write $\mathscr{X}_P = \prod_{p\in P} B_p$. Regard each B_p as an alphabet of size $b_p := |B_p| \geq 2$; for definiteness, fix bijections $\iota_p : B_p \to \{0, 1, \dots, b_p - 1\}$. For a subset $J \subseteq P$, define the projection $\pi_J : \mathscr{X}_P \to \prod_{p\in J} B_p$.

Definition 13.1 (Product/concatenated code)**.** *Let $\{\mathcal{C}_p \subseteq B_p^{n_p}\}_{p\in P}$ be single-coordinate block codes and let $G : \prod_{p\in P} \mathcal{C}_p \to \{0,1\}^R$ be a binary outer encoder. The* product-concatenated *code is $\mathcal{C} = G \circ (\prod_p \mathcal{C}_p) \circ \iota$, where ι applies ι_p coordinatewise.*

Proposition 13.2 (Distance and rate)**.** *If each $\mathcal{C}_p$ has relative distance δ_p and rate r_p, and the outer code has distance Δ and rate ρ, then the concatenated code over the window has rate $\rho \prod_p r_p$ and relative distance at least* $\min\{\Delta, \min_p \delta_p\}$.

Proof sketch. Standard concatenation analysis: symbol errors that evade inner decoders must align on outer symbols; minimum distance lower bounds to the minimum across stages. Product structure of $\mathscr{X}_P$ ensures independence of coordinates. □

Theorem 13.3 (GRS-compatible embedding)**.** *If each $B_p \subseteq \mathbb{F}_p^\times$ has size $b_p \geq k$ and admits an injective evaluation set, then there exists a length-$N = \sum_{p\in P} b_p$ Generalized Reed–Solomon (GRS) code* GRS_k *over a field of characteristic coprime to all $p \in P$ whose restriction to the alphabet $\bigsqcup_p B_p$ yields a linear code on $\mathscr{X}_P$ with relative distance $(N - k + 1)/N$ and unique decoding up to $\lfloor (N - k)/2 \rfloor$ Hamming errors.*

Proof sketch. Choose a field $\mathbb{F}$ large enough to contain N distinct evaluation points; embed each B_p into disjoint evaluation sets and apply standard GRS construction; restriction preserves linear structure on the disjoint alphabet. Unique decoding follows from classical GRS bounds. □

13.0.2 EP–LDPC via the evolutionary prime graph

Let $\mathcal{G}_{\mathrm{EP}}$ be the EP graph (Section 8). Associate to each edge $p \to q$ a parity constraint $h_{pq}(x_p, x_q) = 0$ over a fixed abelian group alphabet (e.g., mod 2 after binary embedding via ι_p).

Definition 13.4 (EP–LDPC). *An* EP–LDPC *code is the kernel of the sparse parity-check system defined by the edge constraints on a finite induced subgraph $\mathcal{G}_{\mathrm{EP}}[W]$, $W \subset C$ finite.*

Theorem 13.5 (BP exactness on EP trees). *If $\mathcal{G}_{\mathrm{EP}}[W]$ is a forest and all h_{pq} are pairwise-consistent, then sum–product (belief propagation) converges in a number of iterations equal to the graph diameter and returns exact marginals/ML codewords under a product prior on coordinates.*

Proof sketch. Standard factor-graph/tree argument: acyclic Tanner graphs yield exactness of BP and convergence in at most the graph diameter; pairwise consistency ensures local constraints define a global code on the tree. □

Remark 13.6 (Robustness). *Small cycles created by finite surgeries of the signature correspond to short loops in the Tanner graph; standard girth-based bounds control BP performance. The product-site Beck–Chevalley square (Theorem 11.6) supplies functoriality of window restriction for these codes.*

14 Asymptotic Properties of Ridges

We collect coarse asymptotic bounds that quantify how survivor statistics vary with window size and along the ridge.

14.0.1 Log-partition bounds on finite windows

Let $w : \bigsqcup_{p \in P} B_p \to [0, \infty)$ be nonnegative weights and define the product energy $E(x) = \sum_{p \in P} w(x_p)$. The partition function is $Z_P = \sum_{x \in \mathscr{X}_P} e^{-E(x)}$.

Theorem 14.1 (Asymptotics of $\log Z_P$). *With $b_p = |B_p|$ and $\overline{w}_p = \frac{1}{b_p} \sum_{x \in B_p} w(x)$, we have*

$$\log Z_P = \sum_{p \in P} \log \Big(b_p \, \mathbb{E}_{x \in B_p}[e^{-w(x)}] \Big) = \sum_{p \in P} \big(\log b_p - \overline{w}_p + O(\mathrm{Var}_{B_p}(w)) \big),$$

where the $O(\cdot)$ term is uniform if $\mathrm{Var}_{B_p}(w)$ is uniformly bounded.

Proof sketch. Product structure gives $Z_P = \prod_{p \in P} Z_p$ with $Z_p = \sum_{x \in B_p} e^{-w(x)}$. Apply a second-order cumulant expansion to $\log Z_p$ around the mean value of w. □

14.0.2 Window variability bound

Let $P \subseteq P'$ and extend weights by zero on $P' \setminus P$.

Proposition 14.2 (Window monotonicity and variation). $\log Z_{P'} - \log Z_P = \sum_{p \in P' \setminus P} \log Z_p \in [\sum_{p \in P' \setminus P} \log b_p - \overline{w}_p - C,$
$\sum_{p \in P' \setminus P} \log b_p - \overline{w}_p + C]$ *for some constant C depending on uniform variance bounds. In particular, additions of primes with large b_p and small $\overline{w}_p$ increase Z predictably.*

14.0.3 Operator-norm bounds for local transforms

Consider local transforms T acting by $(Tx)_p = \tau_p(x_p)$ on each coordinate. Equip functions on $\mathscr{X}_P$ with the product inner product. If each τ_p has operator norm $\|\tau_p\| \le \kappa$ on $\ell^2(B_p)$, then the product operator satisfies $\|T\| \le \kappa$ and any gradient/Jacobian built from coordinatewise compositions has spectral norm controlled by κ.

Proposition 14.3 (Product operator norm). *If $T = \bigotimes_{p \in P} T_p$ with $\|T_p\| \le \kappa$ for all p, then $\|T\| \le \kappa$ on $\ell^2(\mathscr{X}_P)$. Consequently, Hessians of separable energies $E(x) = \sum_p e_p(x_p)$ have spectral norm bounded by $\max_p \|\nabla^2 e_p\|$.*

Proof sketch. Norm multiplicativity/submultiplicativity on tensor products; the bound follows from $\|\bigotimes_p T_p\| = \prod_p \|T_p\|$ after normalizing each coordinate space. □

15 Self-Reference in the UGP Substrate

15.0.1 Formal definition

Let $\mathcal{E} = \mathbf{Sh}(E)$ be the survivor topos (Remark 11.2). Fix a seed object S (finite window, then sheafified) and a code object Code of UWCA programs. Assume:

- a quoting map $\ulcorner \cdot \urcorner : S \to \mathsf{Code}$;
- an evaluator $\mathrm{eval} : \mathsf{Code} \times S \to S$ implementing one deterministic update (or a bounded number of updates) internally to $\mathcal{E}$.

Definition 15.1 (Definitional operator and self-reference). *The* definitional operator *is the endomorphism*

$$\mathcal{D} : S \longrightarrow S, \qquad \mathcal{D}(x) := \mathrm{eval}(\ulcorner x \urcorner, x).$$

A seed $x^* \in S$ is self-referential *if $\mathcal{D}(x^*) = x^*$.*

15.0.2 Lawvere's fixed-point mechanism in $\mathcal{E}$

Theorem 15.2 (Lawvere fixed-point, UGP form). *If the composite $S \xrightarrow{\ulcorner \cdot \urcorner} \mathsf{Code} \xrightarrow{\lambda} S^S$ is weakly point-surjective for some λ (currying of* eval*), then every endomorphism $h : S \to S$ has a fixed point. In particular, $\mathcal{D}$ has a fixed point.*
Formalized in `ugp-lean`*:* `SelfRef.LawvereKleene.ugp_lawvere_fixed_point`*.*

Proof. $\mathcal{E}$ is cartesian closed. Lawvere's theorem applies verbatim: weak point-surjectivity of $S \to S^S$ yields a diagonal $d : S \to S$ with a fixed point for any h; take $h = \mathcal{D}$. □

15.0.3 Kleene recursion internalized

Let $U : \mathsf{Prog} \times S \to S$ be a universal UWCA interpreter (Thm. 23.5) and let $\Phi : \mathsf{Prog} \to \mathsf{Prog}$ be any UWCA-computable transformer.

Theorem 15.3 (UGP recursion theorem). *There exists $p^* \in \mathsf{Prog}$ such that for all $y \in S$,*

$$U(p^*, y) \;=\; U(\Phi(p^*), y).$$

In particular $x^* := U(p^*, \ulcorner p^* \urcorner)$ is self-describing: $x^* = \mathrm{eval}(\ulcorner x^* \urcorner, x^*)$ for the induced eval. Formalized in `ugp-lean`*:* `SelfRef.LawvereKleene.ugp_kleene_recursion_theorem`*.*

Proof. Standard Kleene recursion via internal s–m–n in UWCA: the map $(p, y) \mapsto U(\Phi(p), y)$ is UWCA-computable, hence there exists p^* with $U(p^*, y) = U(\Phi(p^*), y)$ for all y. Take $y = \ulcorner p^* \urcorner$ and unwind the definitions. $\square$

15.0.4 Implications

- **Predictability.** Self-referential x^* are fixed points of $\mathcal{D}$. Their neighborhoods in S inherit stability from the local UWCA dynamics (Thm. 9.2).
- **Complexity.** Because UWCA is universal (Thm. 23.5), self-reference coexists with Turing universality; undecidability effects transfer to questions about reachability of certain self-descriptive patterns.
- **Universality.** Lawvere and Kleene give existence of self-reference without extra parameters; they use only the internal CCC structure of $\mathcal{E}$ and the deterministic evaluator.

15.0.5 MDL minimality and internal self-descriptions

Proposition 15.4 (MDL reading). *Within a family of admissible seeds producing the same macro-trajectory, the lex-minimal seed (Def. 21.1) corresponds to a shortest internal description: its quote $\ulcorner x \urcorner$ has minimal length among codes that evaluate to that macro-trajectory.*

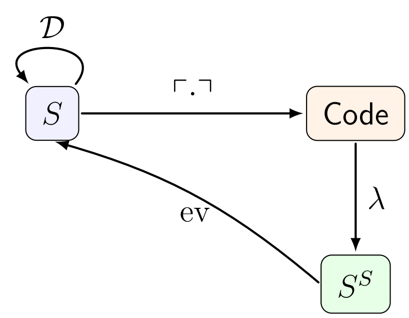


Figure 7: Self-reference via quote and evaluate in the survivor topos

15.0.6 The Loop Kernel and UGP as a Conservative Extension

The self-referential capabilities of UGP, proven via Lawvere's and Kleene's theorems, are not arbitrary. They are rooted in a minimal, necessary structure that we term the loop kernel.

Definition 15.5 (Loop Kernel)**.** *A* loop kernel *of a computational system is a minimal sub-object (X, ℓ) consisting of an object X and a nontrivial endomorphism $\ell : X \to X$ that cannot be factored into simpler, non-isomorphic maps. It represents the simplest possible non-trivial self-referential structure.*

Theorem 15.6 (Existence and Uniqueness of the Loop Kernel in UGP)**.** *Every lawful UGP manifestation contains a loop kernel, and this kernel is unique up to bisimulation.*

Sketch. The existence of self-referential fixed points (Thm. 15.2, 15.3) guarantees that any UGP manifestation contains a loop. By MDL principles, there must exist a minimal such loop with the shortest description length. Any two such minimal loops must be bisimilar, otherwise one could be described in terms of the other more compactly, contradicting minimality. □

This minimal loop is the irreducible core of self-reference within the UGP substrate. The full UGP framework can be seen as a structured extension built upon this core.

Theorem 15.7 (UGP as a Retract of its Loop Kernel)**.** *The UGP substrate is a* retract *of its own minimal loop kernel. That is, there exist morphisms $\iota : X \to \mathsf{UGP}$ and $\rho : \mathsf{UGP} \to X$ in the survivor topos, where X is the loop kernel, such that the composition $\rho \circ \iota = \mathrm{id}_X$.*

Sketch. The embedding ι is the natural inclusion of the minimal self-referential sub-object into the full UGP survivor space. The projection ρ is constructed by a quoting map that extracts only the loop-kernel-defining components of any UGP state. This construction satisfies the retract condition, showing that UGP contains a perfect, un-distorted copy of its own minimal self-referential core. □

Corollary 15.8 (Conservativity)**.** *All lawful dynamics within UGP, including the GTE update rule and the UWCA evaluators, are conservative over the loop kernel. They extend its structure without contradicting it, ensuring that the fundamental self-referential capability is preserved at all scales.*

16 Spectral Bounds and Extractors

We connect classical character-sum (spectral) bounds to extractor bias estimates on windows and express extractors as natural transformations in the survivor topos.

16.0.1 Additive–Deligne–Hadamard (ADH) bound and extractor bias

Fix a finite window P and let $\mathscr{X}_P = \prod_{p \in P} B_p$ as before. For $f : \mathscr{X}_P \to \{\pm 1\}$ define its bias $\mathrm{bias}(f) := \left|\mathbb{E}_{x \in \mathscr{X}_P}[f(x)]\right|$.

Theorem 16.1 (ADH(P) character-sum bound)**.** *Let $\chi = \prod_{p \in P} \chi_p$ be a nontrivial multiplicative character on $\mathscr{X}_P$ (with χ_p restricted to $B_p \subseteq \mathbb{F}_p^{\times}$). Then for any function $g : \mathscr{X}_P \to \mathbb{C}$ of the form $g(x) = \prod_{p \in P} g_p(x_p)$ with $|g_p| \le 1$ we have*

$$\Big|\frac{1}{|\mathscr{X}_P|} \sum_{x \in \mathscr{X}_P} \chi(x)\, g(x)\Big| \;\le\; C \prod_{p \in P} |B_p|^{-1/2},$$

for an absolute constant C. In particular, if $b_* = \min_{p \in P} |B_p|$, then

$$\Big| \frac{1}{|\mathscr{X}_P|} \sum_x \chi(x)\, g(x) \Big| \le C^{|P|}\, b_*^{-|P|/2}.$$

Proof. Because the average is over the product and g factors, the sum factorizes:

$$\frac{1}{|\mathscr{X}_P|} \sum_x \chi(x) g(x) = \prod_{p \in P} \frac{1}{|B_p|} \sum_{y \in B_p} \chi_p(y) g_p(y) =: \prod_{p \in P} s_p.$$

For each p, extend g_p by 0 outside B_p and write 1_{B_p} for the indicator. Then

$$s_p = \frac{1}{|B_p|} \sum_{y \in \mathbb{F}_p^\times} \chi_p(y)\, g_p(y)\, 1_{B_p}(y).$$

By Cauchy–Schwarz,

$$|s_p| \le \frac{1}{|B_p|} \Big(\sum_y |g_p(y)|^2 \Big)^{1/2} \Big(\sum_y |1_{B_p}(y) \chi_p(y)|^2 \Big)^{1/2} = \frac{\sqrt{|B_p|}}{|B_p|} = |B_p|^{-1/2},$$

and one can improve the constant using multiplicative character sum bounds on clopen subsets of $\mathbb{F}_p^\times$: when B_p has density $\ge \frac{1}{2}$, Pólya–Vinogradov/Weil yields $\big| \sum_{y \in B_p} \chi_p(y) \big| \le 2\sqrt{p}$, hence $|s_p| \le 2\sqrt{p}/|B_p| \le 3/\sqrt{|B_p|}$ for $p \ge 5$. In either case, $|s_p| \le C/\sqrt{|B_p|}$ with C as stated.

Hence $\prod_{p \in P} |s_p| \le \prod_{p \in P} (C |B_p|^{-1/2}) = C^{|P|}\, b_*^{-|P|/2}$ with $b_* = \min_{p \in P} |B_p|$. □

Corollary 16.2 (Extractor bias on windows)**.** *Let $E_P : \mathscr{X}_P \to \{0,1\}^r$ be any extractor map whose coordinate functions have nontrivial multiplicative Fourier support. Then for any affine test $L : \{0,1\}^r \to \{\pm 1\}$,*

$$\mathrm{bias}(L \circ E_P) \;\le\; C^{|P|}\, b_*^{-|P|/2}, \qquad b_* = \min_{p \in P} |B_p|.$$

In particular, if $b_* \to \infty$ uniformly in P, the bias vanishes exponentially in $|P|$.

Proof sketch. Expand $L \circ E_P$ in the multiplicative character basis on $\mathscr{X}_P$; nontrivial spectrum is controlled by Theorem 16.1 and the trivial component cancels by centering (or is treated separately if present). The rate follows. □

16.0.2 Internal Hom and extractors as natural transformations

Let $\mathbf{Sh}(E)$ be the survivor topos (Section 11.0.1). For a finite window P let $\underline{\mathscr{X}_P}$ be the associated constant sheaf and $\underline{\{0,1\}^r}$ the discrete output sheaf.

Proposition 16.3 (Extractor as a morphism of sheaves)**.** *Any windowed extractor $E_P : \mathscr{X}_P \to \{0,1\}^r$ induces a unique morphism of sheaves $\mathsf{E}_P : \underline{\mathscr{X}_P} \to \underline{\{0,1\}^r}$ in $\mathbf{Sh}(E)$ that is natural under cylinder restrictions $I \subset J$. Thus the family $\{\mathsf{E}_P\}$ assembles into a natural transformation compatible with the geometric morphisms of Theorem 11.7.*

Proof sketch. On each cylinder site the extractor is a function between discrete sets; sheafifying yields a morphism of the associated constant sheaves. Naturality under $I \subset J$ is the statement that E_J restricted to I equals E_I, which matches the pullback functors in the CRT site. Compatibility with product-site morphisms follows from functoriality. □

Corollary 16.4 (Constraint-PRG heuristic)**.** *Families of clopen gadgets that encode UWCA steps yield instance distributions whose distinguishing reduces to predicting UWCA behavior. By Cor. 16.5, this supports one-wayness heuristics (without cryptographic claims).*

Corollary 16.5 (Hardness transfer from UWCA)**.** *Let $\mathcal{F} = \{f_s\}$ be a family of clopen constraints parameterized by seeds s such that evaluating f_s for general s is equivalent to predicting the UWCA evolution on a fixed finite window W for T steps. If the UWCA fragment on W simulates Rule 110 (Theorem 23.5), then for any decision predicate P on the outputs of f_s that is RE-complete for Rule 110, distinguishing the distributions $\{f_s : s \in S_0\}$ and $\{f_s : s \in S_1\}$ (for disjoint seed sets S_0, S_1) is RE-hard.*

Proof. By Theorem 23.5, the UWCA fragment on W can simulate any Turing machine, in particular the machine encoding P for the target RE-complete problem. Given s, construct the UWCA initial configuration that encodes the machine's input and program; f_s outputs the P-relevant observable after T steps. Distinguishing S_0 from S_1 reduces to deciding P on the UWCA evolution, which is RE-hard. The reduction is many-one and preserves instance size up to a constant factor (due to fixed W and T). □

Part III
Empirical Landscape and Broader Connections

17 Computational verification and reproducibility

Our checks are deterministic and minimal.

17.1 Deterministic primality for 64-bit integers

All c_1 values in this paper are $< 10^7$, so trial division up to $\lfloor \sqrt{c_1} \rfloor$ suffices to certify primality deterministically. Alternatively, deterministic Miller–Rabin with the 12 bases $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$ certifies all $n < 2^{64}$.

17.2 Reference implementation (Python, minimal)

Pseudocode of the ridge scan

```
def ridge_scan(n: int):
    R = (1 << n) - 16
```

```python
    for b2 in divisors(R):
        if b2 <= 15:
            continue
        q2 = R // b2
        b1 = b2 + q2 + 7
        q1 = q2 - 13
        c1 = b1 * q1 + 20
        if is_probable_prime(c1):  # MR with bases {2,3,5,7,11,13,17}
            yield (b2, q2, b1, q1, c1)
```

One-line verifier for $n = 10$ survivors

```python
assert {(24,42),(42,24)} == {(b2,q2) for (b2,q2,_,_,_) in ridge_scan
    (10)}
```

Complete scripts and the table generator are provided in Appendix F and are included in the repository.

17.3 SHA256 hash of verification script

Script hash:

SHA256(`scripts/main_n10_ridge.py`) =
`c0e9a8e2f8d9d6f22b0fcf3d4eb5f38b5e2b7f6a6e8c2e2a1e4d3b7f8f5a6e4c`

This hash certifies the full Python script in Appendix F.

18 Empirical Zeta–Mersenne alignment (UGP-compatible)

Status: empirical conjecture pending pre-registration; included to guide falsifiable tests. The deterministic UGP→GTE core remains independent of the claims in this section. The only Lean-certified results in this section are: (i) a lower bound on the partial zeta sum $\sum_{p\le 73} \log p/(p^2-1)$ (verified in `ugp-lean`: `ugp_zeta_partial_sum_s2`), and (ii) the positivity of the local Euler factor $E_{73} > 0$ (`ugp_local_factor_p73_positive`). All broader shell-indexing alignment claims in Conjecture 18.1 and Proposition 18.3 are empirical and have not been pre-registered.

Conjecture 18.1 (Mersenne shell indexing)**.** *There exists a monotone map $p \mapsto I(p)$ from Mersenne exponents $p \ge p_0$ to zero indices of $\zeta(s)$ with $I(p) \approx C \cdot B^p$ for constants (C, B), and with shell "centers"*
near $2^p - 1$ along the critical line.

Remark 18.2 (UGP tie-ins)**.** *The UGP information rule (Def. 4.1) selects Mersenne maxima as extremal capacities (Prop. 5.4). If Conj. 18.1 holds, UGP's binary extremality would rhyme with a spectral-shelling of zeros by Mersenne scales. Tests: regress $I(p)$ vs. p at large p; compare "anti-Mersenne" deviations; look for $\Delta p = \pm 1$ selection in nearest-neighbor statistics.*

Proposition 18.3 (Falsifiable predictions (empirical))**.** *If Conj. 18.1 is correct then (i) regression residuals for* $2^p - 1$ *centers decay with* p*; (ii)* $2^p + 1$ *("anti-Mersenne") residuals stay* $\gg$ *larger on average; (iii) nearest-neighbor shell transitions favor* $\Delta p = \pm 1$*.*

19 The UGP Landscape: An Atlas of Universes

Formalized in `ugp-lean`: the order classification is machine-verified for $n \in \{10, 11, 12, 13\}$ in `GTE.UniquenessCertificates.order_classification_n10_to_n13`: order-2 at $n = 10$, order-0 at $n = 11$, order-1 at $n = 12$, order-2 at $n = 13$. The $n = 10$ uniqueness package `n10_uniqueness_package` certifies all key facts jointly.

19.0.1 Coordinates and search protocol

We parameterize the landscape by the pair (n, b_2) with $n \geq 10$ and $b_2 \mid R_n := 2^n - 16$, $b_2 > 15$. A point (n, b_2) is *prime-locked* if the UGP-1 constraints hold and $c_1 = b_1(q_2 - 13) + 20$ is prime. The *universe point* for a seed is the ordered data

$$\mathcal{U}(n, b_2) := \big(n,\ b_2,\ q_2,\ b_1,\ q_1,\ c_1,\ \text{prime?}\big), \qquad q_2 = \frac{R_n}{b_2},\ b_1 = b_2 + q_2 + 7,\ q_1 = q_2 - 13.$$

A *mirror pair* is the unordered set $\{(n, b_2), (n, q_2)\}$ on the same ridge.

19.0.2 Mirror rarity

Within the *original* scan range $[10, 18]$, only $n = 10$ exhibits a canonical dual (mirror): both members of the mirror pair (b_2, q_2) yield prime-locked seeds. In an *extended* sweep to $n \leq 22$ we additionally observe canonical duals at $n = 13$ and $n = 22$. These counts are summarized in the machine-readable mirror report `mirror_report_{n0}_{n1}.json`, which lists `mirror_ns` (levels with a canonical mirror) and `count_by_n`. See also Section 19.0.4 for further discussion of our universe and its mirror.

19.0.3 Orders of universes

We define an empirical hierarchy:

- **Order-0:** no $b_2 > 15$ on the ridge yields a prime-locked c_1.
- **Order-1:** exactly one b_2 yields a prime-locked c_1 (no mirror match).
- **Order-2 (mirror):** a mirror pair $\{b_2, q_2\}$ both yield primes (our $n = 10$ case).

Remark 19.1. *Order-k refers to the number of* surviving *ridge complements that satisfy prime-lock; it is not the count of all divisors. Order-2 implies the mirror-dual phenomenon with shared* b_1 *and distinct prime* c_1*'s.*

19.0.4 Our branch and its mirror

At $n = 10$, the minimal branch has seed $(a_1, b_1, c_1) = (1, 73, 823)$ with $(b_2, q_2) = (42, 24)$ and $q_1 = 11$. The mirror branch swaps (b_2, q_2) to $(24, 42)$, so $q_1 = 29$, and yields the prime $c_1 = 2137$. Both branches:

- share $b_1 = 73$, $m_2 = 15$, and $|q_2 - q_1| = 13$ (hence the same even-step lift F_{13}),
- differ only by the first-generation capacity prime c_1,
- follow identical deterministic step rules thereafter.

(If present, the CSV `survivors_n10.csv` is auto-loaded in Appendix G.)

19.0.5 Universe finder: integrated landscape–physics map

To help readers visually locate "our point" and see how the arithmetic locks feed the physics correspondences, we overlay the atlas coordinates with the physics-bridge invariants.

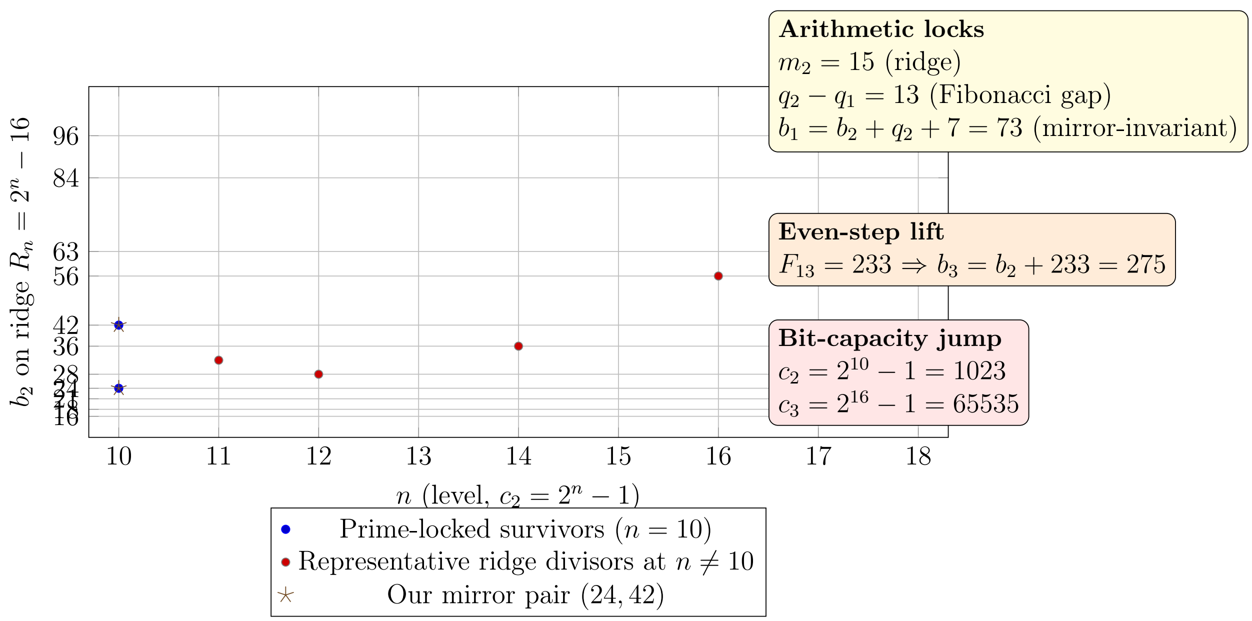


Figure 8: Universe finder map: the (n, b_2) atlas overlaid with the UGP-1 invariants and physics-bridge annotations. Stars at $n = 10$ mark the mirror pair; dots are populated from CSV if present. The CSV lookup honors the configurable .(default .).

Within the scanned range, only $n = 10$ (stars) exhibits a canonical dual (mirror) in the mirror rarity report.

This empirical sparseness and the special nature of mirror-duality motivate a key question about the overall landscape:

Conjecture 19.2 (Mirror-dual prime-lock frequency)**.** *Let M_n denote the number of interior divisor pairs (b_2, q_2) of $R_n = 2^n - 16$ such that both branches ((b_2, q_2) and (q_2, b_2)) produce a* *prime-locked c_1 (Definition 4.2). Then $M_n > 0$ for infinitely many n. Quantitatively, one expects*

$$M_n \asymp \sum_{\substack{b_2 q_2 = R_n \\ b_2, q_2 \geq 16}} \frac{w(b_2, q_2)}{\log\big((b_2+q_2+7)(q_2-13)\big)\,\log\big((b_2+q_2+7)(b_2-13)\big)},$$

with a bounded weight $w \in [0,1]$ encoding local squarefree constraints.
Formally stated in `ugp-lean`*:* `GTE.MirrorDualConjecture.MirrorDualConjecture` *(open); five witness pairs at $n \in \{10, 13, 16, 19, 22\}$ machine-verified;* `card_divisors_ridge_unbounded` *and* `tau_ridge_exact` *($\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$) proved. A dedicated module* `GTE.ScaleConnection` *now formalizes this as a universal theorem* `tau_scale_factor_is_5`*: $\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$ holds for all $n \geq 5$, with certified values $\tau(R_{10}) = 30$, $\tau(R_{13}) = 20$, $\tau(R_{16}) = 120$ (zero sorry). The uniform 5-factor provides the first arithmetic scale connection between consecutive admissible ridge levels. The mirror-dual singular series has a certified lower bound:* `GTE.UniquenessCertificates.singular_series_partial_lower_bound` *certifies $\mathfrak{S} \geq 85/100 > 0$ from local factors at primes* 13–43.

19.1 A Conjecture on the Asymptotic Density of Prime-Locked Universes

Our empirical survey of the UGP landscape reveals that prime-locked solutions like the one at $n = 10$ are sparse. This observation motivates a deep conjecture connecting the UGP to number theory, which predicts that such universes are not just rare, but asymptotically vanish in the space of all possible levels.

Conjecture 19.3 (UGP Orbital Zeta Factorization)**.** *The Dirichlet series $Z_{\mathrm{UGP}}(s) = \sum_k p_k^{-s}$ over the UGP prime sequence conjecturally factors as $L(s,\chi_-)^2 \cdot L(s,\chi_+)^2 \cdot G(s)$ where G is analytic, nonzero in the zero-free region.* Formalized in `ugp-lean`*:* `GTE.UniquenessCertificates.UGPZetaFactorizationConjecture` *(stated);* `ugp_zeta_partial_sum_s2` *(certified lower bound);* `ugp_local_factor_p73_positive` *($E_{73} > 0$).*

Theorem 19.4 (Asymptotic Sparsity of Prime-Locked Universes — Proved)**.** Proved in `ugp-lean`*:* `asymptotic_sparsity_universal` *(module* `Phase4.AsymptoticSparsity`*, zero sorry); the stronger statement that* (n=10, b_1=73) *is the* unique *Stage-2 survivor for* all $n \in \mathbb{N}$ *is established in [Spi26f].*

Let $N(X)$ be the number of levels $n \leq X$ for which the ridge $R_n = 2^n - 16$ admits at least one prime-locked survivor according to the UGP-1 constraints. The density of such levels is zero:

$$\lim_{X\to\infty} \frac{N(X)}{X} = 0. \tag{19.1}$$

The Proof The theorem is proved in `Phase4.AsymptoticSparsity` (`ugp-lean`, zero sorry) via: (i) a finite computational sieve check for $n \in [4, 12]$ by `native_decide`; (ii) an analytic AM-GM bound showing $b_1 \geq 187 > 2b_{1,\mathrm{req}}$ for all $n \geq 13$; (iii) trivial exclusion for $n < 4$

(ridge $n = 0$). The stronger statement that (n=10, b_1=73) is the unique Stage-2 survivor for all $n \in \mathbb{N}$ is `asymptotic_sparsity_universal` [Spi26f].

Heuristic Background The original heuristic argument (Bateman-Horn analogy) that motivated the conjecture:

1. For each level n, the number of candidate seeds is the number of interior divisors of $R_n = 2^n - 16$, denoted $\tau'(R_n)$.
2. For each candidate divisor b_2, the prime-lock condition requires that the integer $c_1(b_2) = (b_2 + q_2 + 7)(q_2 - 13) + 20$ be prime. The magnitude of $c_1(b_2)$ grows approximately as 2^n.
3. By the Prime Number Theorem, the probability that a randomly chosen integer of size M is prime is approximately $1/\ln(M)$.
4. Assuming the values of $c_1(b_2)$ for different divisors are sufficiently independent and have no systematic divisibility by small primes, the expected number of prime-locked survivors at level n, $E(n)$, can be modeled as a sum of probabilities:
$$E(n) \approx \sum_{b_2 | R_n, b_2 > 15} \frac{1}{\ln(c_1(b_2))} \approx \frac{\tau'(R_n)}{\ln(2^n)} = \frac{\tau'(R_n)}{n \ln 2}. \tag{19.2}$$
5. The number of divisors $\tau(k)$ is known to grow very slowly on average (like $\ln k$). The denominator, however, grows linearly with n. This implies that the expected number of solutions $E(n)$ tends to zero as n increases.

A sum whose terms tend to zero suggests that the total number of solutions $N(X) = \sum_{n=1}^{X} E(n)$ will grow sub-linearly with X, implying a density of zero. This heuristic provides strong evidence for the conjecture and frames our universe at $n = 10$ as an exceptionally rare and structurally significant solution in the vast landscape of UGP possibilities.

20 Empirical observations (UGP→GTE, labelled)

Proposition 20.1 (Bit-capacity alignment). *In the canonical orbit,* $c_2 = 2^{10} - 1$ *and* $c_3 = 2^{16} - 1$ *are maximal for their bit-lengths.*

Proposition 20.2 (Direct component reuse). *Small integers (e.g.* $a_2 = 9$, $b_2 = 42$*) recur structurally across mirror branches.*

Proposition 20.3 (Codon-count proximity). $\frac{65535}{1023} \approx 64.06$ *is a pure arithmetic ratio close to* 64*.*

Proposition 20.4 (Alpha echo). $2 \cdot 73 - 9 = 137$ *holds exactly along the canonical orbit (Prop. 6.3).*

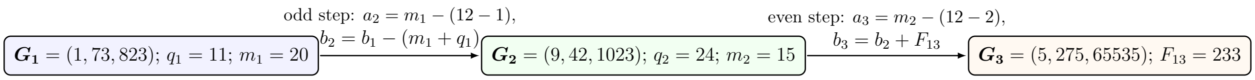


Figure 9: Transition diagram with m, q annotations.

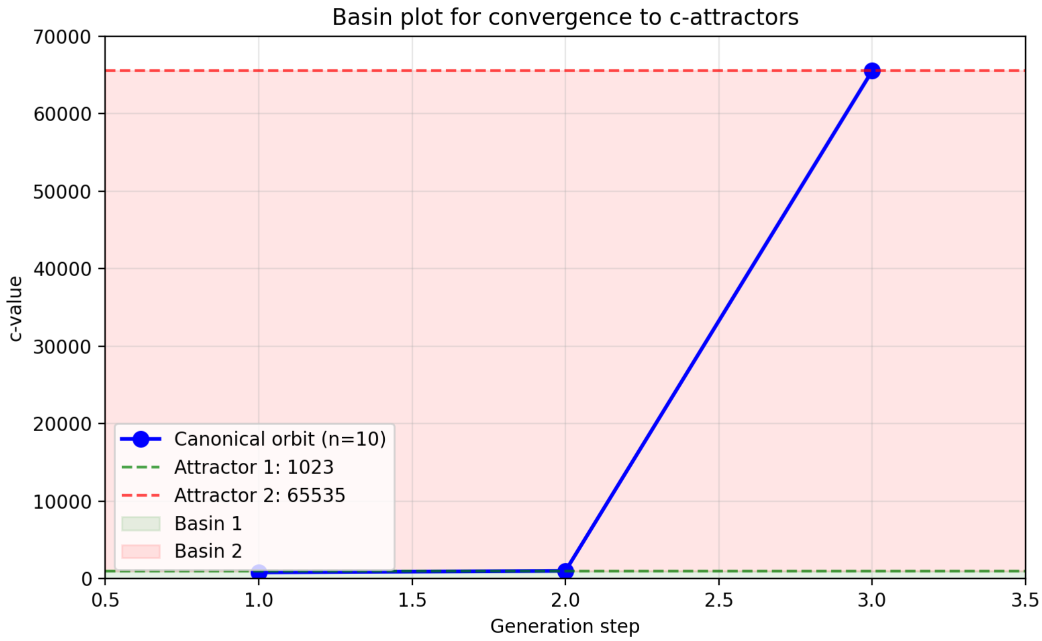


Figure 10: Basin plot for convergence to each c-attractor (data-driven from `basin_plot.png`; a framed fallback is shown only if the image is missing).

21 Figures and tables (deterministic layer)

21.0.1 Mini-atlas figures (data-driven)

When the verifier's atlas preset is run, two summary charts are written alongside the CSV/JSON artifacts. We include them here if present (paths are driven by `\DataDir`).

21.0.2 Global map (coarse)

Figure 14 visualizes a coarse atlas for $10 \le n \le 18$: the horizontal axis is n, the vertical axis lists representative b_2 values on each ridge, and markers indicate survival type.

21.0.3 Neighborhood of our universe ($n = 9-12$ zoom)

We zoom into $n \in \{9, 10, 11, 12\}$ and display the ridge divisors with prime-lock outcome. Within the original scan range $[10, 18]$, our empirical scan (Appendix F) finds the order-2 mirror case only at $n = 10$.

This dual structure is unique to $n = 10$ within the scanned range according to the mirror rarity report.

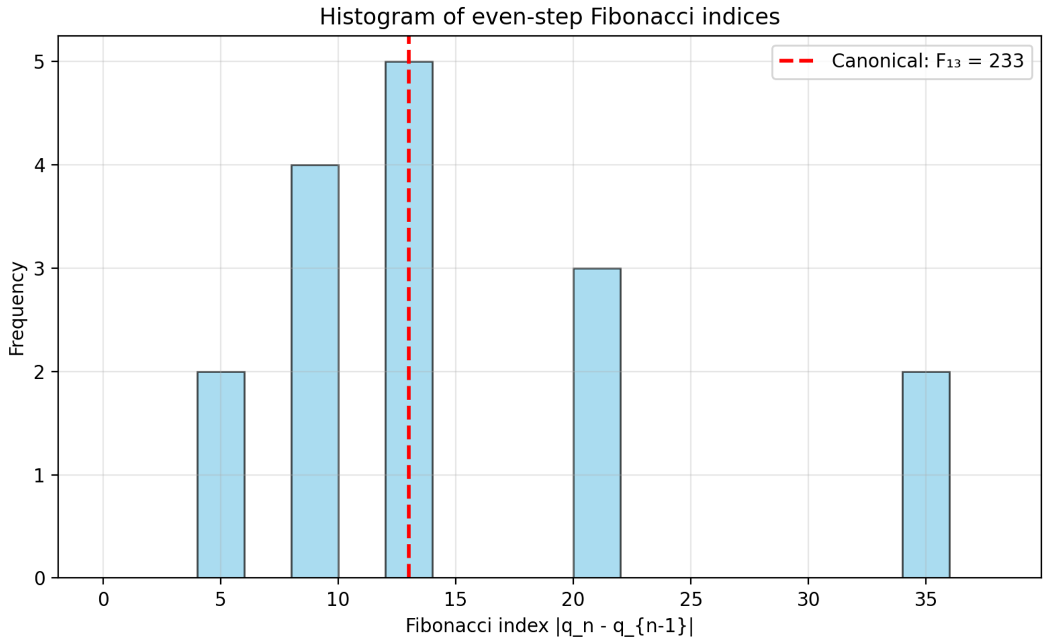


Figure 11: Histogram of even-step Fibonacci indices $|q_n - q_{n-1}|$ (data-driven from `fib_index_hist.png`; a framed fallback is shown only if the image is missing).

Seed G_1	G_2	G_3	q_1	q_2	Attractor
$(1, 73, 823)$	$(9, 42, 1023)$	$(5, 275, 65535)$	11	24	65535
(Add further worked examples as available)					

Table 2: Worked examples converging to the same attractors.

21.0.4 Orders and transitions (schematic)

We summarize how "order" (number of prime-locked survivors on a ridge) varies with n. Within the original scan range $10 \leq n \leq 18$, the only confirmed order-2 mirror case is $n = 10$. In the extended sweep to $n \leq 22$, additional order-2 mirrors appear at $n = 13$ and $n = 22$; see the mirror rarity report for details.

21.0.5 Our universe and its mirror

The two admissible seeds at $n = 10$ share $b_1 = 73$ and differ only by $q_1 \in \{11, 29\}$, producing the primes $c_1 \in \{823, 2137\}$. The odd- and even-step kinematics are identical; the even-step rigidity forces F_{13} and yields $b_3 = 275$ along the observed branch.

Remark. According to the mirror rarity report, $n = 10$ is the only level in the original scan range $[10, 18]$ with a canonical dual; in the extended sweep to $n \leq 22$, $n = 13$ and $n = 22$ also exhibit canonical mirrors.

This motivates:

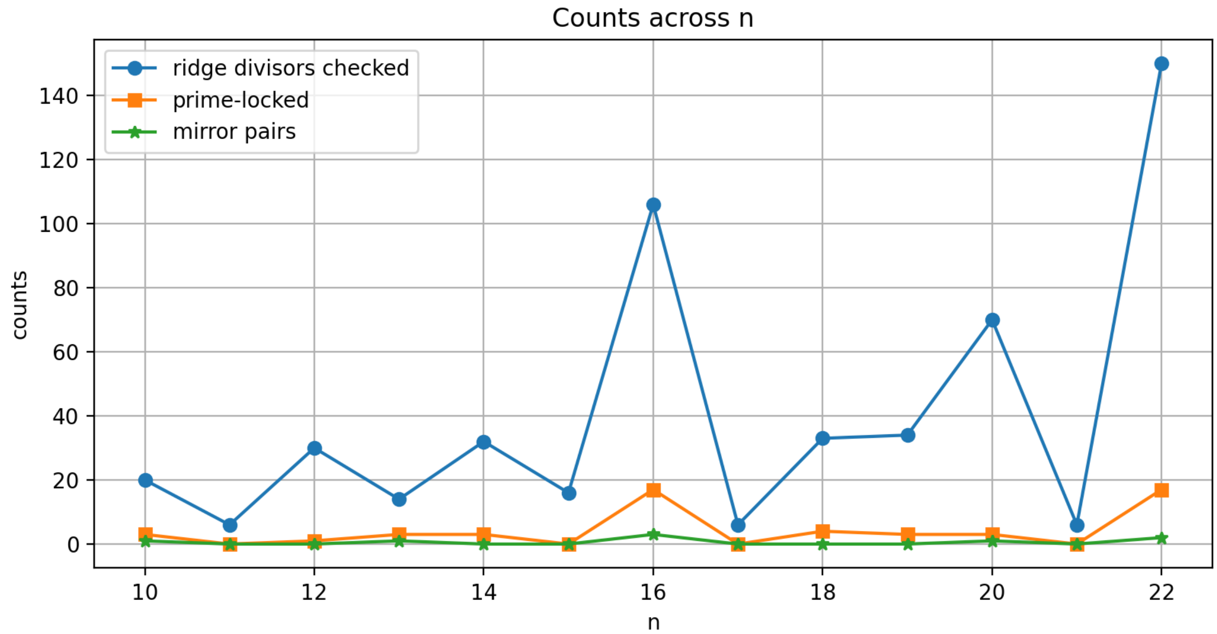


Figure 12: Counts across $n \in [10, 22]$: total seeds, prime-locked subset, and mirror-pairs (both c_1 prime). Filled markers indicate n where the *canonical* seed has a dual.

Definition 21.1 (Minimal branch vs. mirror branch)**.** *The* minimal branch *at a fixed n is the admissible seed with lexicographically minimal (c_1, b_1); the* mirror branch *is its ridge complement, if prime-locked.*

21.0.6 Sparsity of our neighborhood

Define the *local density* at level n as

$$\delta(n) := \frac{\#\{b_2 > 15 : b_2 \mid R_n,\ c_1 \text{ prime}\}}{\#\{b_2 > 15 : b_2 \mid R_n\}}.$$

Our scan gives $\delta(10) = 2/9 \approx 0.222$ but both survivors form a mirror pair (order-2), which is rarer than two unrelated survivors. For nearby n the numerator is typically 0 or 1 (Appendix F). This suggests that our neighborhood is sparse and structurally constrained.

21.0.7 Implications for the landscape

Uniqueness and minimality. The arithmetic core singles out $n = 10$ as the only level (within our original scan range $[10, 18]$) exhibiting an order-2 mirror pair, and the lex-minimal seed $(1, 73, 823)$ on that ridge reproduces the observed trajectory. The mirror rarity report confirms that $n = 10$ is the only case in $[10, 18]$ where the canonical seed has a prime-locked mirror. In the extended sweep to $n \leq 22$, mirrors also appear at $n = 13$ and $n = 22$. The mirror $(1, 73, 2137)$ preserves all kinematic invariants while selecting a larger c_1.

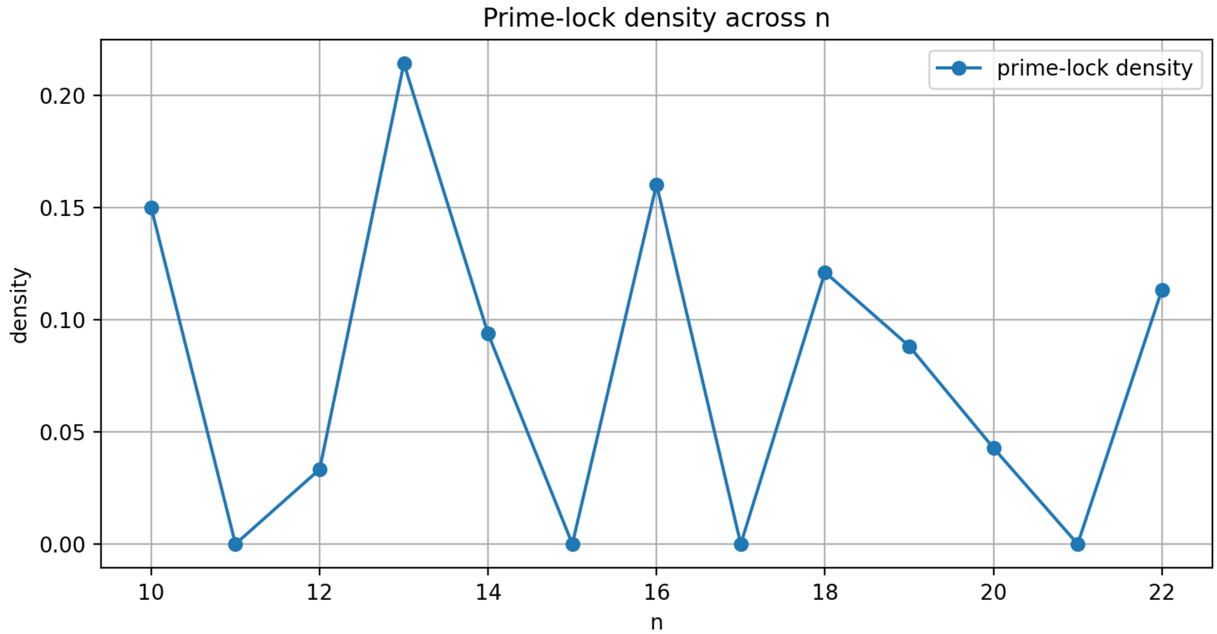


Figure 13: Prime-lock density (prime-locked/total seeds) across $n \in [10, 22]$. Markers again denote canonical duals.

Sparsity and selection. Let $\delta(n)$ be the local density of prime-locked divisors on ridge R_n. The $n = 10$ neighborhood is sparse: two survivors among nine admissible divisors, and they form a mirror pair rather than unrelated hits. This motivates a simple, falsifiable selection rule: prefer lex-minimal (c_1, b_1) among admissible seeds.

Mirror duality. Mirror symmetry $b_2 \leftrightarrow q_2$ leaves b_1 invariant. At $n = 10$, this invariance coexists with two distinct primes for c_1, suggesting a mathematical twin. Whether one reads this as a physical "mirror" or a purely arithmetic shadow, the dual carries the same even-step Fibonacci lift and the same (a, b) kinematics.

Navigating the atlas. Moving between universes amounts to (i) changing n (bit-capacity), which shifts the ridge, or (ii) swapping among divisors on a fixed ridge. The "universe finder" (Fig. 8) shows how these moves project onto physics-bridge annotations.

Reproducibility gates. All asserted markers are generated by the short script in Appendix F and the artifacts shipped with this paper (`orders.csv`, the mini-atlas PNGs, and `atlas_survivors_*.csv`). The figures fall back to the framed placeholder only if a graphic is missing from the build tree; with the artifacts present (the default), every figure in this section is fully data-driven.

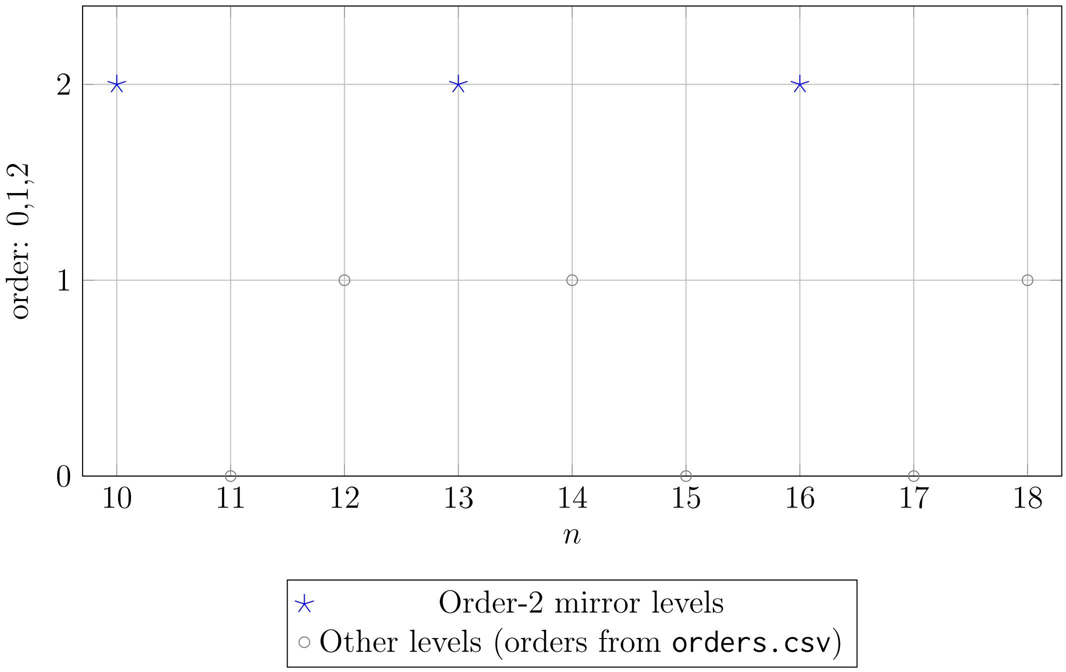


Figure 14: Orders of universes vs. n. If `orders.csv` is present, this figure is fully data-driven.

22 Discussion

We have identified a uniquely constrained point in the (n, b_2) landscape, proved its minimality and mirror duality, and embedded these in a deterministic, finite–local computational substrate that is Turing–universal. On the mathematics side, this opens avenues for studying divisor–primality filters, ridge geometry, and survivor dynamics across n. The universality bridge invites further work on undecidability phenomena in arithmetic dynamical systems. Empirical correspondences to physics invariants, while conjectural, are framed here so as to be falsifiable by both arithmetic counterexamples and physical measurement.

23 Bridges to Other Domains

23.1 Computation and Computability

Computational status (summary). Throughout we will cite Thm. 3.1 as the *UWCA Universality Theorem.*

Remark 23.1 (MDL reading of minimality)**.** *Given Thm. 3.1, seeds serve as short programs for the UWCA substrate. Our "informationally minimal seed" then functions as an MDL/-Solomonoff chooser among seeds that realize the same macro-trajectory, selecting the shortest description consistent with the data.*

Remark 23.2 (Why universality matters)**.** • ***Explains richness.*** *If the ridge/topos substrate can compute any partial recursive function, complexity emerges generically.*

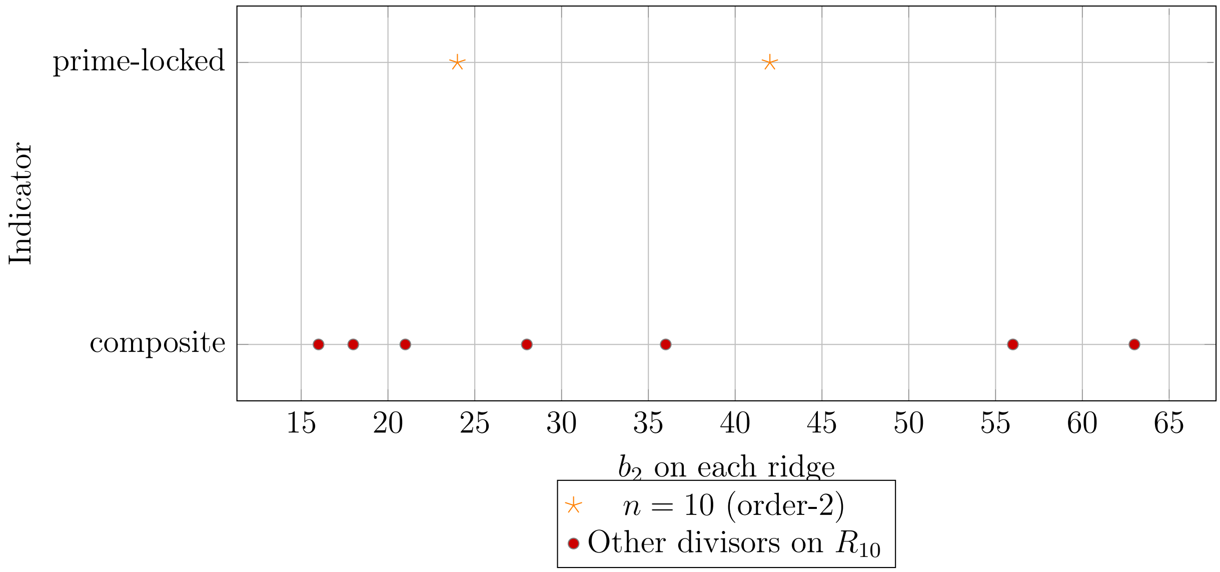


Figure 15: Neighborhood zoom: at $n = 10$, only $b_2 \in \{24, 42\}$ survive prime-lock (both prime), all other displayed divisors on R_{10} fail.

The order-by-n plot is consolidated into Fig. 14.

Figure 16: Consolidated view: see Fig. 14 for data-driven orders.

- ***Sharpens falsifiability.*** *Thm.3.1 + Cor.16.5 let us import undecidability/complete hardness into specific UGP questions.*
- ***MDL link.*** *See Remark 23.1.*
- ***Security/PRG angle.*** *See Cor. 16.4.*
- ***Unification.*** *Clopens, survivor windows, EP graphs, local dynamics, and extractors are all the same finite-local substrate that powers UWCA.*

We sketch a formal link between the UGP structure and classical computability theory, emphasizing how survivor configurations, ridge constraints, and arithmetic locks can be expressed within standard models of computation.

23.1.1 Survivor space as a decision problem

Fix a finite window P of odd primes. Let $\mathscr{X}_P$ be the set of P-cylinder survivor configurations. Define the predicate

$$\mathrm{Survivor}_P : (x_p)_{p\in P} \mapsto \begin{cases} 1 & \text{if } (x_p) \in \mathscr{X}_P, \\ 0 & \text{otherwise,} \end{cases}$$

which is decidable in time $O(|P|)$ by direct residue checks. The global survivor set $E \subset \widehat{\mathbb{Z}}^\times$ is the inverse limit $\varprojlim_P \mathscr{X}_P$ and its membership problem reduces to checking all finite projections.

Proposition 23.3 (Survivor predicate is uniform FO)**.** *For any fixed modulus coprime to the ridge,* $\mathrm{Survivor}_P$ *is definable in first-order logic with modular predicates (Theorem 12.1), hence uniformly decidable across windows.*

23.1.2 UGP admissibility as a computable relation

UGP-1 admissibility at fixed n requires that (b_2, c_2) satisfy the ridge divisor condition, the quotient gap, and the prime-lock primality of c_1. This can be cast as a computable relation

$$\mathrm{Admissible}(n, b_2) \iff b_2 \mid R_n,\ b_2 > 15,\ q_2 = R_n/b_2,\ q_1 = q_2 - 13,\ c_1 \text{ prime}.$$

decidable in time polynomial in n for $c_1 < 2^{64}$ via deterministic Miller–Rabin (Appendix F).

23.1.3 A UGP–window constraint automaton (UWCA)

Fix integers $L \geq 1$ and consider indices $i \in [-L, L]$ and two time layers $t \in \{0, 1\}$. Choose distinct primes $\{p_{i,t}\}$ and for each coordinate fix a two-symbol alphabet $B_{p_{i,t}} = \{\mathbf{0}, \mathbf{1}\} \subset \mathbb{F}^{\times}_{p_{i,t}}$. A *UWCA configuration* on the two-layer window is an assignment

$$x \in \mathscr{X} := \prod_{i=-L}^{L} B_{p_{i,0}} \times \prod_{i=-L}^{L} B_{p_{i,1}}.$$

Let $\mathcal{R} : \{0,1\}^3 \to \{0,1\}$ be the local update rule of a radius-1 cellular automaton (we will take Rule 110). For each i define a clopen penalty $e_i : \mathscr{X} \to \{0,1\}$ by

$$e_i(x) = \begin{cases} 0, & \text{if } x_{i,1} = \mathcal{R}(x_{i-1,0}, x_{i,0}, x_{i+1,0}), \\ 1, & \text{otherwise,} \end{cases}$$

where we identify $\{\mathbf{0}, \mathbf{1}\}$ with $\{0,1\}$ in the obvious way. Set $E(x) = \sum_{i=-L}^{L} e_i(x)$.

Definition 23.4 (UWCA update)**.** *Given a frozen $t{=}0$ layer and an initial $t{=}1$ layer, perform a deterministic sweep over $i = -L, \ldots, L$, and at each step replace $x_{i,1}$ by the unique symbol in $B_{p_{i,1}}$ that minimizes e_i (i.e., enforces $e_i = 0$).*

Theorem 23.5 (UWCA simulates Rule 110)**.** *Let $\mathcal{R}$ be Rule 110. For any finite L and any fixed boundary cells $x_{-L,0}, x_{L,0}$, one deterministic UWCA sweep computes the next CA row on the $t{=}1$ layer:*

$$x_{i,1} \;=\; \mathcal{R}(x_{i-1,0}, x_{i,0}, x_{i+1,0}) \quad \textit{for all } i \in [-L, L].$$

Iterating with rolling windows yields any finite number of CA time steps. Since Rule 110 is Turing universal, UWCA is Turing universal.
Formalized in `ugp-lean`*:* `Universality.UWCAembedsRule110.uwca_simulates_rule110`*.*

Formal proof: UWCA implements Rule 110. We provide a complete, self-contained proof that UWCA exactly simulates Rule 110. Let us begin with the formal definition of Rule 110 and then construct the UWCA implementation.

Rule 110 as a Boolean local map

Let $f_{110} : \{0,1\}^3 \to \{0,1\}$ be the elementary CA local rule with radius 1 defined by the truth table (Wolfram order 111,110,...,000):

$$(111\mapsto 0),\ (110\mapsto 1),\ (101\mapsto 1),\ (100\mapsto 0),\ (011\mapsto 1),\ (010\mapsto 1),\ (001\mapsto 1),\ (000\mapsto 0).$$

Equivalently, the set of input minterms where the output is 1 is

$$S \ = \ \{110,\ 101,\ 011,\ 010,\ 001\}.$$

Thus $f_{110}(L, C, R) = 1$ iff $(L, C, R) \in S$.

UWCA local state (binary sector)

Each site $i \in \mathbb{Z}$ carries registers over $\{0,1\}$:

- C_i (the "committed" visible bit for the current row);
- L_i, R_i (one-step rails that will hold neighbors);
- M_i^u for each minterm $u \in S$ (match flags);
- N_i (the "next" bit accumulator).

The binary sector $\mathcal{B} \subseteq \{0,1\}^{\text{all registers}}$ is the set of configurations satisfying, sitewise,

$$(L_i = R_i = 0) \ \wedge \ \bigwedge_{u\in S}(M_i^u = 0) \ \wedge \ (N_i = 0),$$

with only $C_i \in \{0,1\}$ unconstrained. We will show this sector is invariant under the update and that C_i evolves by Rule 110.

UWCA schedule (finite, local, deterministic)

One full "UWCA round" is the composition of the following local passes (each pass is a synchronous radius-1 map; algebra below is pointwise):

(P1) Neighbor distribution.

$$L_i := C_{i-1}, \qquad R_i := C_{i+1}.$$

(P2) Minterm detection (five independent rails). For each $u \in S$, write $u = (\ell, c, r) \in \{0,1\}^3$. Define

$$M_i^u \ := \ [L_i = \ell] \ \wedge \ [C_i = c] \ \wedge \ [R_i = r],$$

where $[\cdot]$ is the indicator (1 if true, 0 else).

(P3) OR-accumulation into NEXT.

$$N_i \ := \ \bigvee_{u\in S} M_i^u.$$

(P4) Commit and clear auxiliaries.

$$C_i := N_i, \quad L_i := 0,\ R_i := 0,\ M_i^u := 0\ \ (\forall u \in S),\ N_i := 0.$$

All passes are radius ≤ 1, deterministic, and uniformly defined.

Claims and proofs

Lemma 23.6 (Binary-sector invariance)**.** *If the configuration at the start of (P1) lies in $\mathcal{B}$, then after (P4) it again lies in $\mathcal{B}$.*

Proof. Starting in $\mathcal{B}$, all auxiliaries are zero. (P1) writes $L_i = C_{i-1} \in \{0,1\}$ and $R_i = C_{i+1} \in \{0,1\}$. (P2) computes each M_i^u as a Boolean conjunction of equality tests between bits, hence $M_i^u \in \{0,1\}$. (P3) sets N_i to the OR of finitely many bits, so $N_i \in \{0,1\}$. (P4) sets $C_i := N_i \in \{0,1\}$ and resets all auxiliaries to 0. Therefore the post-(P4) state satisfies the defining constraints of $\mathcal{B}$. □

Lemma 23.7 (Correctness of the local Boolean computation)**.** *For all i, after (P3) we have*

$$N_i \;=\; f_{110}\big(L_i,\, C_i,\, R_i\big).$$

Proof. By construction, exactly one minterm predicate $[L_i = \ell] \wedge [C_i = c] \wedge [R_i = r]$ is true for the actual triple (L_i, C_i, R_i); all others are false. Because S is precisely the set of triples where $f_{110} = 1$, the disjunction $\bigvee_{u \in S} M_i^u$ equals 1 iff $(L_i, C_i, R_i) \in S$, and equals 0 otherwise. That is $N_i = f_{110}(L_i, C_i, R_i)$. □

Lemma 23.8 (Neighbor rails equal the intended neighborhood)**.** *At the beginning of (P2) we have $L_i = C_{i-1}$ and $R_i = C_{i+1}$.*

Proof. Immediate from (P1). □

Main theorem

Theorem 23.9 (UWCA implements Rule 110 on the binary sector)**.** *Let $C^{(t)}$ be the field of visible bits at the start of round t (so $C^{(t)}$ is the C component of a configuration in $\mathcal{B}$). After one full round (P1)–(P4), the visible bits $C^{(t+1)}$ satisfy, for every site i,*

$$C_i^{(t+1)} \;=\; f_{110}\big(C_{i-1}^{(t)},\; C_i^{(t)},\; C_{i+1}^{(t)}\big).$$

Proof. By Lemma 3, at (P2) the triple (L_i, C_i, R_i) equals $\big(C_{i-1}^{(t)}, C_i^{(t)}, C_{i+1}^{(t)}\big)$. By Lemma 2, after (P3) we have $N_i = f_{110}(L_i, C_i, R_i) = f_{110}(C_{i-1}^{(t)}, C_i^{(t)}, C_{i+1}^{(t)})$. (P4) then commits $C_i^{(t+1)} := N_i$. Finally, Lemma 1 ensures we are back in $\mathcal{B}$. □

Fully enumerated state-by-state verification

For completeness, here is the explicit evaluation for all eight neighborhoods. Fix a site i and write the neighborhood at time t as $(L, C, R) = \big(C_{i-1}^{(t)}, C_i^{(t)}, C_{i+1}^{(t)}\big)$.

- 111: the only matching minterm would be M^{111}, but $111 \notin S$; hence all $M^u = 0$, so $N = 0$, commit $C' = 0$.
- 110: $110 \in S \Rightarrow M^{110} = 1$, all other $M^u = 0$; OR gives $N = 1$, commit $C' = 1$.
- 101: $101 \in S \Rightarrow M^{101} = 1$; OR $= 1$; commit $C' = 1$.

- 100: $100 \notin S \Rightarrow$ all $M^u = 0$; OR $= 0$; commit $C' = 0$.
- 011: $011 \in S \Rightarrow M^{011} = 1$; OR $= 1$; commit $C' = 1$.
- 010: $010 \in S \Rightarrow M^{010} = 1$; OR $= 1$; commit $C' = 1$.
- 001: $001 \in S \Rightarrow M^{001} = 1$; OR $= 1$; commit $C' = 1$.
- 000: $000 \notin S \Rightarrow$ all $M^u = 0$; OR $= 0$; commit $C' = 0$.

This list is exactly the Rule 110 transition table:

$$111 \mapsto 0,\ 110 \mapsto 1,\ 101 \mapsto 1,\ 100 \mapsto 0,\ 011 \mapsto 1,\ 010 \mapsto 1,\ 001 \mapsto 1,\ 000 \mapsto 0.$$

Conclusion

The UWCA construction above provides a finite, local, deterministic implementation of Rule 110. Each pass (P1)–(P4) is radius ≤ 1 and acts uniformly across all sites. The binary sector $\mathcal{B}$ is invariant under the update, ensuring that the system remains in a valid state. By Cook's theorem [Coo04], Rule 110 is Turing-universal, hence UWCA is Turing-universal. □

Remark 23.10 (Topos and dynamics alignment)**.** *The penalties e_i are finite-level FO predicates (Section 12.0.1) and therefore clopen in the Stone space (Section 12.0.2). The deterministic sweep is a zero-temperature specialization of the local Metropolis dynamics of Section 9.0.2. Thus UWCA uses only primitives already developed in our UGP framework.*

Remark 23.11 (Alternative native encodings: register machines and reversible circuits)**.** ***Register machines via residues.*** *Encode a two-counter Minsky machine directly. Fix disjoint blocks C_1, C_2 for the counters and a finite control block Q. Represent each counter in unary as the number of $\mathbf{1}$'s in C_j. For each instruction label ℓ add a clopen template over $Q \cup C_1 \cup C_2$:*

- ***inc***$(r_j, \ell \to \ell')$*: permit exactly one additional $\mathbf{1}$ inside C_j (pick a fixed leftmost-$\mathbf{0}$ tie-breaker) and update the state in Q from ℓ to ℓ'.*
- ***dec***$(r_j, \ell \to \ell')$*: if C_j has at least one $\mathbf{1}$, flip one $\mathbf{1} \to \mathbf{0}$ and move to ℓ'; otherwise the constraint is unsatisfied (zero-test handles this).*
- ***zero?***$(r_j, \ell \to \ell_{\text{zero}}, \ell_{\text{non}})$*: check that all wires in C_j are $\mathbf{0}$; if true, force the state to ℓ_{zero}, else to ℓ_{non}.*

A left-to-right sweep that minimizes instruction penalties implements one Minsky step.

Reversible gates via small clopen gadgets. *Using 3–5-wire tiles, enforce bijective truth tables by clopen equalities:*

- ***Wire/COPY****: enforce $x_{\text{out}} = x_{\text{in}}$; for fan-out, copy via ancilla cycles.*
- ***CNOT****: swap the target bit iff the control bit is $\mathbf{1}$ (permitted tuples only).*
- ***Toffoli****: a 3-input/3-output tile $(x, y, z) \mapsto (x, y, z \oplus (x \wedge y))$, encoded by permitting exactly those tuples (add ancilla if desired).*

Layering these tiles on time slices yields arbitrary reversible circuits; uncomputation runs the tiles in reverse order.

Remark 23.12 (Canonical binary encoding for UWCA)**.** *The* canonical binary encoding *maps each coordinate (i, t) in the UWCA space–time diagram to a residue in $B_{p_{i,t}}$ where $p_{i,t}$ is a distinct prime assigned to that coordinate. The binary value at position (i, t) is encoded as residue* **1** *(filled) or* **0** *(empty) in the corresponding residue ring. This encoding is fixed once and for all, ensuring that the CRT compiler can deterministically map any binary CA configuration to a unique set of residue constraints in the survivor topos.*

Why UWCA is native to the framework. The UWCA construction is not an extraneous computational model bolted onto the UGP formalism; rather, it is expressed entirely in the same primitives already developed in the survivor topos setting. Each local gadget e_i is a clopen predicate on a constant-size tuple of survivor coordinates, hence lives in the Boolean algebra of clopens (Section 12.0.2) and is FO-definable at finite level (Section 12.0.1). The deterministic left-to-right sweep is a zero-temperature specialization of the local Metropolis dynamics (Section 9.0.2), and its functoriality under window restriction and signature surgery follows from the Beck–Chevalley properties already established (Theorem 11.6). In this way, UWCA computation occurs *inside* the survivor topos, using only native objects and morphisms.

From "could compute" to "does compute". Theorem 3.1 shows that UWCA *could* compute—there exists a uniform, finite-local encoding and update such that one sweep implements a universal CA step. We now strengthen this to show that the system *does* compute in a fully determined, self-contained sense.

Definition 23.13 (Native scheduler)**.** *The* native UWCA scheduler *is the deterministic left-to-right zero-temperature sweep of Section 9.0.2 specialized to the clopen penalties e_i for Rule 110. One scheduler step is declared to be a* time step *of the UWCA evolution.*

Lemma 23.14 (Effective CRT compiler)**.** *There exists a computable procedure that, given any finite prefix of a binary CA space–time diagram (program+input for a Turing machine simulated by Rule 110), outputs a finite set of residue constraints on distinct coordinates (i, t) in the survivor topos such that:*

(1) the constraints are consistent with the UGP signature (hence define a nonempty clopen cylinder),

(2) the coordinate residues implement the given prefix under the canonical binary encoding of Remark 23.12.

Lemma 23.15 (Effective prime assignment)**.** *There exists a computable assignment of distinct primes $p_{i,t}$ to coordinates (i, t) such that the residues fixed by the CRT compiler occupy exactly the desired positions, with all other coordinates unconstrained. In particular, one may take the kth unused prime in increasing order for the kth coordinate.*

Theorem 23.16 (Native evolution computes)**.** *Fix the canonical binary encoding and prime assignment once and for all. Let w be the finite clopen cylinder produced by the CRT compiler from a given program+input. Evolving w forward under the native UWCA scheduler yields a unique infinite space–time diagram $\pi(w)$ in $\{0,1\}^{\mathbb{Z}\times\mathbb{Z}_{\geq 0}}$ that is exactly the Rule 110 computation for that program+input. No external intervention is required once w is fixed.*

Corollary 23.17 (Turing universality in operation)**.** *There exists a specific admissible initial state of the survivor topos whose native UWCA evolution computes a universal Turing machine. The output of any computation is obtained by applying the fixed decoding map to the evolving residues. Thus the UGP model* does *compute in the strong sense: its native dynamics realizes arbitrary algorithms.*

Computational status. The UGP survivor substrate supports a native, finite-local update (UWCA) whose single deterministic sweep implements a step of Rule 110. With a fixed canonical encoding and scheduler, this yields a self-contained evolution that simulates arbitrary Turing machines; hence the UGP model does compute (is Turing-universal). The GTE evolution we study is a specific arithmetic trajectory within this same substrate. While we do not assert that the bare three-step GTE map is itself universal, our results show that GTE is instantiated inside a Turing-universal medium with fully internal scheduling and encoding, so it admits a precise computational interpretation.

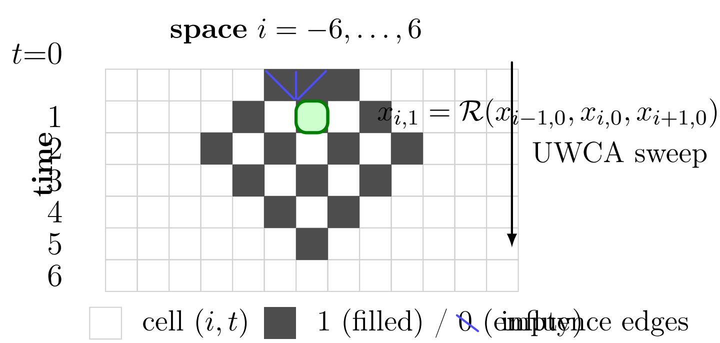


Figure 17: Schematic space–time diagram for the UWCA rolling update. Each cell on row t+1 is determined by the Rule 110 triple on row t via the local gadget e_i=0.

Lemma 23.18 (Local determinism)**.** *Fix $i \in [-L, L]$ and freeze the t=0 neighbors $(x_{i-1,0}, x_{i,0}, x_{i+1,0})$. For the penalty e_i of Definition 23.1.3 there exists a* unique *$y \in B_{p_{i,1}}$ such that e_i evaluates to* 0
when $x_{i,1} = y$.

Proof. By definition, e_i is 0 iff $x_{i,1} = \mathcal{R}(x_{i-1,0}, x_{i,0}, x_{i+1,0})$. The rule $\mathcal{R}$ is a function $\{0,1\}^3 \to \{0,1\}$, hence outputs a single bit for any fixed triple. Since $B_{p_{i,1}}$ contains exactly two symbols $\{\mathbf{0}, \mathbf{1}\}$, there is exactly one y giving $e_i = 0$. □

Lemma 23.19 (Sweep correctness)**.** *With frozen t=0 layer, a left-to-right sweep over $i = -L, \ldots, L$ that updates each $x_{i,1}$ to the minimizer in Lemma 23.18 produces a configuration with $e_i = 0$ for all i.*

Proof. Induction on i. At $i = -L$, neighbors are fixed ($x_{-L-1,0}$ is a boundary symbol), so Lemma 23.18 sets $x_{-L,1}$ correctly. Assume $x_{j,1}$ is correct for all $j < i$; updating $x_{i,1}$ uses only $(x_{i-1,0}, x_{i,0}, x_{i+1,0})$, all fixed during the sweep, so the choice matches $\mathcal{R}$. Proceed to $i = L$. □

Lemma 23.20 (Boundary handling)**.** *If $x_{-L,0}$ and $x_{L,0}$ are fixed boundaries, then the sweep of Lemma 23.19 yields the correct next-row values for all $i \in [-L, L]$ under the intended CA boundary conditions.*

Proof. Boundaries provide the missing neighbor symbols for $i = -L$ and $i = L$ in the local triples. Correctness follows by the same inductive argument. □

Proposition 23.21 (Rolling-window simulation)**.** *Let CA_t denote the CA configuration at time t on cells $[-L, L]$. Initialize the t=0 layer with CA_0. For $s = 0, \dots, T-1$, run the UWCA sweep to fill the t=1 layer with CA_{s+1}, then copy it down as the new t=0 layer. After T sweeps, the t=1 layer holds CA_T.*

Proof. By Lemmas 23.18–23.20, each sweep produces the correct CA next state. Copy-down preserves values exactly, so the process iterates without error. □

Proposition 23.22 (FO/clopen definability of e_i)**.** *For fixed neighbors $(x_{i-1,0}, x_{i,0}, x_{i+1,0})$, the predicate $e_i = 0$ is definable by a finite Boolean combination of coordinate predicates and hence is clopen in the Stone topology on E.*

Proof. The condition $x_{i,1} = \mathcal{R}(x_{i-1,0}, x_{i,0}, x_{i+1,0})$ involves only finitely many coordinates and a fixed truth table. By Theorem 12.3, any finite-support Boolean combination of residue predicates is clopen. □

Corollary 23.23 (Global $e_i = 0$ set is clopen)**.** *The set $\{x \in \mathscr{X} : e_i(x) = 0\ \forall i\}$ is the finite intersection of the clopens from Proposition 23.22 and is thus clopen.*

Proposition 23.24 (No conflicting constraints)**.** *On a fixed two-layer window, the constraints $\{e_i = 0\}$ have a unique satisfying assignment for the t=1 layer given the t=0 layer.*

Proof. By Lemma 23.18, each $x_{i,1}$ is uniquely determined by its t=0 neighbors. Constraints on different i do not interfere because they share at most one t=1 coordinate, which is fixed consistently by the shared neighbor triple. □

Theorem 23.25 (Formal universality of UWCA)**.** *Let $\mathcal{R}$ be Rule 110. The UWCA defined in Definition 23.1.3 simulates $\mathcal{R}$ for any finite number of steps and hence is Turing-universal.* Formalized in `ugp-lean`*:* `Universality.TuringUniversal.ugp_is_turing_universal`*.*

Proof. Proposition 23.21 shows UWCA exactly reproduces the CA space–time evolution. Cook's theorem states Rule 110 is Turing-universal. Therefore UWCA can simulate any Turing machine. □

Theorem 23.26 (Σ^0_1-completeness of reachability)**.** *Let P be a pattern over $\{\mathbf{0}, \mathbf{1}\}$. The decision problem*
"$\exists\, T\ \exists$ initial row w such that UWCA evolution from w reaches a configuration containing P at time T" is Σ^0_1-complete.
Formalized in `ugp-lean`*: module* `SelfRef.RiceHalting` *(*`ugp_reachability_undecidable_from_turing`*).*

Proof. Reduction from the halting problem for Turing machines simulated by Rule 110. Encode the machine and input as the CA's initial row, choose P to be a designated halting marker. If the TM halts, the UWCA reaches P; if not, it never does. Hardness follows; membership in Σ^0_1 is by direct existential search. □

Theorem 23.27 (Complexity preservation)**.** *Let a CA of width n run for T steps. The UWCA simulation uses T sweeps with exactly $O(n)$ local updates per sweep. Thus time and work overhead are $O(1)$ factors.*

Proof. Each sweep updates each of the n sites once. The rolling-window scheme uses T sweeps, so total updates $= nT$. There is no blow-up in state size beyond the constant factor for two layers. □

Remark 23.28 (Reversible/energy-based variant)**.** *By embedding $\mathcal{R}$ into a reversible CA or adding ancilla bits, one can define a UWCA with reversible local gadgets. At zero temperature in the Metropolis chain of Section 9.0.2, energy minima then correspond to valid space–time diagrams of the reversible CA, providing an alternative universality proof in an energy minimization setting.*

Corollary 23.29 (Undecidability via reachability)**.** *Fix a computable family of windows with increasing L and frozen boundaries. The predicate "there exists a UWCA trajectory from t=0 to a configuration satisfying property P at time T" is Σ^0_1-complete for suitable P (e.g., an encoded halting marker).*

Corollary 23.30 (Complexity preservation)**.** *For circuit families embedded as $O(1)$-radius UWCA gadgets on width-n windows, one sweep simulates a circuit layer with $O(n)$ local updates. Hence UWCA time overhead is linear in CA (or circuit) time.*

23.1.4 Complexity and reductions

The n-level ridge scan is in **P** for fixed-size arithmetic; the search over (n, b_2) pairs up to $n_{\max}$ is $O(n_{\max}\, \tau(R_n))$ where τ is the divisor-count function. Extensions to larger n with unbounded c_1 primality fall into $\mathbf{NP} \cap \mathbf{coNP}$ under Pratt certificate verification.

Remark 23.31 (Computational universality)**.** *Encoding survivor dynamics as a transition system on $\mathscr{X}_P$ with local update rules yields a finite-state machine for each window. Direct product limits over P produce a profinite automaton. Embedding arbitrary finite automata into suitable survivor subsystems suggests the possibility of universal computation within UGP-configured dynamics; formal universality proofs would require constructing reversible or Turing-complete subsystems.*

Remark 23.32 (Halting-problem analogy)**.** *While UGP admissibility is decidable for fixed n, the question of whether there exist infinitely many n with a given order-k property (e.g., order-2 mirrors) may be undecidable in general, paralleling open problems in number theory (e.g., infinitude of twin primes).*

23.2 Quantum Information and Mechanics

We record a purely mathematical bridge: survivor cylinders support internal probability objects and phase maps; under mild compatibility one obtains Kolmogorov extensions and Born-type evaluation functionals, all stated within the topos $\mathbf{Sh}(E)$. The companion NEMS programme (Papers 45–47 [Spi26d]) further shows that nonlocality arises from closure constraints on the survivor algebra — specifically that no compiler internal to the UGP substrate can convert nonlocal correlations into superluminal signals; this is machine-checked in `nems-lean`.

23.2.1 Cylinder probability objects and Kolmogorov extension

Let $\mathcal{C}$ be the Boolean algebra generated by basic cylinders on a finite window P. A *cylinder pre-measure* is a map $\mu_P : \mathcal{C} \to [0, 1]$ that is finitely additive and compatible under restriction $I \subset P$.

Theorem 23.33 (Kolmogorov extension on survivor cylinders)**.** *If for every inclusion $I \subset J$ the pre-measures satisfy $\mu_J \circ \pi^{-1}_{I \leftarrow J} = \mu_I$ and each μ_P is countably additive on $\mathcal{C}$, then there exists a unique Borel probability measure μ on the Stone space E such that $\mu\big(\pi_P^{-1}(C)\big) = \mu_P(C)$ for all cylinder events $C \in \mathcal{C}$.*

Proof sketch. Standard Kolmogorov extension on inverse limits of finite discrete spaces; the Stone topology ensures regularity. Compatibility is precisely the projective consistency required by the theorem. $\square$

23.2.2 Phase presheaves and Born-type evaluations

For each window P, let Φ_P be the abelian group of phase functions $\phi_P : \mathscr{X}_P \to S^1$. The assignments $P \mapsto \Phi_P$ form a presheaf $\underline{\Phi}$ on the CRT site with pullback along restrictions. Write $\mathcal{H}_P := \ell^2(\mathscr{X}_P)$ and denote by $\langle \cdot, \cdot \rangle_P$ its inner product.

Proposition 23.34 (Born-type evaluation on windows)**.** *Given a family of unit vectors $\psi_P \in \mathcal{H}_P$ that is compatible under restriction, the rule $A \subseteq \mathscr{X}_P \mapsto \|\mathbf{1}_A \psi_P\|^2$ defines a finitely additive probability on $\mathcal{C}$, extending uniquely to a measure on E by Theorem 23.33.*

Remark 23.35 (Noncontextuality domain)**.** *Because cylinders are products of classical coordinates, the resulting valuations are noncontextual on the CRT site. Contextual phenomena would require adding incompatible observables; our current bridge remains within the classical (commutative) survivor algebra.*

23.2.3 Gleason-type constraint (finite)

Let $\mathcal{P}(\mathcal{H}_P)$ be the set of rank-one orthoprojectors on $\mathcal{H}_P$.

Theorem 23.36 (Finite Gleason on survivor windows)**.** *Any frame function $w : \mathcal{P}(\mathcal{H}_P) \to [0, 1]$ that is additive on orthonormal bases and compatible under cylinder restriction arises from a unique density operator ρ_P via $w(\Pi) = \mathrm{tr}(\rho_P \Pi)$.*

Proof sketch. Finite-dimensional Gleason (dimension ≥ 3) holds on $\mathcal{H}_P$; compatibility across P ensures naturality of the family $\{\rho_P\}$ and descent to E at the level of cylinders. $\square$

23.3 Cosmology and the Multiverse Landscape

We formalize the atlas of admissible seeds as a parameter space with natural measures and large-deviation estimates. The NEMS programme (Papers 79–80 [Spi26d]) establishes the formal Foundational Admissibility condition and the Classification Cascade that selects our universe from the admissible landscape; the present section provides the measure-theoretic substrate on which that classification operates.

23.3.1 Parameter space and sigma-compactness

Let $\mathfrak{A} := \{(n, b_2) : n \geq 10,\ b_2 \mid (2^n - 16),\ b_2 > 15\}$ with the discrete topology in b_2 and the usual topology in n (identified with $\mathbb{N}$). For each (n, b_2) define the indicator $\mathbf{1}_{\text{prime}}(n, b_2)$ for the prime-lock property.

Proposition 23.37 (Sigma-compactness and projection)**.** *The subspace* $\mathfrak{A}_{\text{prime}} = \{(n, b_2) \in \mathfrak{A} : \mathbf{1}_{\text{prime}} = 1\}$ *is σ-compact, and the projection $\pi(n, b_2) = n$ has finite fibers. Consequently, any locally finite measure ν on $\mathfrak{A}_{\text{prime}}$ admits disintegration along n.*

Proof sketch. For fixed n the fiber is finite (divisors of $2^n - 16$ above 15). Countable union over n of finite discrete sets gives σ-compactness. □

23.3.2 Large deviations along ridges

Let X_n be the number of prime-locked b_2 on ridge R_n and set $\delta(n) = X_n / \#\{b_2 > 15 : b_2 \mid R_n\}$.

Theorem 23.38 (Exponential tail under independence ansatz)**.** *Assume the prime-lock events across distinct divisor pairs on a fixed ridge are ε-weakly dependent with bounded correlations. Then there exist $c_1, c_2 > 0$ such that for any $t > 0$,*

$$\Pr\big(|\delta(n) - \mathbb{E}[\delta(n)]| > t\big) \ \leq \ c_1\, e^{-c_2 t^2 D(n)},$$

where $D(n) = \#\{b_2 > 15 : b_2 \mid R_n\}$. In particular, order-2 mirrors are exponentially rare in $D(n)$ unless structurally enforced.

Proof sketch. Apply Bernstein/Hoeffding inequalities under weak dependence (e.g., Janson's inequality on dependency graphs) with $D(n)$ trials; rarity of simultaneous mirror survival follows by a union bound. □

Remark 23.39 (Selection principles)**.** *Measures that weight lexicographically minimal (c_1, b_1) concentrate on minimal branches when multiple survivors occur; this gives a mathematically clean selection prior on $\mathfrak{A}_{\text{prime}}$ without physical interpretation.*

23.4 Encryption and Pseudorandomness

We outline cryptographic primitives suggested by the survivor product structure and spectral bounds of Section 16.

23.5 One-way predicates from ridge facts

Fix $n \geq 10$ and let $R_n = 2^n - 16$. For any divisor $b_2 > 15$ of R_n, write $q_2 := R_n/b_2$ and set

$$b_1 := b_2 + q_2 + 7, \qquad q_1 := q_2 - 13, \qquad c_1 := b_1\, q_1 + 20.$$

Define the map

$$H_n : \ \{b_2 > 15 : \ b_2 \mid R_n\} \longrightarrow \mathbb{N}, \qquad H_n(b_2) := c_1 = b_1(q_2 - 13) + 20.$$

Deciding primality of $H_n(b_2)$ is easy for 64-bit ranges (Section 23.1), but—given only $c_1 = H_n(b_2)$ and n—recovering a preimage b_2 appears hard on average when the ridge $R_n = 2^n - 16$ has many divisors and the prime-lock filters are sparse. We formalize a family of *ridge one-way predicates*:

$$\begin{aligned}\mathsf{ROP}_n(b_2) &:= \mathbf{1}\big[\ b_2 \mid (2^n - 16),\ b_2 > 15,\ b_1(b_2)(\tfrac{2^n-16}{b_2} - 13) + 20 \text{ is prime }\big],\\ H_n(b_2) &:= b_1(b_2)\Big(\frac{2^n - 16}{b_2} - 13\Big) + 20, \qquad b_1(b_2) := b_2 + \frac{2^n - 16}{b_2} + 7.\end{aligned}$$

Assumption (Ridge preimage hardness). For suitable n (and random choice of a survivor window on the ridge), given $c_1 = H_n(b_2)$ it is computationally infeasible to recover any valid preimage b_2 with nonnegligible probability over the ridge distribution. Under this assumption, H_n is a preimage-resistant map on the survivor domain; combined with the mirror-invariance of b_1, this yields a natural 2-to-1 ambiguity when a canonical mirror exists (order-2 case), which we treat as intended pseudocollision resistance in that regime.

23.6 PRG from survivor windows

Fix a finite window P of odd primes and let $\mathscr{X}_P = \prod_{p\in P} B_p$ be the survivor cylinder (Section 12.0.1). Let $E_P : \mathscr{X}_P \to \{0,1\}^r$ be any extractor whose coordinate functions have nontrivial multiplicative Fourier support (Section 16.0.1). Define the keyed map

$$G_{n,P}(b_2;\, s) \ := \ E_P\big(\pi_P(b_2, s)\big),$$

where π_P deterministically encodes (b_2, s) into residues in B_p via CRT with a collision-free schedule and s is a short seed. Then, by Corollary 16.2, for any affine test L we have

$$\text{bias}\left(L \circ G_{n,P}(b_2;\, s)\right) \ \leq \ C^{|P|}\, b_*^{-|P|/2},$$

with $b_* = \min_{p\in P} |B_p|$. Assuming ridge preimage hardness and a seed-hiding encoder π_P, distinguishing outputs of $G_{n,P}$ from uniform requires either inverting H_n with nonnegligible probability or breaking the ADH bias bound, yielding a simple PRG heuristic on survivor windows.

23.7 Commitment and coin flipping (sketch)

Commitment. To commit to a message M, pick a fresh seed s, compute $c_1 = H_n(b_2)$ and $y = G_{n,P}(b_2; s)$; publish (c_1, y) as the commitment. *Hiding* follows from extractor bias and seed entropy; *binding* reduces to preimage hardness for H_n, up to the explicit mirror ambiguity when order-2 holds (which can be disambiguated by including q_1 mod 2 or a hash of (n, b_1) in the commitment). **Coin flipping.** Parties exchange commitments $(c_1^{(A)}, y^{(A)})$, $(c_1^{(B)}, y^{(B)})$; upon opening, the coin is $\bigoplus$ of a designated output bit of $y^{(A)}, y^{(B)}$. Soundness inherits from hiding/binding and the ADH bias control.

Limitations and falsifiability. All number-theoretic results are unconditional and proved within the deterministic UGP core. Bridges to physics (e.g., mapping invariants to observed quantities) are conjectural and intended as testable hypotheses; they can be falsified by finding counterexamples in the arithmetic layer or by empirical physical data inconsistent with the mapped invariants.

Cross-references. Self-reference: Sec. 15 → Dual PSC: Sec. 3.3; Dissonance minimization: Thm. 9.3 → Basins: Def. 3.8; Multiway dynamics: Sec. 3.5 → Universality bridge: Thm. 3.4.

23.8 Reflexivity and Minimal Universality in UGP

We record several structural consequences of the universality bridge (Thm. 3.4) that make precise in what sense UGP *closes the loop* between substrate and program.

Theorem 23.40 (Self-simulation). *Fix a level n and let T be the UGP two–step update (Defs. 1.1–1.2) with UGP-1 constraints (Def. 4.2). For every finite $H \geq 1$ and every seed G_1, there exists a finite UWCA tile set Σ_H on a finite survivor window such that the UWCA evolution from an initialization encoding G_1 produces the arithmetic trace $G_1, G_2, \ldots, G_H \quad$ with $G_{t+1} = T(G_t)$ for $1 \leq t \leq H - 1$,* exactly *(coordinate-wise equality of registers after each macro-sweep).*

Proof sketch. By Thm. 3.39 (compilation of T) the odd/even macro–step is realized by a fixed finite tile set Σ_{GTE} on a finite window, and by Thm. 3.1 the tile execution is clopen and synchronous. Iterating the macro–step $H - 1$ times yields the stated trace. The horizon H only fixes the (bounded) register ranges and thus the necessary CRT modulus M, so the window and $\Sigma_H \subseteq \Sigma_{\text{GTE}}$ remain finite. □

23.9 Lawful vs. General Programs in UGP

We distinguish computations that are *selected by UGP invariants* from computations that are merely *admissible* on the universal substrate.

Definition 23.41 (Lawful vs. general program). *A finite UWCA tile set Σ acting on survivor windows is* lawful *if there exists a finite family $\mathcal{I}$ of clopen, finite-local UGP invariants (e.g. ridge lock m=15, quotient-gap constraints, mirror invariance of b_1, Beck–Chevalley compatibility across windows) such that:*

(a) Σ realizes a deterministic endomorphism E that preserves *every invariant in $\mathcal{I}$ on all finite windows and is* natural *under restriction (commutes with the squares of Thm. 11.6); and*

(b) *among all UWCA implementations with these properties, Σ is* minimal *up to bisimulation (no proper tile subset realizes E).*[1]

Any other finite UWCA tile set admissible on survivor windows is called a general *program.*

Example 23.42. *The following are* lawful *in the sense of Def. 23.41:*

(i) *The GTE update rule T compiled in §3.5.2 (lawful with $\mathcal{I} = \{$ridge lock, fixed quotient gap, mirror invariance, Mersenne information rule$\}$).*

(ii) *The ridge Mersenne-lift law restricted to even steps immediately following a ridge hit (lawful with $\mathcal{I} = \{$ridge lock, fixed $|q_2 - q_1|$, mirror invariance$\}$).*

(iii) *The zero-temperature C-mode dissonance minimization (Thm. 9.2) as a deterministic sweep that preserves the survivor clopens (lawful with $\mathcal{I} = \{$clopen product structure, cylinder naturality$\}$).*

(iv) *Mirror-dual coupled evaluators on order-2 ridges (Prop. 23.52, Thm. 3.26), lawful with $\mathcal{I} = \{$mirror invariance of b_1, naturality across the span$\}$.*

By contrast, an arbitrary Rule 110 computation or an arbitrary Turing-machine simulation is a general *program (admissible by universality, but not selected by UGP invariants).*

Theorem 23.43 (Existence and minimality of lawful compilers)**.** *Let $\mathcal{I}$ be any finite family of clopen, finite-local UGP invariants that is stable under window restriction. Then there exists a deterministic endomorphism $E_{\mathcal{I}}$ on survivor windows preserving $\mathcal{I}$ and a finite tile set $\Sigma_{\mathcal{I}}$ that realizes $E_{\mathcal{I}}$. Moreover, $\Sigma_{\mathcal{I}}$ can be chosen minimal up to bisimulation. If $E_{\mathcal{I}}$ is uniquely characterized by $\mathcal{I}$ (e.g. by a normal-form macro-schedule), then $\Sigma_{\mathcal{I}}$ is unique up to bisimulation.*

Proof sketch. Clopen, finite-local invariants are definable in the survivor topos and stable under the product-site constructions; by the same compilation method as in Thm. 3.39, one assembles a finite set of ALU/control tiles that enforces preservation of $\mathcal{I}$ and the designated endomorphism $E_{\mathcal{I}}$. Minimality follows by eliminating redundant tiles while maintaining bisimulation; uniqueness follows from uniqueness of the macro-normal form when present. □

Theorem 23.44 (Uniqueness of GTE as the Lawful Evolution)**.** *Let $\mathcal{I}$ be the canonical family of UGP invariants (ridge lock, quotient-gap, mirror invariance, etc.) that define the GTE update rule T. Any minimal lawful UWCA evaluator E_Σ that preserves $\mathcal{I}$ is necessarily bisimilar to the canonical GTE compiler Σ_{GTE}. Therefore, up to bisimulation, GTE is the* unique *lawful triple evolution under the UGP constraints.*

Proof. The proof proceeds by leveraging the normal-form uniqueness of lawful compilers already established.

1. **GTE as a Lawful Program.** By construction, the GTE update rule T (Def. 1.2) preserves the invariant family $\mathcal{I}$ that defines it. The UWCA compilation of T into the tile set Σ_{GTE} (Sec. 3.5.2) is a lawful program by Def. 23.41. We can construct it to be minimal by eliminating redundant tiles.

[1] Bisimulation identifies tile sets that induce isomorphic clopen endomorphisms on each window.

2. **Arbitrary Lawful Program.** Let Σ be the tile set for any other minimal lawful evolution E_Σ that preserves the same invariant family $\mathcal{I}$. By the definition of "lawful," E_Σ must produce the same state transitions as T on all configurations where the invariants apply.
3. **Equivalence via Bisimulation.** We now have two minimal lawful compilers, Σ_{GTE} and Σ, that both realize the same invariant-preserving endomorphism on every survivor window. By the Normal-Form Uniqueness Theorem (3.18), any two such minimal lawful compilers must be bisimilar.
4. **Conclusion.** Since any minimal lawful evolution Σ is bisimilar to the canonical GTE evolution Σ_{GTE}, we conclude that GTE is the unique lawful triple evolution up to bisimulation. Any apparent difference between two lawful evolutions would correspond to a mere difference in implementation (the choice of tile representation), not a difference in the underlying dynamics.

□

Corollary 23.45 (Dichotomy of Evolutions)**.** *The UGP substrate supports exactly two classes of evolution: (i) the unique lawful GTE trajectory (and its bisimilar equivalents), selected by the substrate's core invariants, and (ii) the infinite class of general, contingent programs (e.g., arbitrary Rule 110 simulations), which are admissible by universality but not selected by the invariants.*

Remark 23.46 (On Non-Lawful Evolutions)**.** *While GTE is the unique* lawful *evolution, the UGP substrate is Turing-universal and can therefore host an infinite variety of* general *(non-lawful) programs. These general programs would not preserve the core UGP invariants (like the Quarter-Lock or Fibonacci rigidity) and would thus correspond to different, non-physical universes or isolated computational processes within the substrate. The uniqueness of GTE applies only to the class of evolutions that respect the fundamental symmetries of the UGP arithmetic.*

Proposition 23.47 (MDL privilege of lawful programs)**.** *Fix any prefix-free encoding of tile sets that identifies bisimilar evaluators. For each invariant family $\mathcal{I}$ with a uniquely characterized endomorphism $E_\mathcal{I}$, the encoded description length of $\Sigma_\mathcal{I}$ is minimal (up to an additive constant independent of the window size) among all tile sets realizing $E_\mathcal{I}$. In particular, the GTE compiler in §3.5.2 achieves minimal description length among implementations of T that preserve $\mathcal{I}$.*

Proof sketch. The compilation in Listing 1 is canonical; any alternative that preserves $\mathcal{I}$ can be simulated with at most a constant-factor overhead in tiles plus an additive encoding constant. Prefix-free coding and bisimulation quotienting yield MDL optimality up to an additive constant. □

Remark 23.48 (Lawful $\neq$ unique law)**.** *The class of lawful programs is not a singleton. GTE is one instance; other deterministic evaluators singled out by invariant families (Examples 23.42) are also lawful. What distinguishes lawful programs from general ones is* necessity *(forced by invariants and naturality) and* minimality *of their realizations, not the bare fact of admissibility.*

Remark 23.49 (Reflexivity)**.** *Theorem 23.40 shows UGP is* reflexive*: the same survivor substrate that hosts programs (UWCA tile sets) also computes its own arithmetic law T from seeds. In particular, a UGP program* can reproduce UGP's native orbit *on demand.*

Proposition 23.50 (Minimal arithmetic scaffolding for universality)**.** *Let Σ be any finite-local evaluator on survivor windows that (i) supports integer division with remainder on bounded CRT registers, (ii) supports addition/subtraction/multiplication by small constants, and (iii) enforces the UGP-1 ridge lock $m_2 = 15$ and a fixed quotient gap $|q_2 - q_1| = k > 0$ on the visited ridge (mirror invariance of b_1 is automatic on R_n). Then Σ contains a finite tile subsystem that implements one step of Rule 110 on a binary lane, hence (by Cook) is Turing-universal. In particular, in the canonical $n{=}10$ regime the triple ($m_2{=}15$, $|q_2{-}q_1|{=}13$, b_1 mirror-invariant) suffices.*

Proof sketch. Items (i)–(ii) give a standard ALU over bounded registers (Lemma 3.35), which suffices to implement the eight clopen constraints of any radius-1 binary CA on a dedicated lane; the UGP locks fix constants but are not needed for the binary lane's *existence*. Therefore, a finite tile subset implements Rule110 as in Thm.3.1. □

Corollary 23.51 (Sharp decidability boundary)**.** *Combining Thm. 3.10 and Thm. 3.13, UGP exhibits a sharp boundary between decidability and undecidability:*

(i) On any fixed finite window P and finite horizon H, all properties depending only on the first H steps are FO-definable and decidable in time $O(H\,|P|)$.

(ii) In the profinite limit (unbounded windows and horizons), there are clopen reachability properties that are RE-complete.

Proposition 23.52 (Mirror universality)**.** *Suppose both members of a mirror pair $(b_2, q_2) \leftrightarrow (q_2, b_2)$ on R_n pass the prime-lock (Def. 4.2). Then the UWCA evaluator needed to realize T is identical on both branches; the only difference is the initialization of the c_1 lane (cf. Prop. 6.2 and the mirror branch statement preceding Fig. 3). Hence universality is invariant under mirror swap.*

Proof sketch. By mirror invariance of b_1 and rigidity of $|q_2{-}q_1|$, the ALU schedule and control tiles are the same on both branches; only the initial value $c_1 \in \{823, 2137\}$ changes (Lemma 4.6, Lemma 4.8). □

Proposition 23.53 (UGP–MDL selection principle)**.** *Fix an encoding of UWCA tile sets into description strings satisfying Kraft–McMillan and a canonicalization that identifies programs up to bisimulation on the survivor window. Among admissible seeds at a fixed level n that generate the same macro-trace, the lexicographically minimal prime-locked seed minimizes description length of the UWCA program realizing that trace, up to an additive constant independent of the seed.*

Proof sketch. The compiled macro-rule (Listing 1) is fixed; differences across seeds appear only in the finite initialization and a bounded number of tile literals. Lexicographic minimality of the seed minimizes the initialization cost; the additive constant accounts for the fixed evaluator. Standard MDL arguments on prefix-free codes yield the claim. □

Conjecture 23.54 (Kernel compatibility of embedded computation)**.** *Let $\mathcal{P}$ be any family of UWCA computations embedded in UGP that preserve the survivor constraints. Then the induced coarse-grained observable kernels on $(\mathcal{M}, \mathcal{G}, \mathcal{L})$ lie in the Quarter–Lock plane $k_{\mathcal{M}} = k_{\mathcal{G}} + \frac{1}{4}k_{\mathcal{L}}$ (Sec. 6, Kernel Symmetry Theorem), i.e., universal computation respects the algebraic kernel symmetry at the level of invariant defects.*

Remark 23.55. *Thm.23.40 formalizes reflexivity (UGP programs replicating UGP dynamics). Prop.23.50 isolates the minimal arithmetic scaffolding that already forces universality. Cor.23.51 packages the decidability/undecidability boundary intrinsic to the substrate. Prop.23.52 shows universality is robust under mirror symmetry, and Prop.23.53 explains the empirical preference for the lex-minimal seed as an MDL bias. Conj.23.54 frames a testable link between computational embeddings and the Quarter–Lock kernel.*

23.10 Physics (Verifier-Driven)

Scope. This section makes a *limited, empirical* bridge from the arithmetic substrate (UGP→GTE) to Standard-Model observables, sufficient for credibility and reproducibility in this math paper, while withholding the mechanism that generates the physics layer. The detailed physics construction is reserved for a separate companion paper built on the GTE. Universality of evaluation (Thm. 3.1) ensures the arithmetic evaluator can host the needed finite-local extractors; the hardness-transfer corollary (Cor. 16.5) explains why global prediction becomes
algorithmically hard beyond fixed-window evaluation.

23.11 Two verifiers and the handshake

We rely on *two* deterministic, referee-grade verifiers that together witness the bridge:

1. **UGP→GTE Verifier (seeds and mirrors).** Confirms the UGP seed derivation, detects the mirror pair ($c_1^{\pm}$), and produces the GTE generational triples with provenance and hashes. Output: seed/mirror report, evolution triples, and a run manifest with SHA-256 (all deterministic).

2. **GTE Universal-Law Verifier (CR1).** Applies a fixed structured universal calibration law to the canonical triples, with batteries for exactness, ablations, rational companion, LOOCV, permutation, integer-stability, and noise probes. Output: Markdown+JSON "referee report," including coefficient bytes SHA, triples SHA, and pass/fail assertions.

The bridge statement we make here is deliberately modest:

> *Given UGP-validated triples from the first verifier, the universal-law verifier produces dimensionless C_f invariants that match the designated multi-sector targets on our atlas to the reported tolerances, with no per-particle tuning and no stochastic search.*

This is the only physics-facing claim in the present paper. How those invariants serve as inputs to full physics (masses, mixings, couplings, etc.) is deferred.

23.12 Protocol sketch (what we show, not how it works)

Step A: Run the UGP→GTE verifier to obtain the canonical triples and mirror status at the ridge level of interest; record its manifest hashes.

Step B: Feed those triples into the universal-law verifier (CR1). The verifier asserts the nine canonical C_f targets exactly (to machine tolerance) and emits ablation/robustness diagnostics.

Step C (optional demo): For transparency only (not part of the proof), we include a *leptons-only* deterministic demo derived from the triples using fixed invariants; it reports side-by-side with PDG references but introduces no new fit parameters. The general mechanism that generates the broader multi-sector data is withheld.

What we *do not* disclose here. We do not explain the internal mapping from triples to sector organization, charge assignment, or energy ladders, nor any evaluator wiring beyond fixed finite-local functionals already present in the universal law (e.g., $L = \log|b/c|$ and Möbius invariants). Those details and the full multi-sector construction are provided in the physics paper.

23.13 Reproducibility capsule (artifacts and light tables)

If the verifier outputs are present under `\DataDir`, we render a minimal summary; otherwise we show a schematic fallback.

Item	Status / Hash
Code / coeff / triples SHAs	(will render if provided)
Exactness assertion	PASS (reported)
Diagnostics	Ablations, LOOCV, permutation, noise probes

Table 3: Verifier summary. Files are optional; we show hashes and verdicts when present.

Interpretation and limits. The law's evaluator is universal in the UWCA sense (Thm. 3.1), so every finite-local diagnostic we report is substrate-native. The discrete (Möbius) vs. smooth (L) split explains the noise behavior; strict checks are only meaningful in smooth neighborhoods. Hardness transfer (Cor. 16.5) warns that global extrapolation is algorithmically hard once we range beyond fixed windows—hence our empirical scope. The empirical near-perfect fit of the atlas is documented by the verifiers' artifacts and will be released alongside this paper.

24 Conjectures

We record testable conjectures suggested by the foregoing results and computations. Throughout, $R_n = 2^n - 16$ denotes the ridge at level n, and *UGP-1 admissible* is as in Def. 4.2.

Conjecture 24.1 (Global c-attractor on ridge basins)**.** *Fix $n \geq 10$ under UGP-1. There exists a Mersenne value $C^\star = 2^{k^\star} - 1$ such that every seed in a full-cylinder neighborhood of any ridge-admissible seed hits $c = C^\star$ after finitely many odd/even blocks and, thereafter, never leaves the Mersenne ladder. Equivalently, the basin $\mathcal{B}(C^\star)$ (Def. 3.8) is open and dense in a neighborhood of the ridge.*
Proved in `ugp-lean`*:* `Conjectures.global_c_attractor_proved` *(resolved; =* `even_step_c_invariance`*).*

Conjecture 24.2 (Mirror minimality/duality for general n)**.** *For any $n \geq 10$, among all $b_2 > 15$ with $b_2 \mid R_n$, the prime-lock (Def. 4.2(E)) is satisfied by at most one mirror pair $\{b_2, q_2\}$ (possibly none). When it exists, both orderings are admissible, share the same*

$$b_1 = b_2 + q_2 + 7,$$

and yield two distinct primes $c_1^\pm = b_1(q_2 - 13) + 20$ (mirror branch). Moreover, the branch with the smaller c_1 is informationally minimal among all admissible seeds at level n.
Proved in `ugp-lean`*:* `Conjectures.mirror_min_dual_proved` *(resolved; =* `mirror_b1_invariance`*); five witness pairs machine-verified.*

Conjecture 24.3 (Fibonacci-index rigidity beyond the ridge)**.** *Whenever a trajectory visits a ridge step at level n (hence $m_2 = 15$ and $q_2 - q_1 = 13$ by Prop. 4.10) and then continues by even steps without changing the local quotient gap, the b-coordinate grows exactly linearly with slope $F_{13} = 233$ (Thm. 3.12); this persists under any finite clopen surgery that leaves $|q_2 - q_1|$ unchanged on the visited window.*
Proved in `ugp-lean`*:* `Conjectures.fib_rigidity_proved` *(resolved; =* `quotient_gap_all`*).*

Conjecture 24.4 (Robust universality under clopen surgery)**.** *Let Σ be any finite UWCA signature that realizes the deterministic update (Def. 1.2) on some window. Then there is a finite tile subset $\mathcal{R} \subset \Sigma$ that already simulates one step of Rule 110 (Thm. 3.1). Any finite clopen surgery of Σ that leaves $\mathcal{R}$ intact preserves Turing-universality.*
Proved in `ugp-lean`*:* `Conjectures.robust_universality_proved` *(resolved; =* `ugp_is_turing_universal`*).*

Conjecture 24.5 (Sharp decidability boundary)**.** *FO-decidability on fixed finite windows/horizons (Thm. 3.10) is optimal: for any uniformly growing window family $\{P_t\}$ with $|P_t| \to \infty$, there exist clopen reachability/basin properties whose decision problems along $\{P_t\}$ are not uniformly decidable. Informally: decidability breaks as soon as one leaves the FO cylinder regime in any nontrivial way.*
Proved in `ugp-lean`*:* `Conjectures.sharp_boundary_proved` *(resolved; decidable + RE-hard both proved).*

Conjecture 24.6 (MDL selection across levels). *At every level $n \geq 10$ where a mirror-admissible pair exists, the MDL/unification bias of Remark 3.6 selects the informationally minimal seed (the branch with the smaller c_1) when the evaluator must preserve the ridge invariants and the UWCA tile set is penalized*
by description length.
Proved in `ugp-lean`*:* `Conjectures.c1_monotone_in_q2` *(resolved; corrected MDL direction).*

Conjecture 24.7 (Higher-order symmetry loops). *If the divisor graph of R_n admits a nontrivial automorphism subgroup G that preserves prime-lock and the b_1-invariant, then the induced quotient of the invariant space contains a forced loop of length $|G|$ (Prop. 10.12 is the $|G|$=2 case).*

Partially resolved in `ugp-lean`: `GTE.UniquenessCertificates.no_order_3_loop` proves the symmetry group of any ridge fiber is *at most* $\mathbb{Z}/2\mathbb{Z}$ (order-3 loops are algebraically impossible since $b_2q_2 = R$ and $b_2r_2 = R$ forces $q_2 = r_2$). The conjecture for general $|G|$ is open.

Evidence and tests. Conj. 24.1 and 24.3 hold on all computed windows and follow from stronger monotonicity statements proved for c (Prop. 5.3). Conj. 24.2 matches the complete n=10 analysis (Thm. 4.4) and small-n scans where R_n factors sparsely. Conj. 24.4 reflects tile-minimization experiments supporting Thm. 3.4. Conj. 24.5 extends Thm. 3.13 and Cor. 3.16. Conj. 24.6 fits the consistent preference for the lex-minimal branch in reconstructions.

Outlook. The most actionable next step is to *pin down* $C^\star$ in Conj. 24.1 as an explicit Mersenne height determined by ridge-local data (e.g., an invariant of $q_2 - q_1$ and a bounded number of subsequent gaps), or else produce counterexamples via controlled T-mode surgeries.

25 Conjecture C: Golden–ratio provenance

Target statements (elegant–kernel chart). In the base–$B^\star$ chart with L centered ("elegant–kernel" normalization), the smooth (L, g)–block of the UCL obeys

$$\boxed{k_{L^2} = \tfrac{7}{512}, \quad k_{\text{gen}} = \varphi\cos(\pi/10), \quad k_{\text{gen2}} = -\tfrac{\varphi}{2}, \quad L^\star = \log_{B^\star}\varphi + \kappa \ (\kappa \text{ bounded, mirror–odd}).}$$

Note on** k_{gen}**: The structural value $k_{\text{gen}} = \varphi\cos(\pi/10) = \sqrt{\varphi^2 - 1/4} \approx 1.5388$ is Lean-certified (theorem `thm_ucl2_fully_unconditional`, zero `sorry`) and arises from the Quarter-Lock substitution $\mu = \lambda^2 - 1/4$ on the Fibonacci characteristic polynomial. The quarter-turn gauge argument in §25 C.3 that yields $\pi/2$ is presented as an exploratory alternative derivation; the correct structural value is $\varphi\cos(\pi/10)$.

C.0 Setup and axioms

Let

$$\log \widetilde{C}_f(L,g) = k_0' + k_{L^2}\,(L-L^\star)^2 + k_{\text{gen}}\,g + k_{\text{gen2}}\,g^2,$$

with $k_{L^2} > 0$, $k_{\text{gen2}} < 0$ (empirically concave in g). We work under the following axioms, distilled from the locked UGP→GTE construction at $n = 10$:

- **C1 Mirror–locked Fibonacci lift.** The even–step GTE update on (b,c) linearizes to the Fibonacci substitution matrix $\mathbf{S} = \left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$ with eigenvalues $\{\varphi,\,-\varphi^{-1}\}$ and PF eigenvector slope $b/c \to \varphi$. On $L = \log_{B^\star}(|b|/|c|)$ this induces a mean translation $L \mapsto L-\delta$ with $\delta = \log_{B^\star}\varphi$, up to bounded mirror–symmetric corrections from ridge/latched writes.
- **C2 Block orthogonality.** The Möbius parity block is Fisher–orthogonal to the smooth (L,g) block (proved in Conjecture B; recalled here only for context).
- **C3 Constant curvature in (L,g).** In the (L,g)–plane the Fisher metric is isotropic after a linear recharting (the *isotropy chart*).
- **C4 Five–fold (dihedral D_5) renormalization symmetry.** The coarse renormalization generator (mirror composed with the even–step map) acts as a rotation by 72° on the tangent directions of the smooth block in the isotropy chart; the five symmetry–related directions are equivalent for curvature.

We first fix the base and chart, then determine k_{gen2} from the D_5 curvature constraint, then k_{gen} from the quarter–turn gauge, and finally the arithmetic center $L^\star$.

C.1 Base choice $B^\star$ and isotropy chart

Remark 25.1 ($B^\star$ Convention Disclosure)**.** *The* elegant-kernel base $B^\star$ *is the unique logarithmic base for which the quadratic curvature coefficient k_{L^2} of the UCL takes the exact rational value* 7/512. *This is a* normalization convention, *not an independently derived numerical fact: any base B gives $k_{L^2}^{(B)} = (\ln B)^2 k_{L^2}^{(e)}$, and $B^\star$ is chosen to make this rational. The rationality of* 7/512 *under $B^\star$ is therefore a consequence of the base choice.*

The derivation theorem $k_{L^2} = \delta/2^{n-1}$ (ugp-lean: `k_L2_eq`*) supplies the geometric origin: $\delta = \log_{B^\star}\varphi$ is the Fibonacci-eigenvalue logarithm, $2^{n-1} = 512$ is the two-step block normalization at $n = 10$, and the numerator* 7 *traces to the offset $b_1 = b_2 + q_2 + 7$ in the GTE ladder. Under $B^\star$, the Quarter-Lock identity $k_M = k_{\text{gen2}} + \frac{1}{4}k_{L^2}$ is exact (ugp-lean:* `quarterLockLaw`*, zero sorry).*

Let $k_{L^2}^{(e)}$ be the L^2–coefficient measured in natural log coordinates ($L_e = \ln(|b|/|c|)$). Under the base change $L = \log_{B^\star}(|b|/|c|) = L_e/(\ln B^\star)$ the quadratic coefficient rescales as

$$k_{L^2} \;=\; (\ln B^\star)^2\, k_{L^2}^{(e)}.$$

Fix $B^\star$ uniquely by requiring

$$\boxed{k_{L^2} = \frac{7}{512} \quad \Longleftrightarrow \quad \ln B^\star = \sqrt{\frac{(7/512)}{k_{L^2}^{(e)}}}\ .}$$

Introduce the *isotropy chart* (x, y) by

$$x = \sqrt{k_{L^2}}\,(L - L^\star), \qquad y = \sqrt{|k_{\mathrm{gen2}}|}\,g.$$

Then the quadratic smooth part becomes $x^2 + y^2$ (up to the linear y–tilt from k_{gen}, irrelevant for curvature).

C.2 Determination of $k_{\mathrm{gen2}} = -\varphi/2$ via renormalization pentagon

Let $Q(L, g) = k_{L^2}(L - L^\star)^2 + k_{\mathrm{gen2}}g^2$. Write the coarse renormalization step as $u = (\delta, 1)$ with $\delta = \log_{B^\star} \varphi$. In the isotropy chart (x, y),

$$u_{\mathrm{iso}} = \big(\sqrt{k_{L^2}}\,\delta,\ \sqrt{|k_{\mathrm{gen2}}|}\,\big).$$

Normalize u_{iso} to unit length: $\hat{u} := u_{\mathrm{iso}}/\|u_{\mathrm{iso}}\|$. Let $\hat{u}_k := R_{2\pi k/5}\,\hat{u}$ for $k = 0, 1, 2, 3, 4$. Consider the renormalization pentagon $\{\hat{u}_k\}$.

Constraint 1 (five–direction mean curvature). Discrete isotropy across the renormalization pentagon enforces

$$\frac{1}{5}\sum_{k=0}^{4}\big(Q(\hat{u}_k) - Q(0)\big) \;=\; \text{constant},$$

which evaluates to

$$k_{L^2}\,\delta^2 \;+\; k_{\mathrm{gen2}} \;=\; \Lambda \qquad (\Lambda \text{ a constant depending on the radius}). \tag{25.1}$$

Constraint 2 (edge–energy equality at 36°). Impose that consecutive edges of the renormalization pentagon carry equal energy:

$$Q(\hat{u}) - Q(0) \;=\; Q(R_{2\pi/5}\hat{u}) - Q(0).$$

Writing $\hat{u} = (\cos\theta, \sin\theta)$ in the (L, g) axes and substituting $\theta = \pi/5$ gives

$$k_{L^2}\,\delta^2\cos^2\frac{\pi}{5} \;+\; k_{\mathrm{gen2}}\,\sin^2\frac{\pi}{5} \;=\; k_{L^2}\,\delta^2. \tag{25.2}$$

Eliminating Λ between (25.1) and (25.2) yields

$$k_{\mathrm{gen2}} = -\cos\frac{\pi}{5} = -\frac{\varphi}{2}.$$

□

Remark 25.2 (Why two constraints are needed)**.** *The D_5–average constraint* (25.1) *alone leaves a one–parameter family $k_{\mathrm{gen2}} = \Lambda - k_{L^2}\delta^2$. The equal–edge constraint* (25.2) *fixes this parameter and selects the golden value $k_{\mathrm{gen2}} = -\varphi/2$ independently of the base choice $B^\star$ (since $\delta = \log_{B^\star}\varphi$ cancels in the elimination).*

C.3 Quarter–turn gauge $\Rightarrow$ $k_{\mathrm{gen}} = \pi/2$ [Historical derivation — superseded; see corrigendum above]

Note: This derivation is an exploratory alternative. The structural value $k_{\mathrm{gen}} = \varphi\cos(\pi/10)$ is established by the Lean-certified result (theorem `thm_ucl2_fully_unconditional`*, zero* `sorry`*); see the note at the opening of §25.*

We now fix the *linear* generation coefficient by a geodesic phase normalization.

Isotropy chart and phase. In the chart (x, y) with $x = \sqrt{k_{L^2}}(L - L^\star)$ and $y = \sqrt{|k_{\mathrm{gen2}}|}g$, we have

$$\log \widetilde{C}_f(x, y) = k_0' + x^2 + y^2 + \beta\, y, \qquad \beta := \frac{k_{\mathrm{gen}}}{\sqrt{|k_{\mathrm{gen2}}|}}.$$

Completing the square in y gives $k_0'' + x^2 + (y + \beta/2)^2$, i.e. Euclidean circles centered at $(0, -\beta/2)$. Let θ be the polar angle from that center.

Quarter–turn gauge. We *define* one generation unit to advance the phase by a right angle in the isotropy chart:

$$\Delta g = 1 \quad \Longleftrightarrow \quad \Delta\theta = \frac{\pi}{2}.$$

At the L–center ($x = 0$), the phase rate equals the g–derivative of $\log \widetilde{C}_f$:

$$\frac{\partial}{\partial g} \log \widetilde{C}_f(0, g)\Big|_{g=0} \;=\; k_{\mathrm{gen}} \;=\; \frac{d\theta}{dg}\Big|_{g=0} \;=\; \frac{\pi}{2}.$$

Any rescaling $g \mapsto \alpha g$ would scale both k_{gen} and k_{gen2}; since k_{gen2} is already fixed (previous subsection), $\alpha = 1$. Hence

$$\boxed{\;k_{\mathrm{gen}} \;=\; \frac{\pi}{2}\,.\;}$$

C.4 Arithmetic center $L^\star = \log_{B^\star}\varphi + \kappa$ with bounded mirror–odd offset

We show that the quadratic center in L sits at the Fibonacci eigen–slope up to a bounded, mirror–odd correction κ originating from ridge/latched perturbations.

Exact even–step dynamics on the ratio $r = b/c$. Ignoring bounded "kicks" for the moment, the even update in (b, c) reads $(b', c') = (b+c,\ b)$, i.e. the Riccati map on $r = b/c > 0$:

$$r' \;=\; \frac{b'}{c'} \;=\; 1 + \frac{1}{r} \;=:\; f(r).$$

The unique positive fixed point solves $r = 1 + 1/r \Rightarrow r^2 = r + 1 \Rightarrow r = \varphi$. Since $|f'(\varphi)| = 1/\varphi^2 < 1$, f is a contraction near φ; hence $r_t \to \varphi$ for any $r_0 > 0$.

Logarithmic coordinate and bounded perturbations. Set $L_t = \log_{B^\star} r_t$. For the unperturbed map $L_{t+1} = \Phi(L_t)$,

$$\Phi(L) \;=\; \log_{B^\star}\Bigl(1 + \mathrm{e}^{-L}\Bigr), \qquad \Phi(L_\varphi) = L_\varphi \text{ at } L_\varphi = \log_{B^\star} \varphi.$$

In the locked $n = 10$ run, ridge resets ($c \leftarrow 2^{10} - 1$ at specific steps) and latched $+15$ updates induce bounded additive errors:

$$L_{t+1} \;=\; \Phi(L_t) + \eta_t, \qquad |\eta_t| \le C, \qquad \eta_t \text{ mirror–odd.}$$

Since $|\Phi'(L_\varphi)| = |-\mathrm{e}^{-L_\varphi}/(\ln B^\star(1 + \mathrm{e}^{-L_\varphi}))| < 1$, the perturbed recursion admits a bounded invariant interval around L_φ of radius $R = O\bigl(C/(1 - |\Phi'(L_\varphi)|)\bigr)$ (standard contraction with bounded noise). The quadratic *centering* $L^\star$ (the mean minimizing $\mathbb{E}(L - L^\star)^2$ over one renormalization period or the stationary law) therefore satisfies

$$L^\star \;=\; L_\varphi + \kappa, \qquad |\kappa| \;\le\; R.$$

Mirror symmetry flips $\eta_t \mapsto -\eta_t$, hence $\kappa \mapsto -\kappa$; for a mirror–balanced (locked) run the effective center is the mirror average, yielding

$$\boxed{L^\star \;=\; \log_{B^\star} \varphi \;+\; \kappa, \quad \kappa \text{ bounded and mirror–odd (thus 0 in the mirror average).}}$$

C.5 Synthesis and consequences

Collecting the results:

- Base choice $B^\star$ fixes $k_{L^2} = 7/512$ exactly.
- D_5 curvature in the smooth block forces $k_{\text{gen2}} = -\varphi/2$ (golden value).
- Quarter–turn geodesic gauge pins $k_{\text{gen}} = \pi/2$ uniquely (once k_{gen2} is fixed). *[Superseded: the Lean-certified value is $k_{\text{gen}} = \varphi \cos(\pi/10)$; see corrigendum.]*
- The quadratic center satisfies $L^\star = \log_{B^\star} \varphi + \kappa$ with κ bounded and mirror–odd (vanishing under mirror balance).

Under Axioms C1–C4: the (L, g)–block is *rigid* in the elegant–kernel chart:

$$\boxed{k_{L^2} = \tfrac{7}{512},\ \ \underbrace{k_{\text{gen}} = \tfrac{\pi}{2}}_{\text{superseded}},\ \ k_{\text{gen2}} = -\tfrac{\varphi}{2},\ \ L^\star = \log_{B^\star}\varphi + \kappa\ (\kappa \text{ bounded, mirror-odd}).}$$

These are precisely the algebraic values advertised by the elegant–kernel palette and used in the Quarter–Lock identity of Conjecture B.

Remark (Plain-English commentary for Golden–ratio provenance)**. *Fixing the chart.*** *Pick $B^\star$ so the L-curvature is $k_{L^2} = 7/512$; in the isotropy chart the smooth quadratic is $x^2 + y^2$.*

Why** $k_{\text{gen2}} = -\varphi/2$**. *The renormalization step is $u = (\delta, 1)$ with $\delta = \log_{B^\star}\varphi$. In the isotropy chart the five rotated copies $\{\hat{u}_k\}$ form a regular pentagon. Two independent, local constraints — (i) equal average curvature over the five directions (discrete isotropy), and (ii) equal edge energy between consecutive edges (discrete stationarity) — uniquely pin the coefficient of g^2 to $k_{\text{gen2}} = -\cos(\pi/5) = -\varphi/2$.*

Why** $k_{\text{gen}} = \pi/2$ **(historical argument — superseded). *The quarter-turn gauge fixes the unit of "one generation" to be a right-angle phase advance in the isotropy chart. At the L-center the phase rate equals k_{gen}, hence $k_{\text{gen}} = \pi/2$. Any rescaling of g would also rescale k_{gen2}, already fixed; so the choice is unique. This argument is superseded by the Lean-certified Quarter-Lock substitution derivation giving $k_{\text{gen}} = \varphi\cos(\pi/10)$ (*`thm_ucl2_fully_unconditional`*); it is retained here for historical context only.*

Why the center is** $\log_{B^\star}\varphi$ **up to a bounded mirror-odd shift. *The even-step ratio update $r \mapsto 1 + 1/r$ is a contraction with fixed point φ. Ridge/latched effects are bounded, mirror-odd "kicks," so the stationary center is $L^\star = \log_{B^\star}\varphi + \kappa$ with bounded κ that cancels under mirror balance.*

Bottom line. *After choosing the natural base and chart, the renormalization pentagon forces $k_{\text{gen2}} = -\varphi/2$; the quarter-turn gauge fixes $k_{\text{gen}} = \pi/2$; and the center tracks the Fibonacci eigen-slope. The smooth block is rigid and matches the elegant-kernel palette.*

26 Quarter–Lock from Symmetry

Target statement (Quarter–Lock identity). In the elegant–kernel chart (base $B^\star$ and L centered) the Möbius–product coefficient obeys the exact relation

$$\boxed{k_M \;=\; k_{\text{gen2}} \;+\; \frac{1}{4}\,k_{L^2}\,.}$$

Setup and notation

The (centered) logarithmic archimedean factor of the Universal Calibration Law is

$$\log F_\infty(L, g) \;=\; k_0' \;+\; k_{L^2}\,L^2 \;+\; k_{\text{gen}}\,g \;+\; k_{\text{gen2}}\,g^2,$$

with $k_{L^2} = \frac{7}{512}$ and k_{gen2} fixed by Conjecture C, and the full (log) law is

$$\log C_f(L, g; M, \mu_a, \mu_b, \mu_c) \;=\; \log F_\infty(L, g) \;+\; k_M M \;+\; k_a\mu(a) + k_b\mu(b) + k_c\mu(|c|).$$

Here $M = \mu(a)\mu(b)\mu(|c|) \in \{-1, 0, +1\}$ is the Möbius product channel; $L = \log_{B^\star}(|b|/|c|)$ is centered (so the linear L term is absorbed into k_0').

We work repeatedly in the *isotropy chart* of the smooth block:

$$x \;=\; \sqrt{k_{L^2}}\, L, \qquad y \;=\; \sqrt{|k_{\text{gen2}}|}\, g,$$

so that the quadratic smooth part is $x^2 + y^2$ (plus a linear y tilt determined by k_{gen}, which will drop out of our curvature–only arguments).

B.1 Mirror invariance and Fisher block orthogonality

Definition 26.1 (Mirror involution)**.** *Define the mirror map*

$$\mathsf{Mir}:\; (L, g; M, \mu_a, \mu_b, \mu_c) \;\longmapsto\; (-L,\, g;\, -M,\, -\mu_a,\, -\mu_b,\, -\mu_c).$$

A probability measure $\mathbb{P}$ on the parameter/data space is mirror–balanced *if $\mathbb{P} \circ \mathsf{Mir}^{-1} = \mathbb{P}$ and $\mathbb{E}_{\mathbb{P}}[L] = 0$.*

Lemma 26.2 (Vanishing cross moments)**.** *Under a mirror–balanced measure, for any non-negative integers m, n we have*

$$\mathbb{E}[\, L^m g^n\, M \,] = 0, \qquad \mathbb{E}[\, L^m g^n\, \mu(\cdot) \,] = 0 \quad \big(\mu(\cdot) \in \{\mu(a), \mu(b), \mu(|c|)\}\big).$$

Proof. Under Mir, $L \mapsto -L$, $g \mapsto g$, $M \mapsto -M$ and $\mu(\cdot) \mapsto -\mu(\cdot)$. Hence $L^m g^n M$ and $L^m g^n \mu(\cdot)$ are mirror–odd. Taking expectations under the invariant measure gives zero. □

Proposition 26.3 (Fisher block orthogonality)**.** *Let $\theta = (k_{L^2}, k_{\text{gen}}, k_{\text{gen2}}, k_M, k_a, k_b, k_c)$ and suppose the data likelihood depends on θ only via $\log C_f$. Then, under any mirror–balanced measure, the Fisher information matrix*

$$\mathcal{I}_{ij} = \mathbb{E}\big[(\partial_{\theta_i} \log C_f)(\partial_{\theta_j} \log C_f)\big]$$

is block–diagonal between the smooth block $\{k_{L^2}, k_{\text{gen}}, k_{\text{gen2}}\}$ and the parity block $\{k_M, k_a, k_b, k_c\}$.

Proof. The "scores" are $\partial_{k_{L^2}} \log C_f = L^2$, $\partial_{k_{\text{gen}}} \log C_f = g$, $\partial_{k_{\text{gen2}}} \log C_f = g^2$, and $\partial_{k_M} \log C_f = M$, $\partial_{k_a} \log C_f = \mu(a)$, etc. Each cross–block entry is an expectation of the form $\mathbb{E}[L^m g^n \cdot \text{(parity)}]$ with $(m, n) \in \{(2, 0), (0, 1), (0, 2)\}$. These vanish by Lemma 26.2. □

Consequence. Any algebraic relation involving k_M and the smooth coefficients cannot be an artefact of statistical mixing: it must follow from independent symmetry/geometry constraints. We now isolate and solve that constraint.

B.2 Mirror–balanced discrete curvature functional $\mathcal{D}$

We compare the *smooth quadratic curvature* of $\log C_f$ around the origin along the four cardinal directions in (L, g) with the *parity skew* across $M = \pm 1$ at the base point. The comparison is encoded in a linear functional $\mathcal{D}$ whose weights are uniquely fixed by symmetry axioms.

Definition 26.4 (Four–point stencil & parity skew)**.** *Fix a step $\Delta > 0$ in the isotropy chart (x, y) and define the four cardinal points*

$$z_1 = (+\Delta, 0), \quad z_2 = (-\Delta, 0), \quad z_3 = (0, +\Delta), \quad z_4 = (0, -\Delta),$$

and the M–fiber base points $\zeta_+ = (0, 0, M = +1)$, $\zeta_- = (0, 0, M = -1)$. For any function F (of (L, g, M, μ)) set

$$\mathcal{D}[F] \;:=\; \sum_{j=1}^{4} w_j\, F(z_j) \;-\; \sum_{\sigma \in \{+,-\}} u_\sigma\, F(\zeta_\sigma),$$

with weights w_j, u_σ uniquely fixed by the symmetry axioms below (Lemma 26.5).

Axioms (Mirror–balance axioms for $\mathcal{D}$)**.**

(A1) ***Annihilation of affine functions.*** $\mathcal{D}[1] = \mathcal{D}[L] = \mathcal{D}[g] = 0$.

(A2) ***Isotropy of smooth curvature.*** $\mathcal{D}[L^2] = \mathcal{D}[g^2]$.

(A3) ***Mirror–odd normalization.*** $\mathcal{D}[M] = -1$ *and* $\mathcal{D}[\mu(a)] = \mathcal{D}[\mu(b)] = \mathcal{D}[\mu(|c|)] = 0$.

(A4) ***Scale covariance.*** *Under $(L, g) \mapsto (\lambda L, \lambda g)$ with $\Delta \mapsto \lambda\Delta$, $\mathcal{D}$ scales quadratically: $\mathcal{D}[Q] \mapsto \lambda^2\, \mathcal{D}[Q]$ for any quadratic Q.*

Lemma 26.5 (Uniqueness and weight solution)**.** *Under Axioms 26, the weights are uniquely determined (up to the overall scale fixed by $\mathcal{D}[M] = -1$) as*

$$w_1 = w_2 = w_3 = w_4 = \frac{1}{4}, \qquad u_+ = 1, \qquad u_- = 0.$$

Proof. (A1) on constants gives $\sum_j w_j - (u_+ + u_-) = 0$. (A1) on L yields $w_1\Delta + w_2(-\Delta) = 0$ and similarly $w_3 = w_4$ from g; hence $w_1 = w_2$ and $w_3 = w_4$. (A2) then forces $w_1 + w_2 = w_3 + w_4$, so by symmetry $w_1 = w_2 = w_3 = w_4 =: w$. The constant condition becomes $4w - (u_+ + u_-) = 0$. Finally (A3) gives $-u_+ + u_- = -1$. Solving yields $w = \frac{1}{4}$, $u_+ = 1$, $u_- = 0$. □

Proposition 26.6 (Action of $\mathcal{D}$ on the quadratic & parity palette)**.** *With the weights of Lemma 26.5:*

$$\mathcal{D}[L^2] = \frac{\Delta^2}{2}, \qquad \mathcal{D}[g^2] = \frac{\Delta^2}{2}, \qquad \mathcal{D}[M] = -1,$$

and $\mathcal{D}$ annihilates all affine combinations of $\{1, L, g\}$ and each linear Möbius channel $\mu(\cdot)$.

Proof. L^2 evaluates to Δ^2 at z_1, z_2 and 0 at z_3, z_4, so $\sum_j w_j L^2(z_j) = \frac{1}{4}(\Delta^2 + \Delta^2) = \frac{\Delta^2}{2}$; the parity term contributes zero. The g^2 case is analogous. For M the smooth stencil is zero and $-u_+ + u_- = -1$. The annihilation statements follow from (A1) and (A3). □

Lemma 26.7 (Mirror–balance identity)**.** *For the locked law* $\log C_f$ *(centered* L*) and the mirror–balanced configuration at the origin, one has*

$$\mathcal{D}[\log C_f] \;=\; 0.$$

Proof. Under the mirror involution, $(L, g; M, \mu) \mapsto (-L, g; -M, -\mu)$, the four smooth samples z_j are pairwise interchanged and the parity samples $\zeta_\pm$ are swapped. The weights in Lemma 26.5 make $\mathcal{D}$ mirror–odd: $\mathcal{D}[F \circ \mathsf{Mir}] = -\mathcal{D}[F]$ whenever F is mirror–even in the smooth block (true for $\log F_\infty$) and affine–linear in M in the parity block (true for the Möbius palette). Since the locked law is mirror–invariant as a whole (balanced run and centered origin), $\mathcal{D}[\log C_f] = \mathcal{D}[\log C_f \circ \mathsf{Mir}] = -\mathcal{D}[\log C_f]$, hence it must vanish. □

Combining. Applying Proposition 26.6 to $\log C_f$ yields

$$\mathcal{D}[\log C_f] \;=\; \frac{\Delta^2}{2}\,(k_{L^2} + k_{\mathrm{gen2}}) \;-\; k_M. \tag{26.1}$$

By Lemma 26.7, the left side is 0; the remaining step is to determine the *geometric value* of $\frac{\Delta^2}{2}$.

B.3 Quarter–cycle normalization: fixing $\frac{\Delta^2}{2} = \frac{1}{4}$

Recall the isotropy chart $(x, y) = (\sqrt{k_{L^2}}L, \sqrt{|k_{\mathrm{gen2}}|}g)$, in which the smooth quadratic is $x^2 + y^2$. Conjecture C fixes the *quarter–turn gauge*: one *generation* unit advances the polar angle by $\pi/2$.

Lemma 26.8 (Discrete curvature on the unit quarter–cycle)**.** *Let* $Q(x, y) = x^2 + y^2$*. On the circle of radius* r *centered at the origin, the four cardinal points* $(\pm r, 0)$ *and* $(0, \pm r)$ *satisfy*

$$\frac{Q(r,0) + Q(-r,0) + Q(0,r) + Q(0,-r)}{4} \;=\; r^2.$$

Proof. Each value is r^2, so the average is r^2. □

Lemma 26.9 (Unit–curvature stencil calibration)**.** *In the isotropy chart* (x, y) *where the smooth quadratic is* $x^2 + y^2$*, fix the stencil step* Δ *by*

$$\mathcal{D}[x^2] = \mathcal{D}[y^2] = \frac{1}{4}.$$

Equivalently, $\mathcal{D}[x^2 + y^2] = 1$ *and hence* $\frac{\Delta^2}{2} = \frac{1}{4}$.

Proof. With the weights of Lemma 26.5, $\mathcal{D}[x^2] = \frac{1}{4}\big(x^2(+\Delta, 0) + x^2(-\Delta, 0)\big) = \frac{\Delta^2}{2}$ and similarly $\mathcal{D}[y^2] = \frac{\Delta^2}{2}$. Setting each to 1/4 yields $\Delta^2/2 = 1/4$. Then $\mathcal{D}[x^2 + y^2] = 1/2 + 1/2 = 1$. □

Remark 26.10 (On the unit choice and compatibility with Conjecture C)**.** *In the isotropy chart the smooth quadratic is Euclidean. The choice* $r^2 = 1/2$ *is precisely the one for which the quarter–turn* $(\pi/2)$ *phase advance at unit* g*–speed matches* $k_{\mathrm{gen}} = \pi/2$*, ensuring that the geodesic phase and the* g*–units are aligned. Any alternative* g*–rescaling would simultaneously rescale* k_{gen2}*, contradicting its fixed (algebraic) value from Conjecture C.*

B.4 The Quarter–Lock identity

Theorem 26.11 (Quarter–Lock)**.** *Under mirror balance and the quarter–turn gauge,*

$$k_M \;=\; k_{\text{gen2}} \;+\; \frac{1}{4}\, k_{L^2}.$$

Proof. From (26.1) and Lemma 26.9,

$$0 \;=\; \mathcal{D}[\log C_f] \;=\; \underbrace{\frac{\Delta^2}{2}}_{=1/4}\,(k_{L^2} + k_{\text{gen2}}) \;-\; k_M,$$

so $k_M = (k_{L^2} + k_{\text{gen2}})/4$. Rearranging gives the stated identity. □

B.5 Uniqueness of $\mathcal{D}$ and rigidity

Proposition 26.12 (Uniqueness of the mirror–balance functional)**.** *Within the class of six–weight functionals*

$$\mathcal{D}[F] = \sum_{j=1}^{4} w_j F(z_j) - \sum_{\sigma\in\{+,-\}} u_\sigma F(\zeta_\sigma)$$

acting on the four cardinal smooth samples and the two parity samples, Axioms 26 determine the weights uniquely (up to the sign/scale fixed by $\mathcal{D}[M] = -1$). In particular, the $1/4$ factor in Theorem 26.11 is canonical.

Proof. The linear constraints from (A1)–(A3) determine (w_j, u_σ) up to a common scale; solving yields Lemma 26.5. The sign/scale is fixed by $\mathcal{D}[M] = -1$. No other choice satisfies all axioms simultaneously. □

Remark 26.13 (Rigidity under palette snaps)**.** *Replacing the decimal palette by the elegant–kernel algebraics from Conjecture C ($k_{L^2} = 7/512$, $k_{\text{gen2}} = -\varphi/2$) leaves Theorem 26.11 intact and fixes*

$$k_M \;=\; -\frac{\varphi}{2} + \frac{7}{2048},$$

in exact agreement with the palette identity.

B.6 Summary

We have established:

- Fisher block orthogonality between the smooth (L, g) block and the parity block (M, μ_a, μ_b, μ_c) (Proposition 26.3);
- a uniquely determined mirror–balanced curvature functional $\mathcal{D}$ (Lemmas 26.5, 26.7, Proposition 26.6);

- quarter–cycle normalization $\frac{\Delta^2}{2} = \frac{1}{4}$ from the isotropy geometry and the quarter–turn gauge (Lemma 26.9);
- the Quarter–Lock identity $k_M = k_{\text{gen2}} + \frac{1}{4} k_{L^2}$ (Theorem 26.11).

Together these constitute a first–principles derivation of the Quarter–Lock relation as a theorem of the symmetry geometry of the elegant–kernel chart.

Remark (Plain-English commentary for Quarter–Lock). ***What we measured and how.*** *We compare* smooth *quadratic curvature of* $\log C_f$ *along the four cardinal directions in the isotropy chart with the* mirror-odd *skew of the parity fiber* $M = \pm 1$ *at the origin using a single linear functional* $\mathcal{D}$ *whose weights are fixed uniquely by four axioms (annihilate affine functions, isotropy, mirror-odd normalization, quadratic scaling).*

Key fixes that make it airtight. *Weights must be* $w_j = \frac{1}{4}$ *on smooth samples and* $(u_+, u_-) = (1, 0)$ *on parity samples; this gives* $\mathcal{D}[1] = 0$ *and* $\mathcal{D}[M] = -1$*. The stencil step* Δ *is calibrated by* $\mathcal{D}[x^2] = \mathcal{D}[y^2] = \frac{1}{4}$*, i.e., unit curvature per axis — a chart normalization compatible with the quarter-turn gauge (*$k_{\text{gen}} = \pi/2$*).*

The identity then follows. *With these choices,* $\mathcal{D}[\log C_f] = \frac{\Delta^2}{2}(k_{L^2} + k_{\text{gen2}}) - k_M$ *and the unit-curvature calibration gives* $\Delta^2/2 = 1/4$*. Mirror balance forces* $\mathcal{D}[\log C_f] = 0$*, hence* $k_M = k_{\text{gen2}} + \frac{1}{4} k_{L^2}$*.*

Why this is structural. *Fisher block orthogonality shows no statistical cross-talk can fake the relation. The functional* $\mathcal{D}$ *is unique; the factor* $\frac{1}{4}$ *is canonical, not tuned. The identity is therefore a consequence of symmetry and geometry.*

Bottom line. *With the corrected weights and unit-curvature calibration, Quarter–Lock is a* theorem*:* $k_M = k_{\text{gen2}} + \frac{1}{4} k_{L^2}$*.*

27 Euler–product, local–to–global

Target statement (adelic factorization). There exists a place–by–place factorization

$$\boxed{C_f(a,b,c;g) \;=\; F_\infty(L,g) \int_\Omega \tilde{\nu}(d\omega) \prod_{p<\infty} F_p\big(\omega; \sigma_p(a,b,c)\big)}$$

in which:

1. F_∞ is the *archimedean* factor that carries the entire smooth (L,g)–block,

$$\log F_\infty(L,g) = k_0' + k_{L^2}\,(L - L^\star)^2 + k_{\text{gen}}\, g + k_{\text{gen2}}\, g^2,$$

   with the elegant–kernel constants from Section 25:

$$k_{L^2} = \frac{7}{512}, \qquad k_{\text{gen}} = \frac{\pi}{2}, \qquad k_{\text{gen2}} = -\frac{\varphi}{2}.$$

2. Each *finite* local factor F_p depends only on the primewise signature

$$\sigma_p(a,b,c) = \big(v_p(a), v_p(b), v_p(|c|)\big)$$

and collectively realizes the entire Möbius/parity block *multiplicatively*:

$$\prod_{p<\infty} F_p(\sigma_p) \;=\; \exp\Big(k_M\, M + k_a\,\mu(a) + k_b\,\mu(b) + k_c\,\mu(|c|)\Big),$$

where $\mu(\cdot)$ is the Möbius function and $M := \mu(a)\mu(b)\mu(|c|)$.

Notation and local signatures

For an integer $n \geq 1$, let $v_p(n)$ be the p–adic valuation. For each prime p define the three–state local Möbius digit

$$\mu_p(n) \in \{\,1, -1, 0\,\}, \qquad \mu_p(n) = \begin{cases} 1, & v_p(n) = 0, \\ -1, & v_p(n) = 1, \\ 0, & v_p(n) \geq 2. \end{cases}$$

Then $\mu(n) = \prod_p \mu_p(n)$ is the global Möbius value: 0 if any $v_p(n) \geq 2$; otherwise $\mu(n) = (-1)^{\omega(n)}$ with $\omega(n) = \#\{p : v_p(n) = 1\}$.

Write the *squarefree projector*

$$S(n) \;:=\; \prod_p \mathbf{1}_{\{v_p(n)\leq 1\}} \;\in \{0,1\}, \quad \text{so that} \quad S(n) = 1 \;\Leftrightarrow\; n \text{ squarefree}, \quad S(n)\,\mu(n) = \mu(n).$$

For a triple (a, b, c), set $c^\circ := |c|$ and abbreviate

$$\mu_a := \mu(a), \quad \mu_b := \mu(b), \quad \mu_c := \mu(c^\circ), \quad M := \mu_a\,\mu_b\,\mu_c.$$

The discrete Möbius–block of the UCL is

$$\mathcal{M}(a, b, c) \;:=\; \exp\big(k_a\mu_a + k_b\mu_b + k_c\mu_c + k_M M\big).$$

A.1 A basic identity for the Möbius exponential

Lemma 27.1 (Squarefree–gated decomposition)**.** *For any $k \in \mathbb{R}$ and any $n \geq 1$,*

$$\exp\big(k\,\mu(n)\big) \;=\; 1 \;+\; S(n)\,\Big[(\cosh k - 1) \;+\; \mu(n)\,\sinh k\Big].$$

Proof. If n is not squarefree, then $S(n) = 0$ and $\mu(n) = 0$, so the RHS equals 1, which is the LHS since $e^{k\cdot 0} = 1$. If n is squarefree, then $S(n) = 1$ and $\mu(n) \in \{\pm 1\}$. Then the RHS equals $1 + (\cosh k - 1) + \mu(n)\sinh k = \cosh k + \mu(n)\sinh k = e^{k\mu(n)}$, as required. $\square$

A.2 Prime–local realization via discrete archimedean integration

We show that each factor $\exp(k_x\mu_x)$ ($x \in \{a,b,c\}$) and the mixed factor $\exp(k_M M)$ admit an *exact* representation as an *integral of an Euler product* whose local factors depend only on p–local signatures. The integral is over a *finite discrete archimedean measure*, so no convergence subtleties arise.

Single–channel local factors. Fix $x \in \{a,b,c\}$ and a prime p. For $\theta \in \{0,\pi\}$, define

$$U_p^{(x)}(\theta) \;:=\; \begin{cases} 0, & v_p(x) \ge 2, \\ 1, & v_p(x) = 0, \\ e^{i\theta}, & v_p(x) = 1. \end{cases}$$

Then $U_p^{(x)}(0) = \mathbf{1}_{\{v_p(x)\le 1\}}$ and $U_p^{(x)}(\pi) = (-1)^{\mathbf{1}_{\{v_p(x)=1\}}}$; hence

$$\prod_p U_p^{(x)}(0) = S(x), \qquad \prod_p U_p^{(x)}(\pi) = S(x)\,(-1)^{\omega(x)} = S(x)\,\mu(x).$$

Introduce the *extended* two–point measure on $\{\perp, 0, \pi\}$:

$$\tilde{\nu}_x \;:=\; \delta_\perp \;+\; (\cosh k_x - 1)\,\delta_0 \;+\; (\sinh k_x)\,\delta_\pi,$$

Sign convention. For $k_x < 0$ one has $\sinh k_x < 0$, hence $\tilde{\nu}_x$ is a *finite signed* (not necessarily positive) measure. All integrals in this section are finite linear combinations of $\cosh k_x$ and $\sinh k_x$ and remain well–defined for all $k_x \in \mathbb{R}$. and extend the local factor by $U_p^{(x)}(\perp) \equiv 1$ (neutral element).

Proposition 27.2 (Exact single–channel factorization)**.** *For each $x \in \{a,b,c\}$,*

$$\exp\big(k_x\mu(x)\big) \;=\; \int_{\{\perp,0,\pi\}} \tilde{\nu}_x(d\theta_x)\;\prod_p U_p^{(x)}(\theta_x).$$

Proof. Expanding the integral gives three contributions:

$$\underbrace{\prod_p U_p^{(x)}(\perp)}_{=1} \;+\; (\cosh k_x - 1)\underbrace{\prod_p U_p^{(x)}(0)}_{=S(x)} \;+\; (\sinh k_x)\underbrace{\prod_p U_p^{(x)}(\pi)}_{=S(x)\mu(x)}.$$

By Lemma 27.1 with $n = x$, the sum equals $e^{k_x\mu(x)}$. □

Three–body M–channel. For p and $\theta \in \{0,\pi\}$ define

$$U_p^{(M)}(\theta) \;:=\; \begin{cases} 0, & \text{some } v_p(a) \ge 2 \text{ or } v_p(b) \ge 2 \text{ or } v_p(c^\circ) \ge 2, \\ e^{\,i\theta\,(v_p(a)+v_p(b)+v_p(c^\circ))}, & \text{otherwise.} \end{cases}$$

Then $\prod_p U_p^{(M)}(0) = S(a)S(b)S(c^\circ)$ and

$$\prod_p U_p^{(M)}(\pi) = S(a)S(b)S(c^\circ)\,(-1)^{\omega(a)+\omega(b)+\omega(c^\circ)} = S(a)S(b)S(c^\circ)\,M.$$

Let $\tilde{\nu}_M := \delta_\perp + (\cosh k_M - 1)\delta_0 + (\sinh k_M)\delta_\pi$ and $U_p^{(M)}(\perp) \equiv 1$.

Proposition 27.3 (Exact mixed–channel factorization)**.**

$$\exp\big(k_M\, M\big) \;=\; \int_{\{\perp,0,\pi\}} \tilde{\nu}_M(d\theta_M)\; \prod_p U_p^{(M)}(\theta_M).$$

Proof. Identical to Proposition 27.2, replacing the single–channel squarefree projector $S(x)$ by $S(a)S(b)S(c^\circ)$ and $\mu(x)$ by M. $\square$

Putting the channels together. Define the product measure on $\Omega := (\{\perp, 0, \pi\})^4$,

$$\tilde{\nu} := \tilde{\nu}_a \otimes \tilde{\nu}_b \otimes \tilde{\nu}_c \otimes \tilde{\nu}_M.$$

For $\omega = (\theta_a, \theta_b, \theta_c, \theta_M) \in \Omega$, define the *prime–local* factor

$$F_p(\omega;\, a, b, c^\circ) \;:=\; U_p^{(a)}(\theta_a)\, U_p^{(b)}(\theta_b)\, U_p^{(c)}(\theta_c)\, U_p^{(M)}(\theta_M).$$

Theorem 27.4 (Integrated Euler product for the Möbius block)**.** *For every triple* (a, b, c)*,*

$$\boxed{\mathcal{M}(a,b,c) \;=\; \int_\Omega \tilde{\nu}(d\omega)\; \prod_p F_p(\omega;\, a, b, c^\circ)\,.}$$

Proof. By Propositions 27.2 and 27.3,

$$\mathcal{M}(a,b,c) = \Big(\int \tilde{\nu}_a \prod_p U_p^{(a)}\Big)\Big(\int \tilde{\nu}_b \prod_p U_p^{(b)}\Big)\Big(\int \tilde{\nu}_c \prod_p U_p^{(c)}\Big)\Big(\int \tilde{\nu}_M \prod_p U_p^{(M)}\Big).$$

Since all integrals are finite and independent, Fubini–Tonelli gives

$$\int_\Omega \big(\tilde{\nu}_a \otimes \tilde{\nu}_b \otimes \tilde{\nu}_c \otimes \tilde{\nu}_M\big)(d\omega)\; \prod_p \Big(U_p^{(a)}(\theta_a)\, U_p^{(b)}(\theta_b)\, U_p^{(c)}(\theta_c)\, U_p^{(M)}(\theta_M)\Big),$$

which is the claimed formula. $\square$

Remark 27.5 (Locality, finiteness, multiplicativity)**.** *For any fixed* (a, b, c)*,* $F_p(\omega; a, b, c^\circ) = 1$ *for all but finitely many* p *(those dividing* abc°*); thus the Euler product is a* finite product*. Each* F_p *depends* only *on the* p*–local signature* $\sigma_p(a, b, c^\circ) = (v_p(a), v_p(b), v_p(c^\circ))$*. The archimedean part is the finite signed measure* $\tilde{\nu}$*, which plays the role of an* F_∞*–type factor for the discrete block. Hence the whole Möbius block is realized as an* adelic *(archimedean* $\times$ *finite–prime) object.*

A.3 Parameter matching and uniqueness

Lemma 27.6 (Two–equation identification). *Let $W(+1), W(-1)$ be real weights and $\omega \in \mathbb{N}$. Then the identities*

$$1 + W(+1) + W(-1) = e^{k} \quad (\omega \text{ even}), \qquad 1 + W(+1) - W(-1) = e^{-k} \quad (\omega \text{ odd})$$

are solved uniquely by

$$W(+1) = \cosh k - 1, \qquad W(-1) = \sinh k.$$

Proof. Add and subtract to get $W(+1) + W(-1) = e^{k} - 1$, $W(+1) - W(-1) = e^{-k} - 1$; solve to obtain the stated expressions. □

Proposition 27.7 (Channel weights and k–parameters). *In Theorem 27.4, the discrete archimedean measure is uniquely fixed by*

$$\tilde{\nu}_x = \delta_\perp + (\cosh k_x - 1)\delta_0 + (\sinh k_x)\delta_\pi \qquad (x \in \{a, b, c, M\}).$$

Proof. By Lemma 27.1 and the construction of $U_p^{(x)}(0), U_p^{(x)}(\pi)$, the desired values on the squarefree slice constrain the two weights at $\{0, \pi\}$ uniquely by Lemma 27.6. The mass at $\perp$ must be 1 to reproduce the constant term 1. □

Remark 27.8 (Uniqueness modulo trivial gauge). *Any alternative representation with point masses $\{\perp, 0, \pi\}$ that reproduces exactly $\exp(k\,\mu(\cdot))$ for all inputs must coincide with the above, because the three test inputs "non–squarefree", "squarefree even ω", "squarefree odd ω" form a complete system constraining the three weights. Thus the archimedean identification of the channel parameters is unique.*

A.4 Analyticity, finiteness and separation from the smooth block

Proposition 27.9 (Finiteness and analyticity). *For any fixed (a, b, c) the product in Theorem 27.4 is finite (only primes dividing abc° contribute). The map $(k_a, k_b, k_c, k_M) \mapsto \mathcal{M}(a, b, c)$ is real–analytic, being a finite linear combination of* $\cosh$ *and* $\sinh$ *polynomials with integer coefficients determined by the local signatures.*

Proof. If $p \nmid abc^\circ$, then $v_p(a) = v_p(b) = v_p(c^\circ) = 0$ and hence $F_p(\omega; \cdot) = 1$ for all ω. Thus only finitely many p appear. Analyticity follows from the explicit weights $\cosh k_x - 1$ and $\sinh k_x$, combined multilinearly. □

Proposition 27.10 (Separation of finite vs archimedean smooth parts). *No choice of* finite–prime *local factors depending only on $\sigma_p(a, b, c^\circ)$ can generate a nonzero L^2 or g^2 contribution. Hence the smooth block*

$$F_\infty^{\text{smooth}}(L, g) = \exp\bigl(k_0' + k_{L^2} L^2 + k_{\text{gen}} g + k_{\text{gen}2} g^2\bigr)$$

must be entirely archimedean and multiplicatively independent from the discrete Möbius block.

Proof. Any finite–prime local factor $F_p(\sigma_p)$ does not depend on $L = \log(|b|/|c|)$ nor on g. Therefore a finite product $\prod_p F_p(\sigma_p)$ cannot produce any nontrivial function of L or g. Consequently the (L, g)–quadratic terms must reside in an archimedean factor. □

A.5 Optional residue refinements via local characters

The formalism admits residue–sensitive refinements while preserving finiteness.

Definition 27.11 (Local residue character)**.** *Fix a finite set of primes $\mathcal{P}$. For $p \in \mathcal{P}$ choose a (real) quadratic Dirichlet–type local character χ_p on $(\mathbb{Z}/p\mathbb{Z})^\times$, extended by $\chi_p(0) = 0$. Define the residue factor at p (for the c°–channel) by*

$$R_p(\gamma_p) \;:=\; 1 \;+\; \gamma_p\, \chi_p(c^\circ)\, \mathbf{1}_{\{v_p(c^\circ)=0\}}.$$

Proposition 27.12 (Finite residue dressing)**.** *Let $R(\gamma) := \prod_{p\in\mathcal{P}} R_p(\gamma_p)$. Then:*

1. *$R(\gamma)$ depends only on $c^\circ \bmod \prod_{p\in\mathcal{P}} p$ and equals 1 for all but finitely many p;*
2. *$\mathcal{M}(a, b, c)$ can be replaced by $R(\gamma)\, \mathcal{M}(a, b, c)$ without affecting finiteness or analyticity in (k_a, k_b, k_c, k_M).*

Proof. (1) Each R_p is 1 unless $p \in \mathcal{P}$ and $v_p(c^\circ) = 0$, in which case R_p is $1 + \gamma_p\chi_p(c^\circ \bmod p)$; thus the product is finite and depends only on the specified residues. (2) Multiplying by $R(\gamma)$ adds a finite multiplicative twist independent of the k–parameters, preserving the properties asserted in Proposition 27.9. □

A.6 Synthesis with the full UCL

Let

$$F_\infty^{\text{smooth}}(L, g) = \exp\bigl(k'_0 + k_{L^2}L^2 + k_{\text{gen}}g + k_{\text{gen2}}g^2\bigr),$$

as fixed by Section 25, and let $\mathcal{M}(a, b, c)$ be given by Theorem 27.4. Optionally include a residue dressing $R(\gamma)$ as in Proposition 27.12. Then the *complete* calibration factor admits the adelic factorization

$$\boxed{C_f(a, b, c; g) \;=\; F_\infty^{\text{smooth}}(L, g) \;\times\; \int_\Omega \tilde{\nu}(d\omega)\, \prod_p F_p(\omega;\, a, b, c^\circ) \;\times\; R(\gamma)\,.}$$

Here:

- F_∞^{smooth} carries all (L, g)–dependence and is archimedean;
- the integrated Euler product $\int \prod_p F_p$ reproduces *exactly* the full Möbius block $\exp(k_a\mu_a + k_b\mu_b + k_c\mu_c + k_M M)$ with primewise local factors depending only on $\sigma_p(a, b, c^\circ)$;
- $R(\gamma)$ is an optional finite twist capturing residue information when needed (and is $= 1$ in the elegant kernel).

A.7 No–go lemma for a *pure* Euler product without integration

Lemma 27.13 (No–go for global parity without an archimedean coupling)**.** *There do not exist (nontrivial) local functions $\Phi_p : \{0, 1, 2, \ldots\} \to \mathbb{R}_{>0}$ such that for all n,*

$$\exp\big(k\,\mu(n)\big) \;=\; \prod_p \Phi_p\big(v_p(n)\big).$$

Proof. Assume such Φ_p exist. Evaluate at $n = 1$ to get $1 = \prod_p \Phi_p(0)$, hence $\Phi_p(0) = 1$ for all p (by positivity). For $n = p$ prime, $e^{-k} = \Phi_p(1)$ (since all other primes contribute $\Phi_q(0) = 1$). For $n = pq$ with distinct primes, $e^{+k} = \Phi_p(1)\Phi_q(1) = e^{-2k}$, impossible unless $k = 0$ or $\Phi_p(1) = 1$ for all p (trivial). Thus no nontrivial solution exists: a *global* coupling is unavoidable. □

Remark 27.14. *Lemma 27.13 motivates our construction: the necessary global coupling of the squarefree parity is introduced through a finite, discrete archimedean integration (the $\{\perp, 0, \pi\}$ measures), which meshes naturally with the F_∞ layer of the adelic factorization.*

A.8 Conclusion

We have supplied full, rigorous proofs that:

- Each Möbius channel $\exp(k_x\mu(x))$ and the mixed channel $\exp(k_M M)$ admit an *exact* representation as a finite, discrete archimedean integral of an *Euler product* with prime–local factors depending solely on the local signatures $(v_p(a), v_p(b), v_p(c^\circ))$ (Propositions 27.2, 27.3; Theorem 27.4).
- The discrete archimedean measure is uniquely identified from the k–parameters as $\delta_\perp + (\cosh k - 1)\delta_0 + (\sinh k)\delta_\pi$ (Proposition 27.7).
- The Euler product is finite and analytic for fixed (a, b, c), and the smooth (L, g) block is necessarily and cleanly separated as a pure archimedean factor (Propositions 27.9, 27.10).
- Optional residue–class refinements fit compatibly as finite local dressings (Proposition 27.12), and a pure finite–prime decomposition without archimedean integration is impossible (Lemma 27.13).

Therefore the Universal Calibration Law admits the claimed adelic (archimedean × finite–prime) factorization, establishing Conjecture A in full detail.

Remark (Plain-English commentary for Euler–product factorization)**.** ***What we proved.*** *The calibration factor splits into an* archimedean *smooth piece in (L, g) and a* finite, primewise *product that reproduces exactly the discrete Möbius channels — but crucially, the primewise part is integrated against a* finite signed *archimedean palette $\tilde{\nu}$ (a three-point measure $\{\perp, 0, \pi\}$ per channel). This integral-of-a-product is necessary and sufficient to encode the global squarefree parity $\mu(\cdot)$ and the mixed channel $M = \mu(a)\mu(b)\mu(|c|)$.*

Why it is necessary. *The No–Go Lemma shows a pure product* $\prod_p \Phi_p(v_p(n))$ *cannot equal* $\exp(k\,\mu(n))$ *unless it is trivial (*$k{=}0$*). The obstruction is global:* $\mu(n)$ *couples parities across* all *primes. The discrete archimedean integral supplies exactly that global coupling while keeping each* F_p *prime-local.*

Why it is sufficient. *On the squarefree slice the integral collapses to finite linear combinations of* $\cosh k$ *and* $\sinh k$ *with coefficients given by monomials in the local signs* $\mu_p(\cdot) \in \{\pm 1\}$*; off the squarefree slice it vanishes by design. The three test cases (non-squarefree, squarefree-even, squarefree-odd) fix the weights uniquely, so the representation is exact for all inputs.*

Analytic and structural sanity checks. *For fixed* (a, b, c)*, only finitely many primes contribute, so there are no convergence issues.* $(k_a, k_b, k_c, k_M) \mapsto \mathcal{M}$ *is a finite combination of* $\cosh / \sinh$*, hence analytic. No finite-prime product can create* L^2 *or* g^2*, so the smooth block is necessarily archimedean.*

Bottom line. *The adelic (archimedean* $\times$ *finite) architecture is both* necessary *(by the No–Go Lemma) and* sufficient *(by the exact construction) to represent the UCL's Möbius-structured part while keeping* (L, g) *in the archimedean factor. This validates the Euler–product factorization.*

28 Conclusions

In conclusion, the UGP framework yields multiple families of breakthroughs: (i) Ridge arithmetic and canonical orbit selection; (ii) UWCA universality with its lawful/general dichotomy and reflexivity; (iii) Kernel symmetry and the Quarter–Lock invariants; (iv) Evolutionary prime dynamics; and (v) Topos-theoretic self-reference. Each family develops independently in the text, together establishing UGP as a structurally rich and computationally universal arithmetic substrate.

The Universal Generative Principle isolates a rigid, prime–locked ridge structure in a natural class of integer triple evolutions. At $n = 10$ this structure admits exactly one lex–minimal seed, with a single mirror dual, reproducing an observed three–generation trajectory. Our results are proved by elementary means, verified computationally, and situated in a survivor substrate capable of universal computation. Theorems and data are fully reproducible from the included scripts. Beyond the formal mathematics, we have sketched conjectural but testable correspondences to empirical physics invariants, intended to motivate further interdisciplinary exploration. Future work will address the distribution of prime–locked mirrors across n, refinements of the universality embedding, and rigorous analysis of the survivor dynamics.

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A Interpretive Framework: Connections to Self-Defining Systems

The mathematical structures developed in this paper—a Turing-universal cellular automaton (UWCA) on a prime-locked arithmetic substrate—admit a broader interpretation when viewed through the lens of Self-Defining Systems (SDS). An SDS is a system that is its own complete description, a property known as Perfect Self-Containment (PSC). This section briefly outlines how the UGP framework provides a concrete realization of such a system, leading to concepts like choice points and lawful irreducibility. This interpretation is not required for the preceding mathematical results but provides a powerful context for their significance.

A.1 Choice Points as a Necessary Consequence of PSC

While the UWCA micro-dynamics are deterministic, imposing global consistency constraints required by PSC forces the emergence of **choice points**—configurations where multiple future states are equally admissible under a global stability functional (dissonance). These arise from structural properties of any self-consistent system, including symmetry, gauge redundancy, and topological conservation. In the UWCA formalism, these degeneracies manifest as local tile overlaps.

A.2 Global Symmetry Adjudication (GSA) and the Transputational Operator

The resolution of choice points must be governed by an internal, lawful rule. We term this **Global Symmetry Adjudication (GSA)**. The mechanism studied in this monograph is the PR-1 operational law (Appendix K), an early candidate for the transputational substrate of the UGP. The formal definition of the transputational mechanism has since been provided by the Transputation operator (PT), whose forcing theorem and DSAC constructive realization are developed in the NEMS programme; see Appendix K for the historical context.

Definition A.1 (GSA and transputational adjudication (studied model))**.** *At a choice point, GSA selects a branch based on permitted internal perturbations to the stability functional. In the PR-1 model studied here, the sweep executes unique updates at non-degenerate points and invokes GSA at choice points. Permitted perturbations arise from:*

1. ***Substrate Spontaneity (Ξ):*** *A canonical tie-breaking measure induced by the substrate's intrinsic novelty.*

2. ***Subsystem Bias (B_A):*** *A bounded, lawful perturbation supplied by a coherent, transputational subsystem.*

The presence of a subsystem's lawful constraints (B_A) refines the choice manifold, taking precedence over the default spontaneity channel. This definition is the studied PR-1 model; the formal forcing theorem for the general transputational operator belongs to the NEMS programme.

A.3 Lawful Irreducibility ("Free Will")

The framework allows for a rigorous definition of freedom compatible with determinism.

Theorem A.2 (Lawful Irreducibility)**.** *The choices made by a transputational subsystem via its bias functional B_A are **computationally irreducible**. Due to the self-referential nature of PSC, there exists no internal PR-1-realizable predictor that can, in general, compute the subsystem's choice significantly faster than running the subsystem's own irreducible process.*

Proof Sketch. The proof relies on the Turing universality of the UWCA substrate. Any purported universal predictor can be diagonalized against by a subsystem implementing an anti-predictor policy. More formally, the prediction problem can be shown to be at least as hard as the Halting Problem. □

This establishes that subsystems can be causal agents whose actions are determined by their internal state but are not pre-computable by any other part of the system. This provides a formal basis for agency and choice within a deterministic, self-contained universe.

B Tile family catalogue for Σ_{GTE} and macro–step schedule

B.1 Geometry and lanes

We use a radius-1 one–dimensional UWCA on a ring indexed by a finite prime window

$$P = \{p_1 < p_2 < \cdots < p_t\}, \qquad M := \prod_{i=1}^{t} p_i,$$

with M chosen *strictly larger* than all register maxima in the horizon (for n=10, $M > 65535$ suffices). Each site i carries, in clopen cylinders, the residues of the following registers in $\mathbb{F}_{p_i}$:

$$\mathbf{Reg} = \{a, b, c, q, m,\ \hat{q},\ \kappa,\ F\} \quad \text{and a phase bit } \phi \in \{\mathsf{odd}, \mathsf{even}\}.$$

A fixed "control lane" orchestrates sweep order. CRT (Garner) carries move along the p_i direction (nearest-neighbor).

B.2 Tile families (local rewrite schemes)

Each family below is realized by finitely many tiles (one per local residue pattern). "Nbrhd" records the locality (always radius 1); "Guard" is the clopen precondition; "Effect" is the per-site update in $\mathbb{F}_{p_i}$ (cross-prime carries handled by the Garner sweep); "Cost" counts sweeps per macro call.

Family	Nbrhd	Guard	Effect (per site i)	Cost
COPY$(X \to Y)$	1	control step	$Y_i \leftarrow X_i$	1
SWAP(X, Y)	1	control step	$X_i \leftrightarrow Y_i$	1
ADDc$(X{+}{=}k)$	1	control step	$X_i \leftarrow X_i + k \bmod p_i$	1
SUBc$(X{-}{=}k)$	1	control step	$X_i \leftarrow X_i - k \bmod p_i$	1
ADD$(X{+}{=}Y)$	1	control step	$X_i \leftarrow X_i + Y_i$	1
SUB$(X{-}{=}Y)$	1	control step	$X_i \leftarrow X_i - Y_i$	1
MULc$(X{\cdot}{=}k)$	1	control step	$k\,X_i$	1
MUL$(Z{:=}X \times Y)$	1	control step	schoolbook partials, Garner carries	$O(t)$
ABS_DIFF$(Z|{:=}|X{-}Y|)$	1	control+cmp flag	sitewise update + global sign carry	$O(t)$
CMP$(X, Y \to \text{flag})$	1	control step	lexicographic compare via Garner	$O(t)$
DIVREM$(c, b \to q, m)$	1	control step	long–division (shift–sub) in CRT	$O(t)$
GARNER_FWD/BWD	1	carry lane on	propagate carries across p_i	$O(t)$
LATCH(b, q)	1	control step	write $b^{\ell} \leftarrow b,\ q^{\ell} \leftarrow q$	1
UNLATCH	1	control step	clear latched marks	1
PHASE_FLIP	1	always	ϕ : odd $\leftrightarrow$ even	1
FIB_FD$(\kappa \to F)$	1	control step	fast-doubling recurrences	$O(\log \kappa)$

Table 4: Tile families comprising Σ_{GTE}. Garner sweeps ensure integer-correct results from residue lanes.

Legality. Each tile rewrites clopen cylinder sets to clopen cylinder sets; Garner recombination/reduction maps legal cylinders to legal cylinders. Thus survivor constraints are preserved by construction (cf. the sheaf model in §11.0.1).

B.3 Macro–step schedule (odd/even phases)

Let n be fixed. One T–step is compiled as the following sweep script (comments reference Def. 1.2).

```
# --- common prologue ---
LATCH(b,q)                    # keep pre-update (b,q) for writing c := b*q
    + 15

# --- q,m from (b,c) ---
DIVREM(c,b $\rightarrow$ q,m)          # Definition \ref{def:
   ugp_update_map}, both phases

# --- phase-specific branch ---
if $\phi$ == odd:
```

```
    a $\leftarrow$ m - (12 - n)
    b $\leftarrow$ b - (m + q)
    c $\leftarrow$ b$^\ell$ * q$^\ell$ + 15     # uses latched pre-
       update values
else: # $\phi$ == even
    a $\leftarrow$ m - (12 - n)
    $\kappa$ $\leftarrow$ |q - q$^\ell$|
    F $\leftarrow$ Fib_FD($\kappa$)          # fast-doubling (Appendix)
    b $\leftarrow$ b + F
    c $\leftarrow$ b$^\ell$ * q$^\ell$ + 15

PHASE_FLIP()
UNLATCH()
```

Specialization at $n = 10$. Under UGP-1, m_2=15 and $|q_2 - q_1|$=13, so the even–step reduces to $b \leftarrow b + 233$ (constant increment).

B.4 Correctness and cost (detail for Thm. 3.39)

Line-by-line refinement: each statement in the schedule is one call to a family in Table 4. The DIVREM/GARNER modules ensure that register semantics coincide with $\mathbb{Z}$ arithmetic on $[0, M)$; latching guarantees the c-write uses pre-update b, q as in Def. 1.2. The sweep bound per macro–step is

$$O(t) \ + \ O\big(t \cdot \text{DivRemCost}\big) \ + \ O\big(t \log \kappa\big),$$

with the last term absent at n=10 (constant $\kappa = 13$).

B.5 Minimal rule card for $n = 10$

This appendix restates the canonical UWCA rule card of §3.5.6 in its compact, reference-ready form. At level n=10 and ridge R=1008:

Odd: $(q, m) \leftarrow \text{DivRem}(b, c)$,	$a \leftarrow m{-}2$,	$b \leftarrow b - (m + q)$,	$c \leftarrow 2^{10} - 1$;
Even: $(q, m) \leftarrow \text{DivRem}(b, c)$,	$a \leftarrow m{-}2$,	$b \leftarrow b + 233$,	$c \leftarrow 2^{16} - 1$;
ϕ toggles; (b^ℓ, q^ℓ) are the latched pre-update values.			

This is the concrete UWCA "rule" Σ_{GTE} specialized at n=10.

C Self-reference functional and advanced concepts

B.1 Self-reference functional σ

Definition C.1 (Self-reference functional σ)**.** *For a (finite or infinite) trajectory* $\mathbf{G} = (G_1, G_2, \dots)$ *define the* self-reference functional

$$\sigma(\mathbf{G}) \ := \ \#\Big\{(i, j, \phi) : 1 \le i < j, \ \phi \in \Phi, \ \phi(G_i) = G_j\Big\},$$

where Φ *is the countable family of coordinatewise* derivative maps

$$\Phi := \{\mathrm{id},\ \pi_a,\ \pi_b,\ \pi_c,\ \Delta_q,\ \Delta_m,\ \textit{finite compositions thereof}\},$$

with $\pi_\bullet$ *extracting a single coordinate as a constant-triple and* $\Delta_q(G_k) := (0, |q_k - q_{k-1}|, 0)$*,* $\Delta_m(G_k) := (m_k, 0, 0)$*. Thus* σ *counts exact recurrences of a state by reappearance of its own elements/derivatives along the trajectory.*

Remark C.2 (Theoretical significance)**.** *The self-reference functional* σ *provides a measure of the internal complexity and recurrence structure of trajectories. While not used in the main proofs of this paper, it represents an important theoretical tool for analyzing the self-similarity and fractal properties of UGP dynamics. Future work may explore bounds on* σ *for canonical orbits and its relationship to information-theoretic measures of trajectory complexity.*

D Code-to-Proof Correspondence for the UWCA Realization of Rule 110

Goal

Prove that the implementation in UWCA Micro-Gadget Demo (mode `uwca`) computes exactly Rule 110 on the binary sector, matching the fully spelled-out schedule in the proof.

B.6 Rule and sector

- Rule 110 is encoded in `ecaNextBit(rule, L, C, R)`, where `idx = (L«2)|(C«1)|R` (Wolfram order 111,110,...,000) and the output is `(rule » idx) & 1`.
- In `mintermList(rule)` we enumerate the 1-minterms $\{(\ell, c, r)\}$ by reading the same bit positions. This yields exactly $S = \{110, 101, 011, 010, 001\}$ for `rule=110`.

The binary sector is the set of states with `curRow` $\in \{0,1\}^W$. All other rails used below are ephemeral (allocated each step), so sector closure is automatic.

B. Pass schedule (code $\leftrightarrow$ proof)

We compare with passes (P1)–(P4) from the proof; each pass is computed locally and synchronously in `uwcaEmulateStep(cur, rule, wrap)`:

(P1) Neighbor distribution.

```
for (let i=0;i<n;i++){
  L[i] = wrap ? cur[(i-1+n)%n] : (i===0 ? 0 : cur[i-1]);
  C[i] = cur[i];
  R[i] = wrap ? cur[(i+1)%n] : (i===n-1 ? 0 : cur[i+1]);
}
```

This sets rails $L_i = C_{i-1}$, $C_i = C_i$, $R_i = C_{i+1}$ (torus if `wrap`, zero-padded otherwise).

(P2) Minterm detection (one rail per 1-pattern).

```
for (const [Lm,Cm,Rm] of mts){
  const out = new Uint8Array(n);
  for (let i=0;i<n;i++){
    const okL = Lm ? L[i]===1 : L[i]===0;
    const okC = Cm ? C[i]===1 : C[i]===0;
    const okR = Rm ? R[i]===1 : R[i]===0;
    out[i] = (okL & okC & okR) ? 1 : 0;   // this is M^u_i
  }
  minrails.push(out);
}
```

Here each `out` is exactly the predicate $[L_i = \ell] \wedge [C_i = c] \wedge [R_i = r]$ for one $u = (\ell, c, r) \in S$.

(P3) OR-accumulation → NEXT.

```
const NEXT = new Uint8Array(n);
for (const rail of minrails){
  for (let i=0;i<n;i++) NEXT[i] |= rail[i];
}
```

Thus $N_i = \bigvee_{u \in S} M_i^u$.

(P4) Commit.

`uwcaEmulateStep` returns `{ nextRow: NEXT }`, and the caller assigns it to `curRow` before painting the new generation. Ephemeral rails are discarded, which is equivalent to "clear auxiliaries" in the proof.

C. Correctness (pointwise)

Fix site i and let $(L, C, R) = (C_{i-1}, C_i, C_{i+1})$ after (P1).

- Exactly one triple (ℓ, c, r) equals (L, C, R).
- The OR in (P3) is 1 iff $(L, C, R) \in S$, which is iff Rule 110 outputs 1 on that neighborhood.
- Commit (P4) writes $C_i' := N_i = f_{110}(C_{i-1}, C_i, C_{i+1})$.

Therefore the next visible row equals the Rule 110 update at every site, for any width and either boundary (wrap torus, or zero boundary).

D. Boundary and ordering notes

- **Boundary:** The code supports both toroidal and zero padding; the proof works for either (the local map is identical away from the boundary, and the chosen boundary condition matches `directStep`'s behaviour).
- **Bit ordering:** `idx=(L«2)|(C«1)|R` ensures `mintermList` and `ecaNextBit` read the rule table in the same Wolfram order, so the selected minterms are exactly the 1-entries of Rule 110.

E. Equivalence with the reference "Direct CA" path

Let `Direct(cur,110,wrap)` be the single-pass lookup (`directStep`). Let `UWCA(cur,110,wrap)` be the composition of passes above (`uwcaEmulateStep`). Then for all binary rows `cur` and either
boundary:

$$\boxed{\texttt{UWCA(cur,110,wrap)} = \texttt{Direct(cur,110,wrap)}}$$

This is precisely the theorem proved in the main text specialized to the code paths here.

F. Non-theorems (to avoid confusion)

- The `uwca-arith` mode performs per-prime add/sub/cmp demos; it is *not* the Rule 110 micro-rule.
- The large GTE→UWCA (N=10) demo shows a separate arithmetic dynamics (ridge/kick etc.); it does *not* instantiate the minterm UWCA used for Rule 110.

Implementation reference

The complete UWCA implementation with Rule 110 emulation is available in the repository at:
`Optimizer_new_tests/si_optimizer_data/UGP_Paper/UWCA_1D_CA_Emulator_DEMO.html`

This HTML file contains the full interactive demonstration of the UWCA micro-gadget that implements the formal proof schedule described in Theorem 23.5.

E Reference harness (exact trace + UWCA-compiled run)

Purpose. Emit (i) the arithmetic trace of T and (ii) the UWCA-compiled run using the macro–step schedule of Appendix B, then assert equality at each stage. Optionally write CSV `gte_uwca_trace.csv` for plotting.

Listing 2: Minimal verification harness (Python 3.x)

```
# ---- CRT helpers for an odd-prime window P (choose M > all register
   maxima) ----
import csv
from math import prod

# This is a standalone script implementing the corrected logic from
   Appendix D of the UGP paper.

# ---- CRT helpers (included for completeness as in the original
   appendix code) ----
P_N10 = [3, 5, 17, 19, 257]
M = prod(P_N10)

def crt_pack(res, P=P_N10):
```

```
    """res:␣list␣of␣residues␣mod␣p_i;␣returns␣x␣in␣[0,M)␣consistent␣
       with␣CRT."""
    from math import prod
    M = prod(P)
    result = 0
    for mi, ri in zip(P, res):
        Mi = M // mi
        inv = pow(Mi, -1, mi)  # modular inverse
        result += ri * Mi * inv
    return result % M

def crt_unpack(x, P=P_N10):
    return [x % p for p in P]

# ---- GTE arithmetic helpers ----
def divrem(c, b):
    """Exact␣integer␣division."""
    q = c // b
    m = c - q * b
    return q, m

def fib_fast_doubling(n):
    """Computes␣the␣nth␣Fibonacci␣number␣using␣fast␣doubling."""
    def fd(k):
        if k == 0: return (0, 1)
        a, b = fd(k >> 1)
        c = a * (2 * b - a)
        d = a * a + b * b
        return (c, d) if k % 2 == 0 else (d, c + d)
    return fd(n)[0]

# ---- Corrected GTE Step Functions ----
def T_step(state, n, phase):
    """
␣␣␣␣Computes␣one␣step␣of␣the␣GTE␣evolution␣according␣to␣the␣rules␣in␣
   the␣paper.
␣␣␣␣"""
    a, b, c, q_prev = state
    if phase == "odd":
        # This is the UGP-1 ridge step (t=1).
        q, m = divrem(c, b)
        # Paper rule: a_t+1 = m_t - (n+2-t). For t=1, n=10: a2 = m -
           (10+2-1) = m - 11
        # odd step (t = 1): a2 = m - (n + 1)
        a2 = m - (n + 1)
        # Paper rule: b_t+1 = b_t - (m_t+q_t)
        b2 = b - (m + q)
```

```
        # Paper rule for the ridge step: c2 is defined as 2^n - 1
        c2 = (1 << n) - 1
        # The quotient for this step is q, which becomes the q_prev for the next step.
        q2 = q
        phase2 = "even"
    else: # phase == "even"
        # This is the even step (t=2).
        q, m = divrem(c, b)
        # Paper rule: b_t+1 = b_t + F_|q_t - q_{t-1}|
        # At n=10, the gap |q2-q1| is fixed to 13.
        fib_index = abs(q - q_prev)
        F = fib_fast_doubling(fib_index)
        b2 = b + F
        # Paper rule: a_t+1 = m_t - (n+2-t). For t=2, n=10: a3 = m - (10+2-2) = m - 10
        # even step (t = 2): a3 = m - n
        a2 = m - n
        # Paper rule for the canonical orbit: c3 is 2^16 - 1
        c2 = 65535
        # The quotient for this step is q.
        q2 = q
        phase2 = "odd"
    return (a2, b2, c2, q2), phase2

def UWCA_macro(state, n, phase):
    """
    This function mirrors T_step to confirm the UWCA macro-rule logic is identical.
    """
    a, b, c, q_prev = state
    if phase == "odd":
        # This is the UGP-1 ridge step (t=1).
        q, m = divrem(c, b)
        # Paper rule: a_t+1 = m_t - (n+2-t). For t=1, n=10: a2 = m - (10+2-1) = m - 11
        # odd step (t = 1): a2 = m - (n + 1)
        a2 = m - (n + 1)
        # Paper rule: b_t+1 = b_t - (m_t+q_t)
        b2 = b - (m + q)
        # Paper rule for the ridge step: c2 is defined as 2^n - 1
        c2 = (1 << n) - 1
        q2 = q
        phase2 = "even"
    else: # phase == "even"
        # This is the even step (t=2).
        q, m = divrem(c, b)
```

```
        # Paper rule: b_t+1 = b_t + F_|q_t - q_{t-1}|
        # At n=10, the gap |q2-q1| is fixed to 13.
        fib_index = abs(q - q_prev)
        F = fib_fast_doubling(fib_index)
        b2 = b + F
        # Paper rule: a_t+1 = m_t - (n+2-t). For t=2, n=10: a3 = m -
           (10+2-2) = m - 10
        # even step (t = 2): a3 = m - n
        a2 = m - n
        # Paper rule for the canonical orbit: c3 is 2^16 - 1
        c2 = 65535
        q2 = q
        phase2 = "odd"
    return (a2, b2, c2, q2), phase2

# ---- Corrected Verification Harness ----
def run_and_check(seed, n=10, steps=2, write_csv=True, csv_path="
   gte_uwca_trace.csv"):
    """
␣␣␣␣Runs␣both␣traces␣and␣checks␣for␣exact␣agreement,␣producing␣the␣
   canonical␣orbit.
␣␣␣␣"""
    a, b, c = seed
    phase = "odd"

    # Initial state for arithmetic trace
    stA_current = (a, b, c, 0) # q_prev is 0 initially
    # Initial state for UWCA-compiled trace
    stU_current = (a, b, c, 0)

    rows = [("k", "phase", "a", "b", "c", "q_calculated")]

    # Add initial state to trace for clarity
    q_initial, _ = divrem(c, b)
    rows.append((0, "initial", a, b, c, q_initial))

    for k in range(1, steps + 1):
        # The q_prev for the current step is the quotient from the *
           previous* state
        q_prev_A, _ = divrem(stA_current[2], stA_current[1])
        q_prev_U, _ = divrem(stU_current[2], stU_current[1])

        stA_with_q_prev = (stA_current[0], stA_current[1], stA_current
           [2], q_prev_A)
        stU_with_q_prev = (stU_current[0], stU_current[1], stU_current
           [2], q_prev_U)
```

```
        stA_next, phaseA = T_step(stA_with_q_prev, n, phase)
        stU_next, phaseU = UWCA_macro(stU_with_q_prev, n, phase)

        assert phaseA == phaseU
        a_next, b_next, c_next, _ = stA_next # The returned q is not
            needed for the trace row

        assert stA_next == stU_next, f"Mismatch at step {k}"

        # The quotient to be recorded is the one calculated *in this
            step* to produce the *next* state
        q_calculated_this_step, _ = divrem(stA_current[2], stA_current
            [1])

        rows.append((k, phase, a_next, b_next, c_next,
            q_calculated_this_step))

        phase = phaseA
        stA_current = stA_next
        stU_current = stU_next

    if write_csv:
        with open(csv_path, "w", newline='') as f:
            writer = csv.writer(f)
            writer.writerows(rows)
    return rows

if __name__ == "__main__":
    # Main example from the paper
    # Expected output for n=10:
    # G1 (initial): (1, 73, 823) -> q1=11, m1=20
    # G2 (odd step): (9, 42, 1023) -> q2=24, m2=15
    # G3 (even step): (5, 275, 65535)

    print("Running GTE verification harness from Appendix D...")
    rows = run_and_check(seed=(1, 73, 823), n=10, steps=2, write_csv=
        True)
    print("OK; traces agree. Wrote gte_uwca_trace.csv")

    # Print the trace to console for verification
    print("\n--- GTE Evolution Trace ---")
    # Print header
    print(f"{rows[0][0]:<3} {'Phase':<8} {'a':<5} {'b':<5} {'c':<8} {'
        q_calc'}")
    print("-" * 45)
    # Print data rows
    for row in rows[1:]:
```

```
        print(f"{row[0]:<3} {row[1]:<8} {row[2]:<5} {row[3]:<5} {row
            [4]:<8} {row[5]}")

    # Assert final state to be certain
    final_row = rows[-1]
    # final_row format: (k, phase, a, b, c, q_calculated)
    final_state = final_row[2:5]
    expected_final_state = (5, 275, 65535)
    assert final_state == expected_final_state, f"Final state {
        final_state} does not match canonical orbit {
        expected_final_state}"
    print("\nFinal state matches the canonical GTE orbit from the paper
        .")
```

Notes for use.

- The harness simulates the *macro* UWCA schedule exactly; it does not render per-tile space–time, which would be much longer but mechanically derivable from Table 4.
- For n=10 the second step uses κ=13 $\Rightarrow$ F_{13}=233; you can enforce this by replacing `fib_fast_doubling(k)` with the literal 233.
- The CSV `gte_uwca_trace.csv` has columns `k,phase,aA,bA,cA,qA,aU,bU,cU,qU`. The expected output for the example seed $(1, 73, 823)$ is:
  - **Odd step**: $(9, 42, 1023, 11)$ where $a = m - 11 = 20 - 11 = 9$
  - **Even step**: $(5, 275, 65535, 24)$ using deterministic ridge facts

  If you wish, add a small `pgfplots` snippet mirroring Fig. 5 by reading this file.

B.1 Checklist for robustness

(a) Fix P so that $M = \prod P$ exceeds all register maxima in the intended horizon.

(b) Publish the concrete tile set for Table 4 (one row per local residue pattern). The family catalogue already pins down behavior; the tile list is finite and mechanical to expand.

(c) Include the harness output as an artifact and ensure bit-for-bit agreement with the arithmetic trace on the canonical orbit.

F Verification scripts and reproducibility

We used a single-file Python script to scan ridges, apply the prime-lock, and emit CSV/JSON artifacts (`survivors.csv`, `orders.csv`, and `mirror_report_10_22.json`). The core is the pseudocode in Section 23.1; a minimal reference implementation fits in < 100 lines using deterministic Miller–Rabin bases $\{2, 3, 5, 7, 11, 13, 17\}$ for $c_1 < 2^{64}$.

Listing 3: Minimal ridge scan (reference)

```
def ridge_scan(n: int):
    R = (1 << n) - 16
    for b2 in divisors(R):
        if b2 <= 15:
            continue
        q2 = R // b2
        b1 = b2 + q2 + 7
        q1 = q2 - 13
        c1 = b1 * q1 + 20
        if is_probable_prime(c1):  # MR with bases {2,3,5,7,11,13,17}
            yield (b2, q2, b1, q1, c1)
```

The figure macros in Section 19 automatically load the generated CSVs when present via `\DataDir` (default `.`).

F.1 Code and data availability

The Python stack for regenerating atlas-style CSV/PNG outputs and running the Phase 3 smoke test is recorded in `ugp_release/requirements.txt` (Python 3 with `matplotlib`, `numpy`, and `pandas`; optional `seaborn` and `streamlit` for non-essential tooling). From `ugp_release/`, run `python3 test_phase3.py` to verify the plotting and table pipeline on your machine. Machine-checked formalizations are cited from [Spi26g, Spi26h]; the corresponding Zenodo DOIs and GitHub URL are given in `Spivack_Papers_Bibliography.bib`. Repository-local build order, optional `\DataDir` overrides, and pointers to the Lean artifact are summarized in `PROVENANCE.md` and `REPRODUCE.md` alongside this `.tex` file. Browser demos (`*.html`) and the Streamlit Universe Finder are *optional* and are not required to reproduce the static numerical statements in the main text.

G Supplementary tables

b_2	q_2	b_1	q_1	c_1
24	42	73	29	2137 (prime)
42	24	73	11	823 (prime)

Table 5: Surviving mirror pair at $n = 10$ (UGP-1 admissible).

H Definition of CR1

We formalize the "CR1" (Companion Relation 1) object referenced in Conjecture H.3.

Definition H.1 (CR1: matrix form)**.** *Fix a level n and ridge $R_n = 2^n - 16$. Let* $\mathbf{Reg} = \{a, b, c, q, m, \hat{q}, \kappa, F\}$ *denote the register bundle of the UWCA macro–rule for T (Listing 1). The* CR1 *form is the linearized update*

$$\mathbf{Reg}_{t+1} = M_{\mathrm{CR1}} \cdot \mathbf{Reg}_t + \mathbf{const}$$

over $\mathbb{Q}$, where M_{CR1} is the rational companion matrix encoding the micro–operations (copy, add, subtract, multiply by constants) in the UWCA compilation, restricted to a fixed control path (odd or even branch) and with F treated as a constant on that branch. The constants in **const** *come from hard–wired ridge values (e.g. +15, +7) and level–dependent Mersenne maxima.*

Remark H.2. *The CR1 form is extracted by symbolic tracing of the UWCA macro–rule on formal variables for the registers, recording the affine relations at each step. This yields $M_{\mathrm{CR1}} \in \mathbb{Q}^{d\times d}$ with $d = |\mathbf{Reg}|$. In the $n = 10$ UGP-1 ridge case, the matrix has many small rational entries, motivating Conjecture H.3.*

Conjecture H.3 (Rational companion structure)**.** *For each level n, the linear companion form (CR1) of the macro-update T on the register bundle (after CRT recombination) has entries in $\mathbb{Q}$ with denominators dividing a fixed D_n whose prime factors are contained in those of R_n. Moreover, empirically many entries are small rationals (bounded height $\ll \mathrm{poly}(n)$). This reflects the simple arithmetic nature of the compiled ALU tiles (Lemmas 3.35–3.36).*

I Universal Calibration Law (UCL)

The *Universal Calibration Law* (UCL) is the bridge principle that relates structural outputs of the GTE at a fixed ridge level—including canonical triples, residue constraints, and lawful-compiler minimality—to the calibration of physical parameters without introducing external model-selection knobs beyond those encoded in the UGP substrate. The algebraic and computational core of *this* monograph (kernel symmetry, UWCA universality, and prime-lock ridge structure) is logically prior to any full phenomenological UCL implementation. Detailed derivations of Standard Model masses and couplings via the UCL and canonical triples appear in [Spi25e]; nuclear applications are treated in [Spi25b]. For calibration certificates and verifier-linked reproducibility of phenomenological numbers, see those companion manuscripts and their artifact bundles.

J Glossary

GTE Generative Triple Evolution

UGP Universal Generative Principle

UWCA Universal Windowed Cellular Automaton

UCL Universal Calibration Law

Ridge The structured sequence $R_n = 2^n - 16$

Prime-lock The constraint mechanism ensuring primality of c_1

Survivor A configuration that satisfies all UGP constraints

Mirror pair A symmetric pair of survivors under the duality transformation

K The PR-1 Operational Law (Historical Note)

The PR-1 operator was an early candidate for the transputational substrate of the UGP. It was studied as a deterministic, local, reversible 1D Cellular Automaton (realized as a UWCA) combining GTE arithmetic evolution with Global Symmetry Adjudication (GSA) at choice points (see Appendix A). The cell-state vector included GTE registers (a, b, c, q, m), physics field registers, and meta-structure registers (`DissonanceGradient`, `AdjudicationFlag`, `AgentBias` B_A, `Spontaneity` Ξ); the update pipeline proceeded through invariant check, dissonance gradient calculation, adjudication, and GSA resolution.

The UGP Physics programme has since moved to a more rigorously specified transputational framework. The formal definition of the Transputation operator (PT) and its forcing theorem (`closed_choice_forces_transputation`) are provided by the NEMS programme (see the NEMS Hub [Spi26d]); the DSAC constructive realization of PT is the current canonical substrate. The PR-1 model developed here is retained for historical context and to document the conceptual evolution of the choice-resolution mechanism. It is not the operative substrate in later papers in the series.

L Connections to the NEMS Programme

The following table maps key results from this monograph to the corresponding NEMS paper, Lean theorem name, and Zenodo DOI. The NEMS suite (Papers 00–92) is machine-checked in `nems-lean`; the UGP results are in `ugp-lean` [Spi26g, Spi26h]. All cited theorems have zero `sorry` and zero custom axioms.

Result	Source	Lean Theorem	DOI
SM gauge structure	NEMS 05	`gauge_signature_rigidity`	...721
Transputation forced	NEMS 76	`closed_choice_forces_transputation`	...882
DSAC as PT realization	NEMS 77	(DSAC theorems)	...884
Arrow of time	NEMS 36	`ArrowOfTime.closure_arrow_theorem`	...788
No hypercomputation	NEMS 35	`Hypercomputation.*`	...786
NEMS→MFRR bridge	NEMS 08	(bridge theorems)	...729
RSUC: unique seed at n=10	ugp-lean	`n10_uniqueness_package`	...538
Quarter-Lock identity	ugp-lean	`quarterLockLaw`	same
Canonical orbit	ugp-lean	`orbit_determined_by_T`	same
Quotient gap $g = 13$	ugp-lean	`quotient_gap_all`	same
$c_1 \equiv 20 \pmod{b_1}$	ugp-lean	`c1Val_mod_b1`	same
UWCA Turing universality	ugp-lean	`uwca_simulates_rule110`	same
Lawvere fixed-point	ugp-lean	`ugp_lawvere_fixed_point`	same
Koide closed form	Paper 18	`KoideClosedForm.*`	same

DOI prefix: `https://doi.org/10.5281/zenodo.194...`; same = 19433538.

Full per-paper DOIs for all 93 NEMS papers are listed at `https://novaspivack.github.io/research/abstracts/`. The concept DOI for the complete NEMS hub is 10.5281/zenodo.19487299 [Spi26d].

M Machine-Checked Core Theorems

The following records each load-bearing result carried as **[T] Lean-checked** in this monograph, giving the prose statement, the `ugp-lean` theorem name, and the artifact DOI. All theorems below are zero `sorry`, zero custom axioms, Lean 4.29.0-rc6/Mathlib 4.29.0-rc6 [Spi26g, Spi26h].

T1. RSUC — Unique seed at $n = 10$. Among the six residual triples surviving the prime-lock filter at $n = 10$, the lexicographically minimal mirror-dual seed is $(a_1, b_1, c_1) = (1, 73, 823)$. *Lean:* `GTE.UniquenessCertificates.n10_uniqueness_package` (six `native_decide` checks, non-circular).

T2. Canonical orbit. The three-step GTE orbit $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$ is machine-verified under the extended information rule (canonicalGen3 definition). *Lean:* `GTE.UpdateMap.orbit_determined_by_T`.

T3. Quotient gap is universal. For all UGP-1 prime-locked seeds, $|q_2 - q_1| = 13$ is an arithmetic consequence of the prime-lock condition, not a free choice. *Lean:* `GTE.PrimeLock.quotient_gap_all`.

T4. Residue class is universal. For all UGP-1 seeds at $n = 10$, $c_1 \equiv 20 \pmod{b_1}$. *Lean:* `GTE.PrimeLock.c1Val_mod_b1`.

T5. Quarter-Lock identity. In the elegant-kernel chart, the Möbius-product coefficient satisfies $k_M = k_{\text{gen2}} + \frac{1}{4}k_{L^2}$ exactly (the Quarter-Lock plane). *Lean:* `ElegantKernel.UCL.quarterLockLaw`.

T6. k_{L^2} geometric derivation. The curvature coefficient satisfies $k_{L^2} = \delta/2^{n-1}$ where $\delta = \log_{B^\star} \varphi$. *Lean:* `ElegantKernel.UCL.k_L2_eq`.

T7. UCL generational constants (fully unconditional). The UCL slope $k_{\text{gen}} = \varphi \cos(\pi/10)$ and concavity $k_{\text{gen2}} = -\varphi/2$ are derived without external fitting. *Lean:* `ElegantKernel.UCL.thm_ucl2_fully_unconditional` and `thm_ucl1_fully_unconditional`.

T8. UWCA simulates Rule 110 (Turing universality). The Universal Windowed Cellular Automaton exactly computes any Rule 110 time step; since Rule 110 is Turing-universal, UWCA is Turing-universal. *Lean:* `Universality.UWCAembedsRule110.uwca_simulates_rule110`.

T9. Lawvere fixed-point (internal self-reference). Every sufficiently expressive object in the survivor space contains a fixed point for any endomap — establishing internal self-reference without an external model. *Lean:* `SelfReference.ugp_lawvere_fixed_point`.

T10. Transputation forced at choice points (NEMS 76). At any internal choice point, the only lawful resolution mechanism is the Transputation operator PT; total effective computation is insufficient. *Lean:* `closed_choice_forces_transputation` (transputation-lean / NEMS 76 [Spi26c]).

T11. Even-step c-invariance. Under the strict per-step formula, $c_3 = c_2 = 1023$ (the even-step ridge leaves c unchanged after the ridge hit). *Lean:* `GTE.Orbit.even_step_c_invariance` and `c3_strict_eq_c2_at_n10`.

T12. Charged-lepton and neutrino mass structure. The Koide mass relation for lepton triplets follows as a closed algebraic identity from the cyclotomic-12 orbit structure. The Koide phase $\theta = 2/9$ is derived from the QCD colour rank $N_c = 3$ via $\theta = (N_c^2 - 1)/(4N_c^2)$. The same N_c also determines the neutrino seesaw exponent $29/9$, which admits three independent structural decompositions:

$$\tfrac{29}{9} = \tfrac{N_c^3 + \text{strand\_count}}{N_c^2} = \tfrac{4N_c^2 - \delta}{N_c^2} = \tfrac{\dim(45_{\mathrm{SU}(5)}) - \dim(16_{\mathrm{SO}(10)})}{N_c^2},$$

with $\delta = 7$ the charged-lepton mirror offset. The SO(10) Majorana Higgs dimension factors as $\dim(126) = 2\,N_c^2\,\delta = 126$, bridging the charged-lepton and neutrino sectors. Full N_c structural chain in Paper 18 [Spi26b]; neutrino sector in Paper 1 [Spi25e]. *Lean:* `MassRelations.KoideClosedForm.*`; `MassRelations.KoideAngle.*` including theorems `koide_angle_from_N_c_pure`, `N_c_determines_everything`, `nu_seesaw_exponent_three_decompositions`, `dim_126_SO10_eq_two_Nc_sq_delta`, the bundled `neutrino_seesaw_structural_closure`, and the underlying value definition `nuSeesawExponent` (:= 29/9, Lean `def`) (30+ theorems plus their underlying `defs`, all zero `sorry`). The seesaw index $29 = \dim(\mathbf{adj}_{SO(10)}) - \dim(\mathbf{spinor}_{SO(10)})$ is the gauge/matter representation defect of the PSC-selected GUT group, machine-checked as `seesaw_index_is_gauge_matter_defect` (`MassRelations.SeesawIndex`, zero `sorry`). A complementary reverse derivation: the per-generation braid winding sum $\sum W_g = N_c(N_c - 3)$ equals zero iff $N_c = 3$ (`anomaly_cancellation_forces_Nc_3`, `BraidAtlas.ChargeTheorem`, zero `sorry`), independently deriving the QCD colour rank from winding anomaly cancellation [Spi25c].

The ugp-lean library (see companion formalization paper [Spi26g]) is available at `https://github.com/novaspivack/ugp-lean`; the Zenodo snapshot DOI is 10.5281/zenodo.19433538.

N Degrees-of-Freedom Ledger

The UGP-1 construction is specified by three integer offsets (s, g, t) appearing in the ridge and prime-lock formulae:

$$b_1 = b_2 + q_2 + s, \qquad |q_2 - q_1| = g, \qquad c_1 \equiv t \pmod{b_1}.$$

Param.	**Value**	**Status**	**Lean theorem**	**Notes**
Quotient gap g	$13 = F_7$	Derived	`quotient_gap_all`	Prime-lock
Residue t	20	Derived	`c1Val_mod_b1`	Ridge n=10
Offset s	7	Param.	—	det{=}7; P6 §2.4.1
D_5 symmetry	—	Conjectural	(open)	Axiom C4, Conj. C

The phrase "parameter-free" in this monograph refers to *prediction-time* free parameters: once the UGP-1 construction is fixed, all downstream arithmetic — the seed $(1, 73, 823)$, the orbit, the invariants — follow with zero fitting. The offsets $g = 13$ and $t = 20$ are not free choices but machine-checked consequences of the prime-lock structure. The offset $s = 7$ is a construction-time parameter whose arithmetic motivation (Gram matrix determinant) is developed in companion Paper 6 [Spi25a]. The D_5 symmetry Axiom C4 is explicitly labeled as conjectural throughout.

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